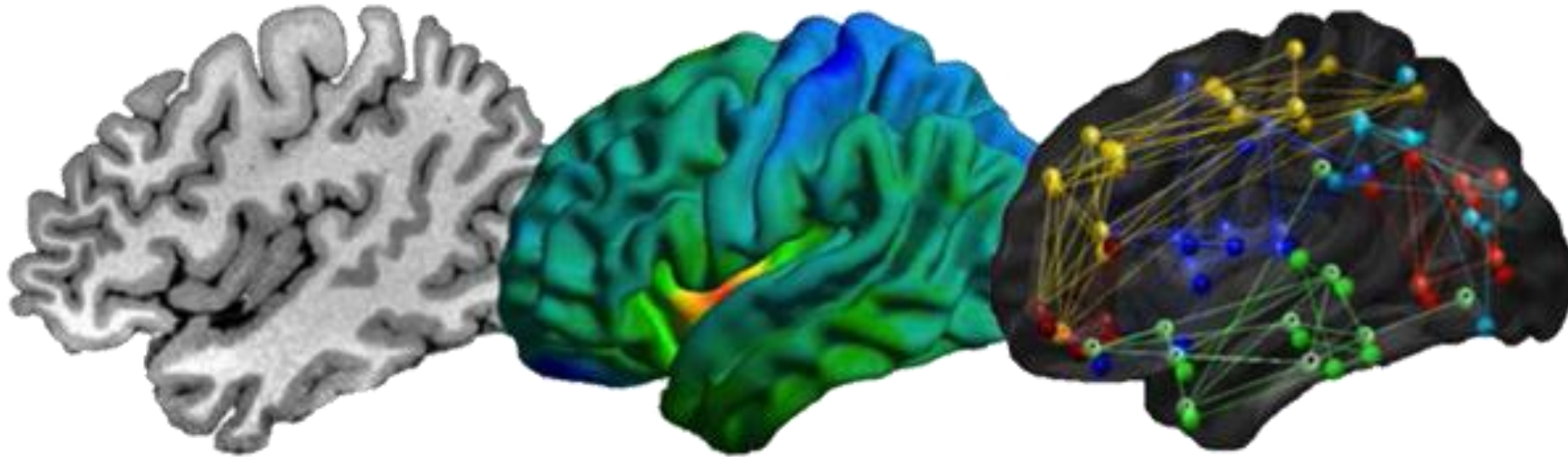


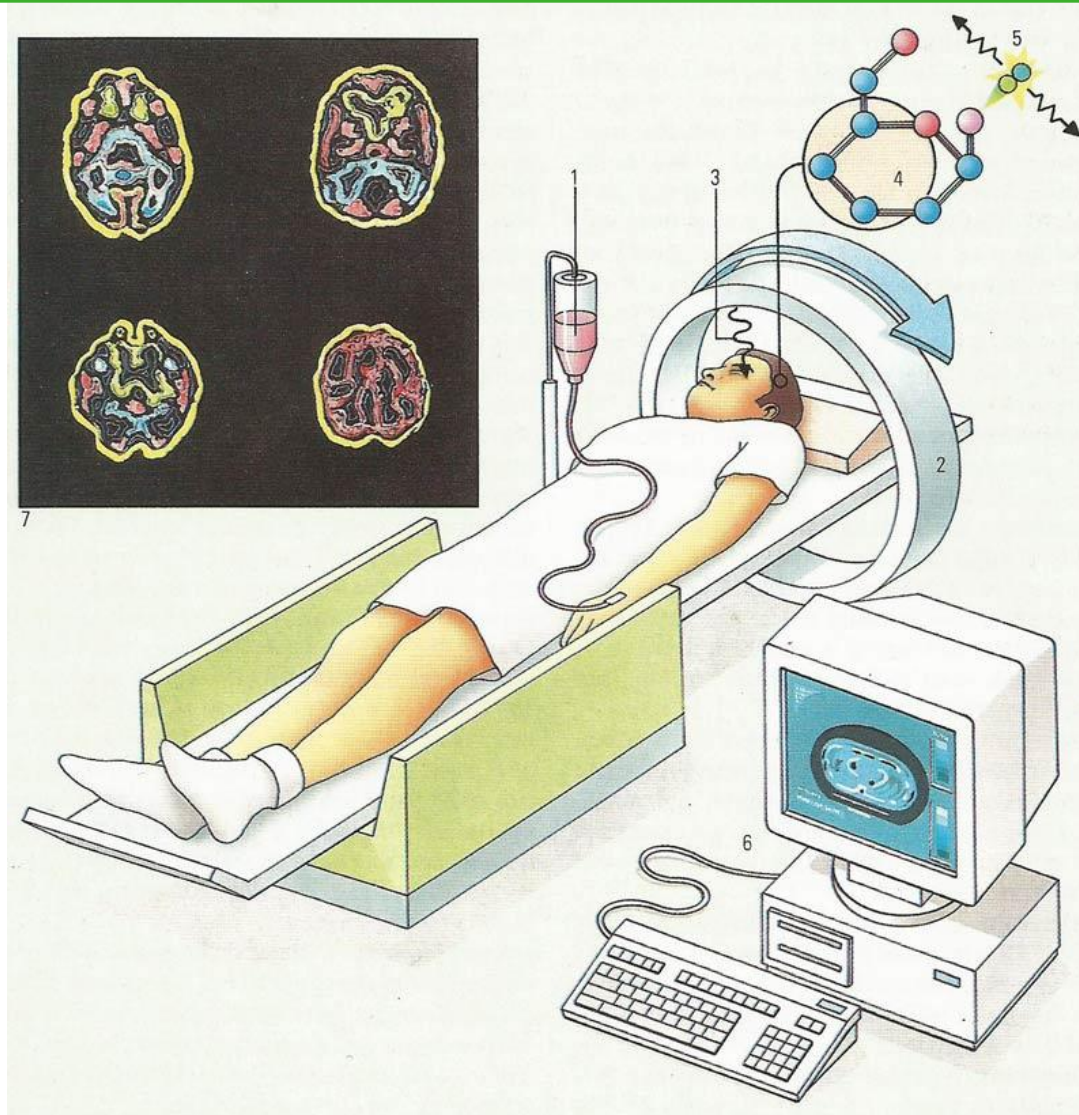


The brain at rest

The Study of Brain Activity In-Vivo

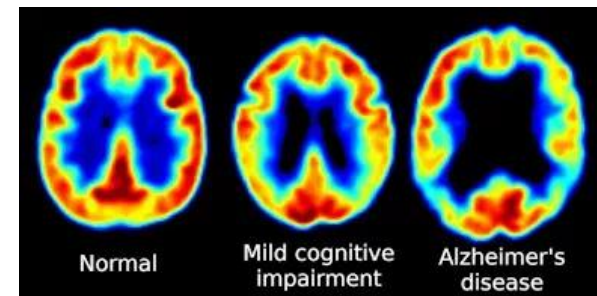


Positron Emission Tomography



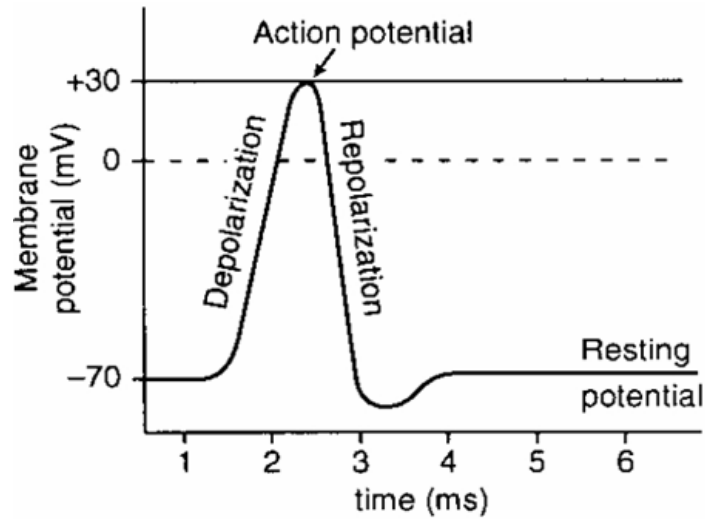
PET measures emissions from radioactively labeled metabolically active chemicals that have been injected into the bloodstream.

Fludeoxyglucose (^{18}F) – FDG-PET

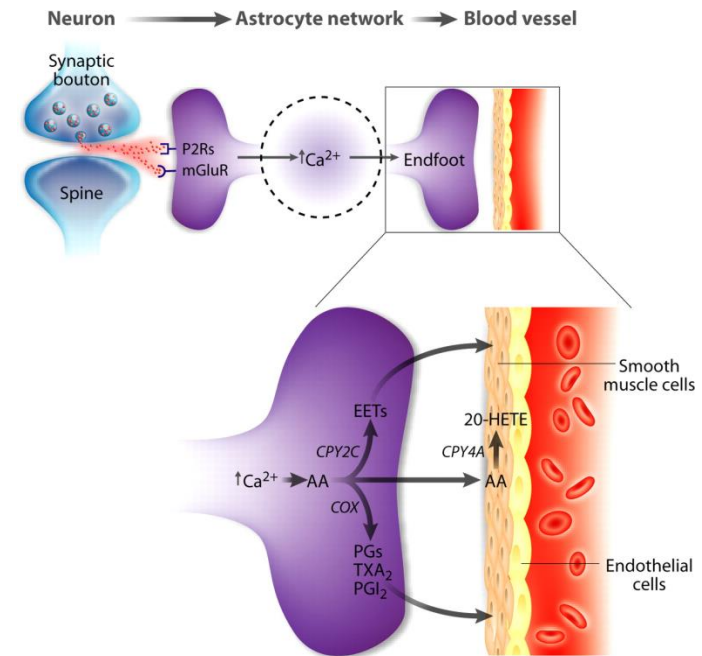


Functional Magnetic Resonance Imaging

Neuronal Activation

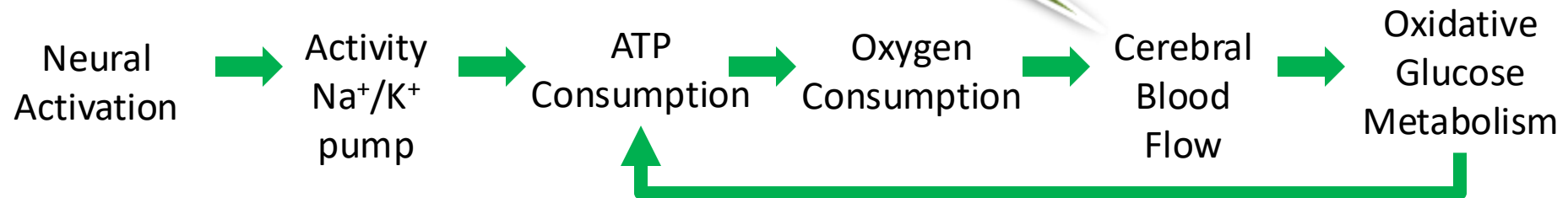


Hemodynamic Response



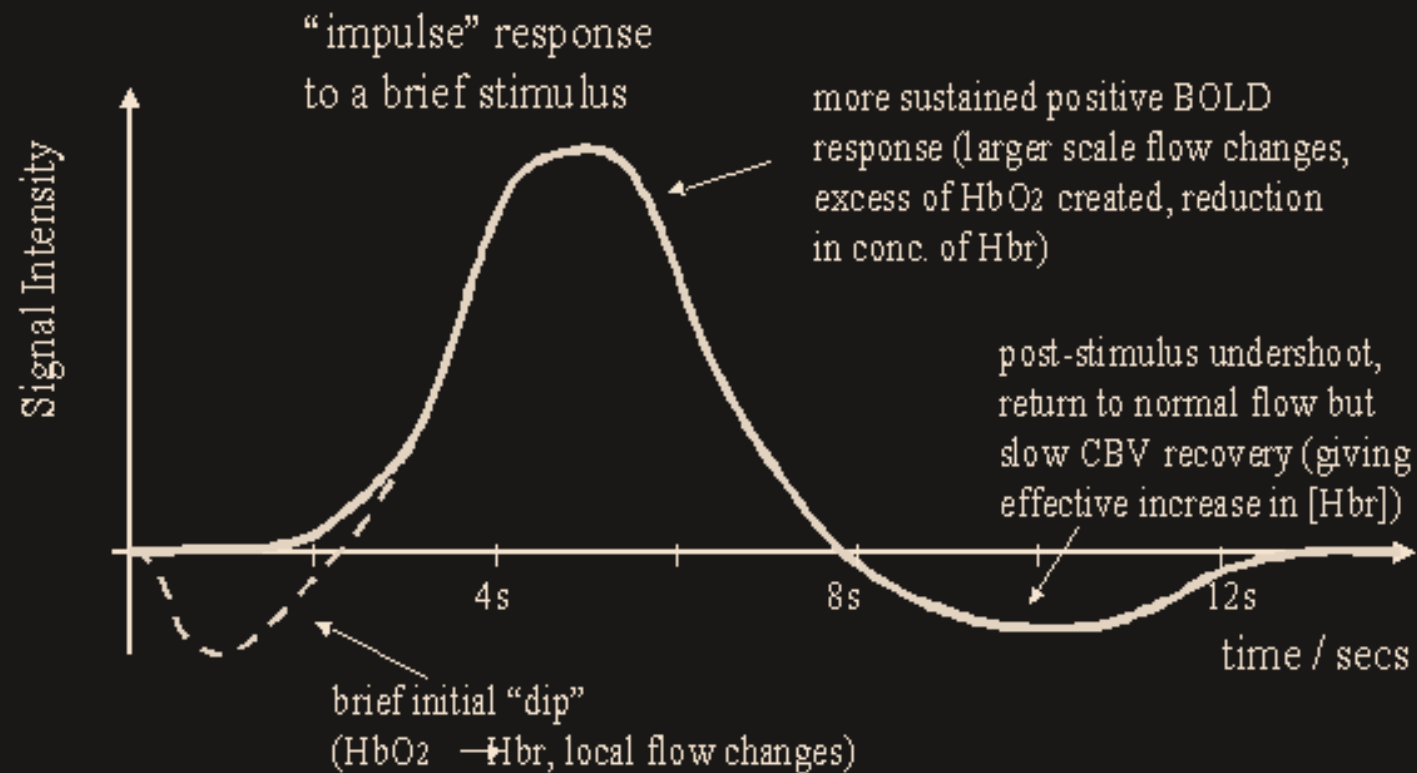
Neurovascular
Coupling

fMRI



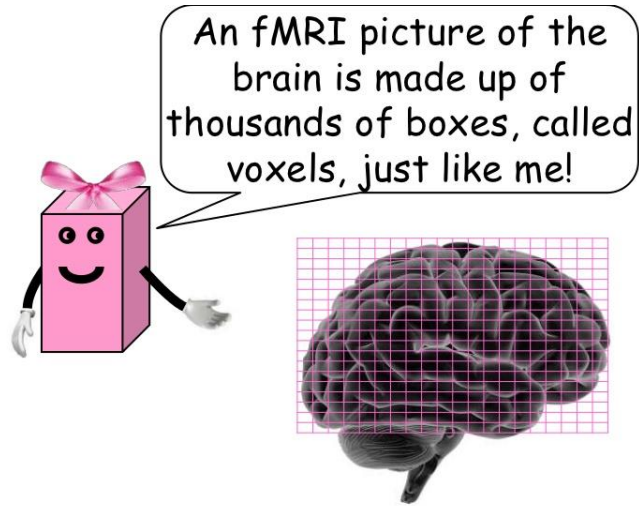
Functional Magnetic Resonance Imaging

BOLD Signal Response

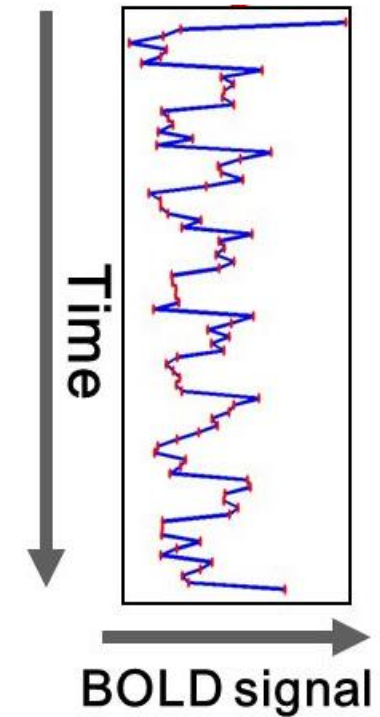
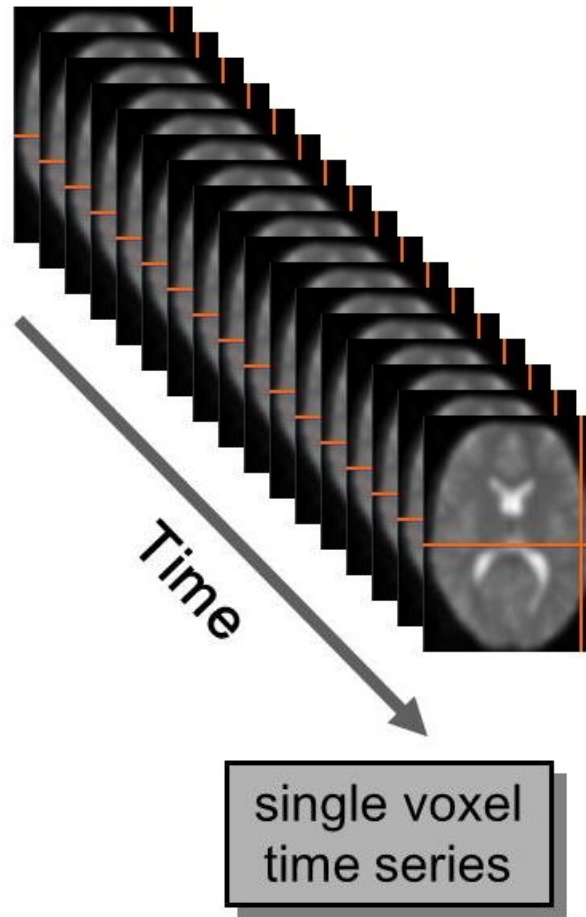


fMRI “dip”: Menon *et al.*, *MRM* 33:453; Ernst & Hennig, *MRM* 32:146; Hu *et al.*, *MRM* 37:877

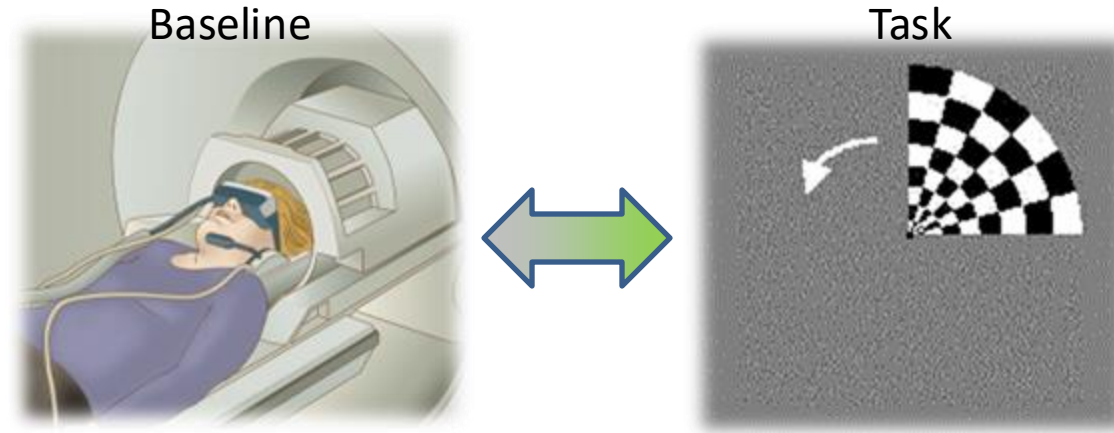
Functional Magnetic Resonance Imaging



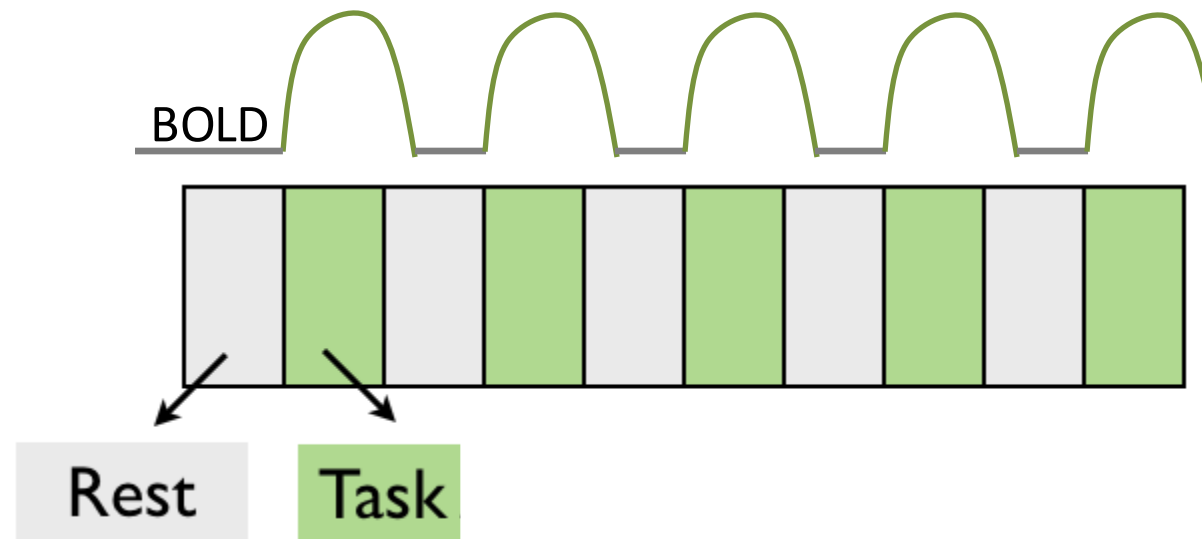
The goal of fMRI data analysis is to detect correlations between brain activation and a task the subject performs during the scan.



Activity Baseline in Brain Functional Studies



Experimental Paradigm – Block Design



*Resting State Brain Activity
and the Default Mode Network*

The Resting State

“Resting State” is an operational definition referring to a constant condition *without imposed stimuli* or other *behaviorally salient events*.

The eyes may be closed or open, with or without visual fixation.

The objective of resting state experiments is to capture the statistical properties of endogenously generated (i.e., spontaneous) neural activity. In contrast, the objective of event-related studies is to measure evoked or induced responses.

History: How active is a «resting» brain?

**OLD
VIEW**



Most neurons are quiet until needed for some activity, such as reading, at which point the brain fires up and expends energy on the signaling needed for the task.

No Activity



High Activity



In 1929 Hans Berger, inventor of the electroencephalogram, deduced from the ceaseless electrical oscillations detected by the device that “we have to assume that the central nervous system is always, and not only during wakefulness, in a state of considerable activity.” But his ideas about how the brain functions were largely ignored, even after noninvasive imaging methods became available in neuroscience laboratories.

High Activity

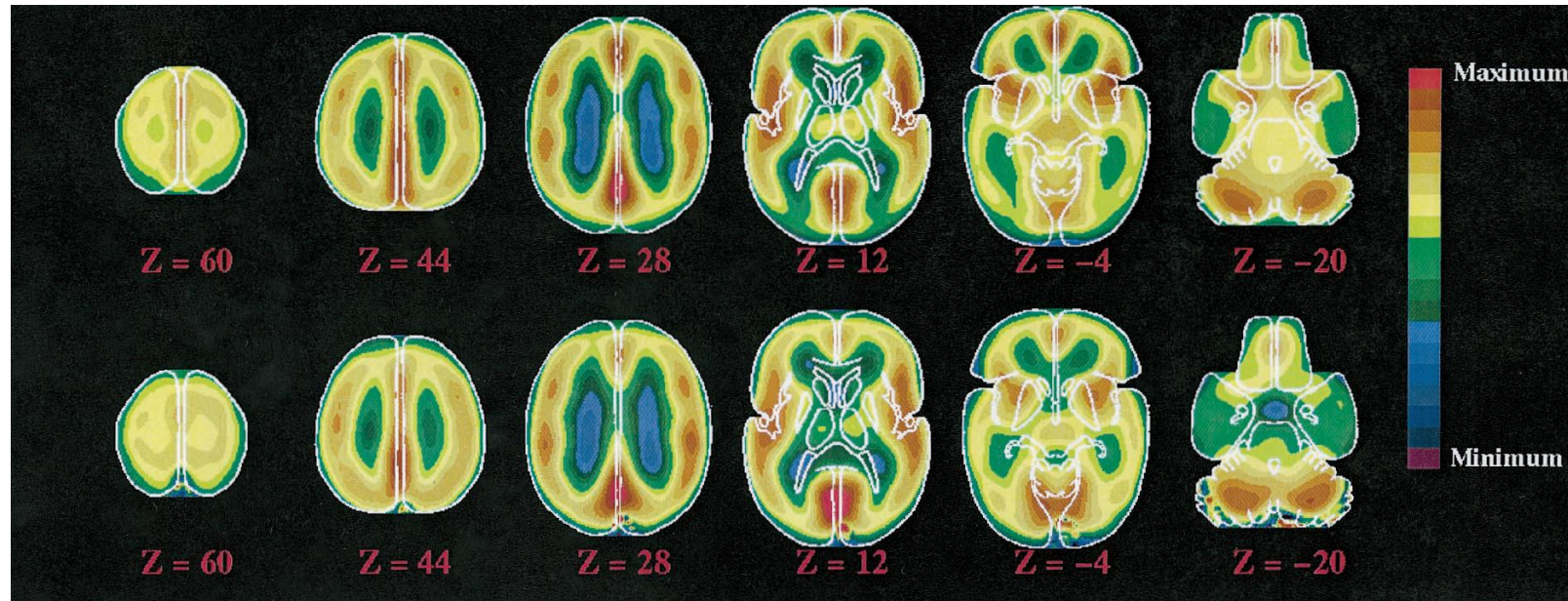


Higher Activity



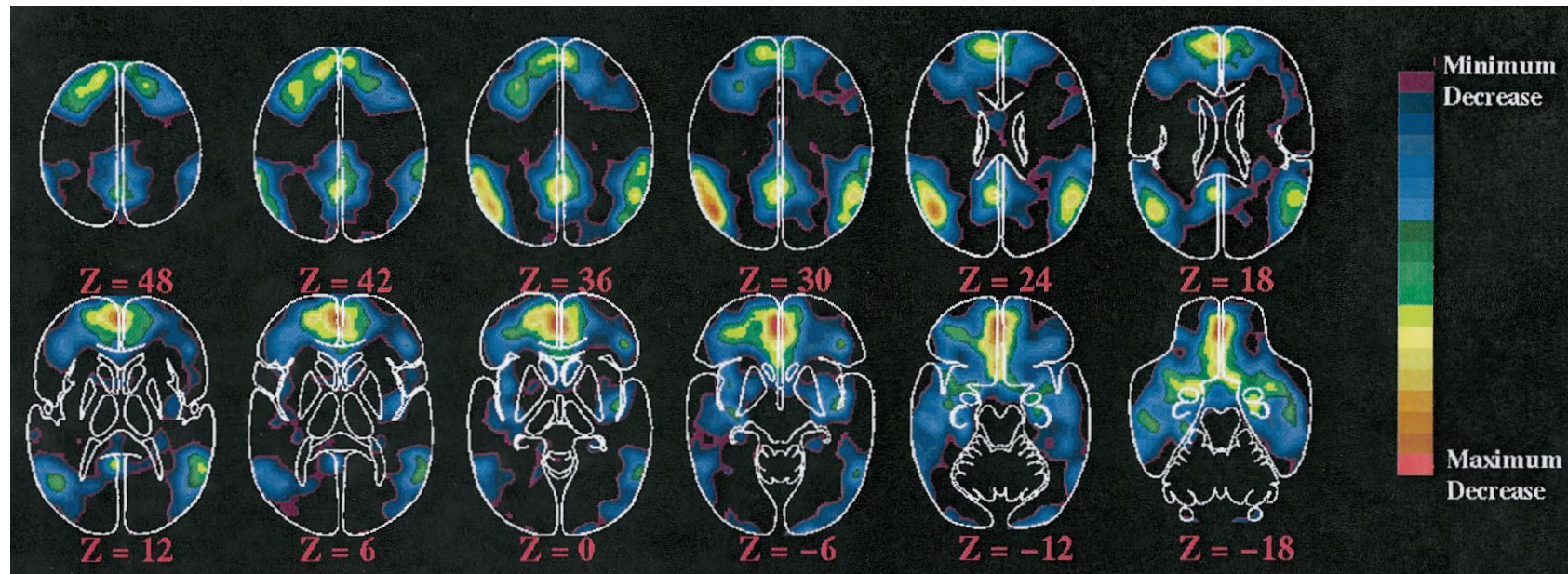
The brain maintains a high level of activity even “at rest.” Behavioral tasks require minimal extra energy (~5% increment) over what is already consumed in the baseline state.

Energy Consumption in the Human Brain at Rest



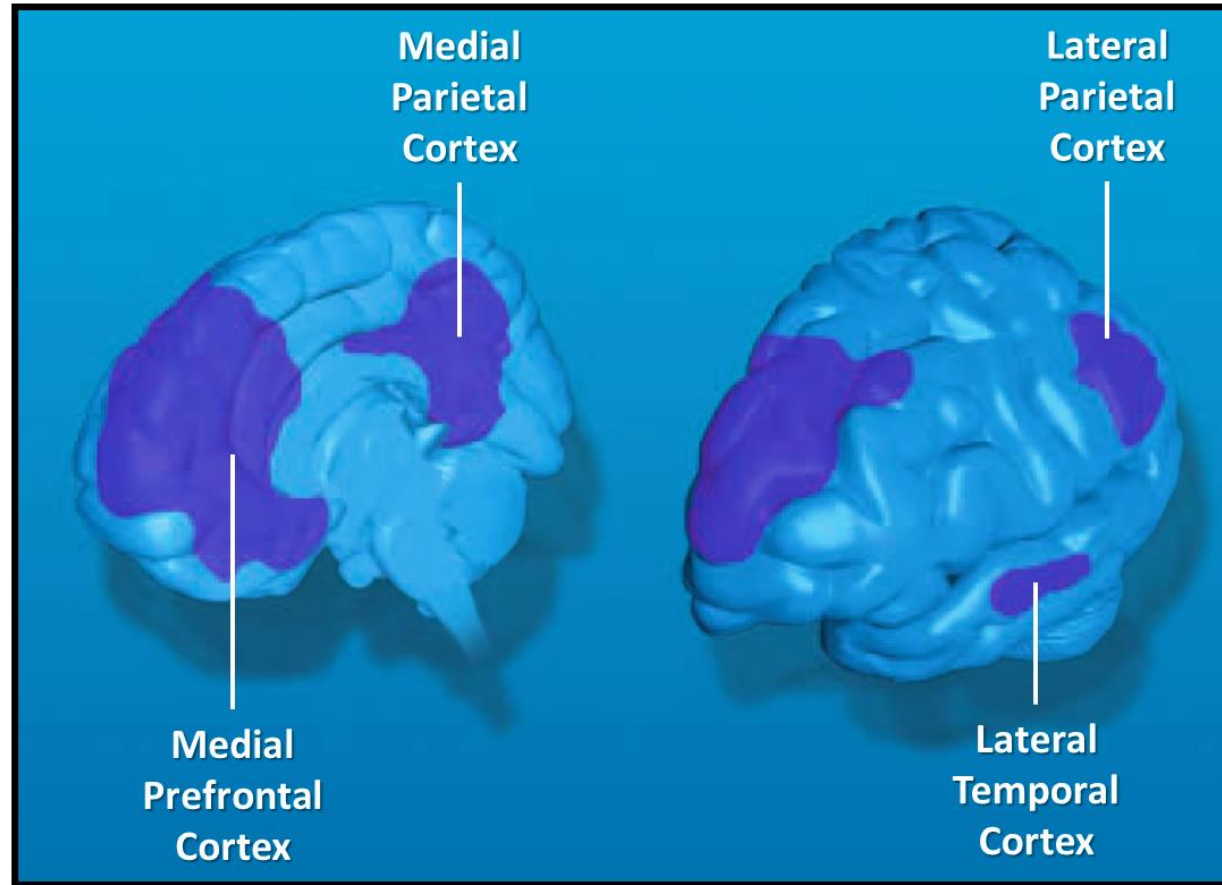
Quantitative maps of blood flow (Upper) and oxygen consumption (Lower) during quiet rest with eyes closed. Note the large variation in blood flow and oxygen consumption across regions of the brain. These vary most widely between gray and white matter. Despite this variation, blood flow and oxygen consumption are closely matched.

History: A Default Mode of Brain Function



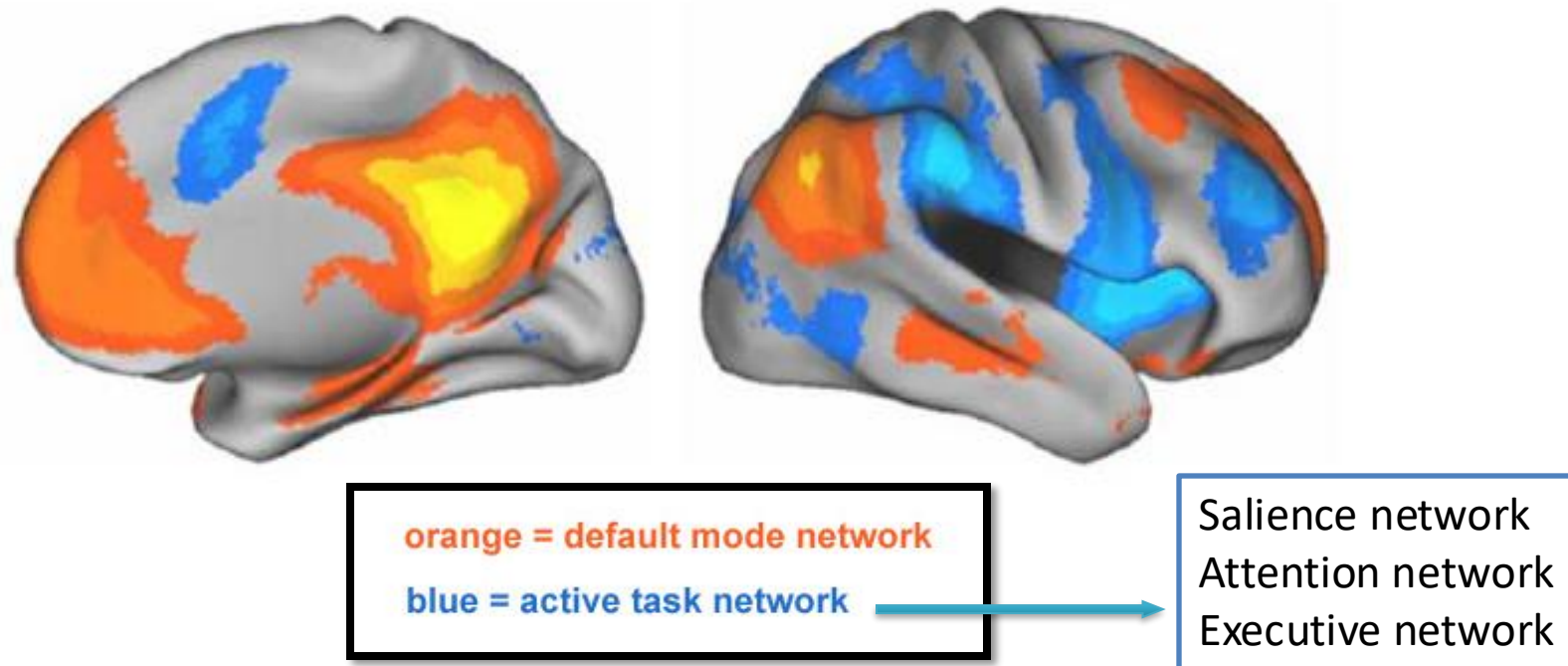
Regions of the brain regularly observed to decrease their activity during attention demanding cognitive tasks. These data represent a meta-analysis of nine functional brain imaging studies performed with PET. In each of the included studies, the subjects processed a particular visual image in the task state and viewed it passively in the control state. One hundred thirty-two individuals contributed to the data in these images. These decreases appear to be largely task independent. The images are oriented with the anterior at the top and the left side to the reader's left.

Default Mode Network – Brain Areas



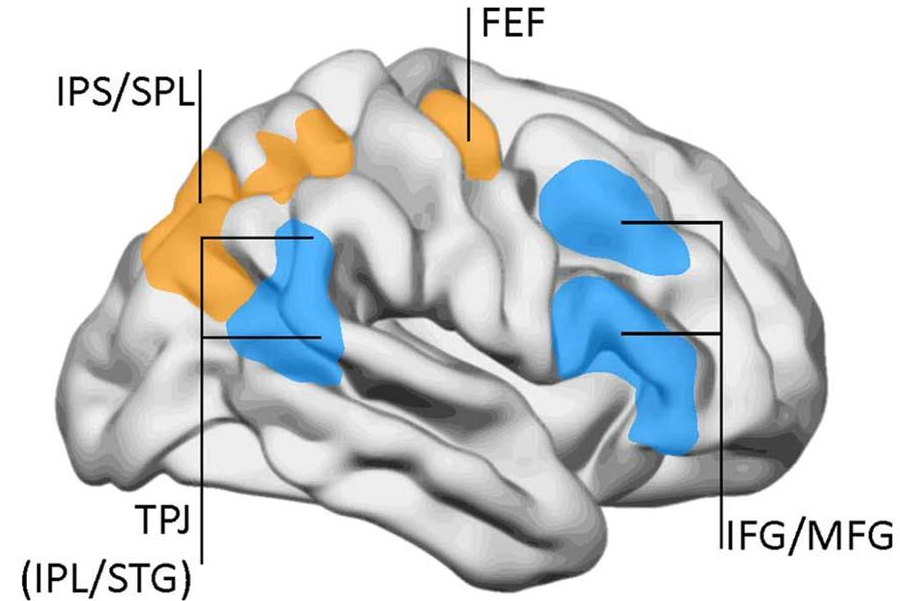
The DMN includes part of the medial temporal lobe, part of the medial prefrontal cortex, and the posterior cingulate cortex, along with the adjacent ventral precuneus and the medial, lateral and inferior parietal cortex.

Default Mode Network – Brain Areas



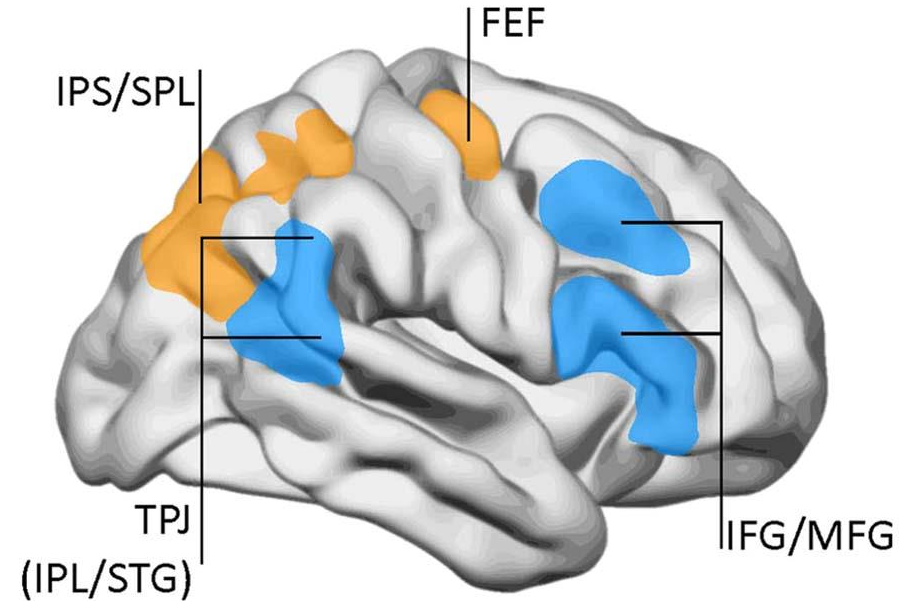
The DMN is negatively correlated (anti-correlated) with brain systems that focus on **external signals**.

Default Mode Network – Brain Areas



Attention Network – Dorsal System: It is composed of the **intraparietal sulcus** (IPS) and **frontal eye field** (FEF) at the junction of the precentral and superior frontal sulcus. It is involved in **voluntary (top-down) orienting** and shows activity increases after presentation of cues indicating where, when, or to what subjects should direct their attention.

Default Mode Network – Brain Areas

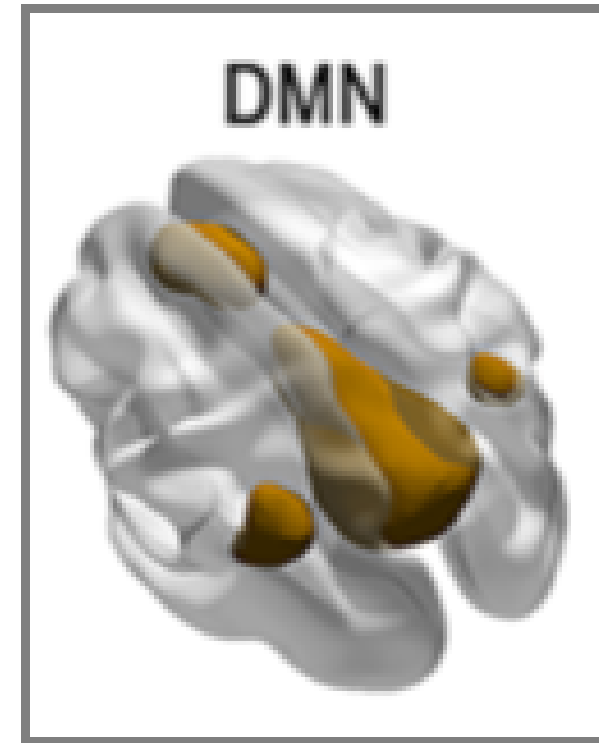


Attention Network – Ventral System: It is right-lateralized and composed of the **right temporo-parietal junction** and the **right ventral frontal cortex**. This system shows activity **increases upon detection of salient targets** and after abrupt changes in sensory stimuli. The attention network is usually identified in paradigms contrasting perceived versus unperceived stimuli.

Default Mode Network – Brain Areas

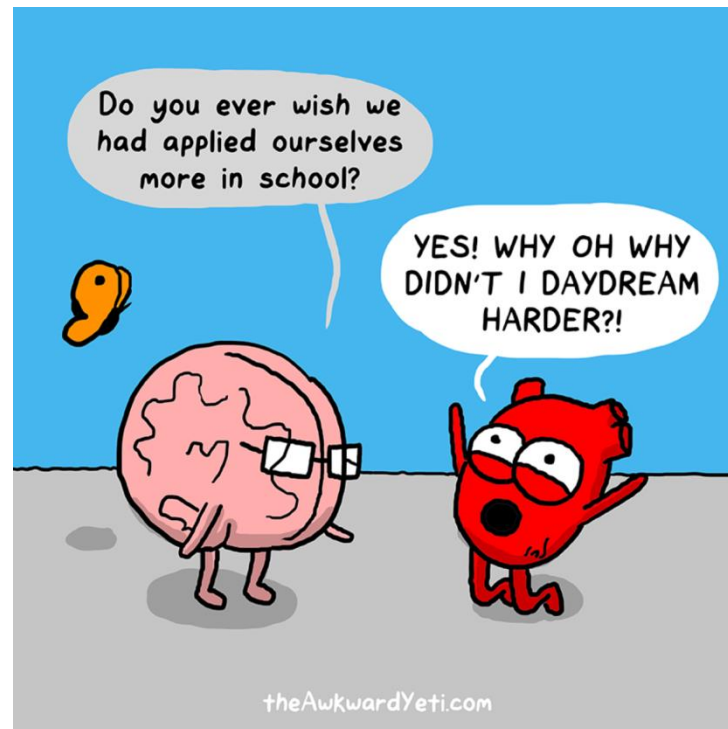
The Default Mode Network (DMN) is a network of brain areas that are active when the individual is not focused on specific tasks requiring attention to external stimuli and the brain is at wakeful rest.

The DMN is characterized by **coherent functional oscillations** at a rate lower than **0.1 Hz** (one every ten seconds).

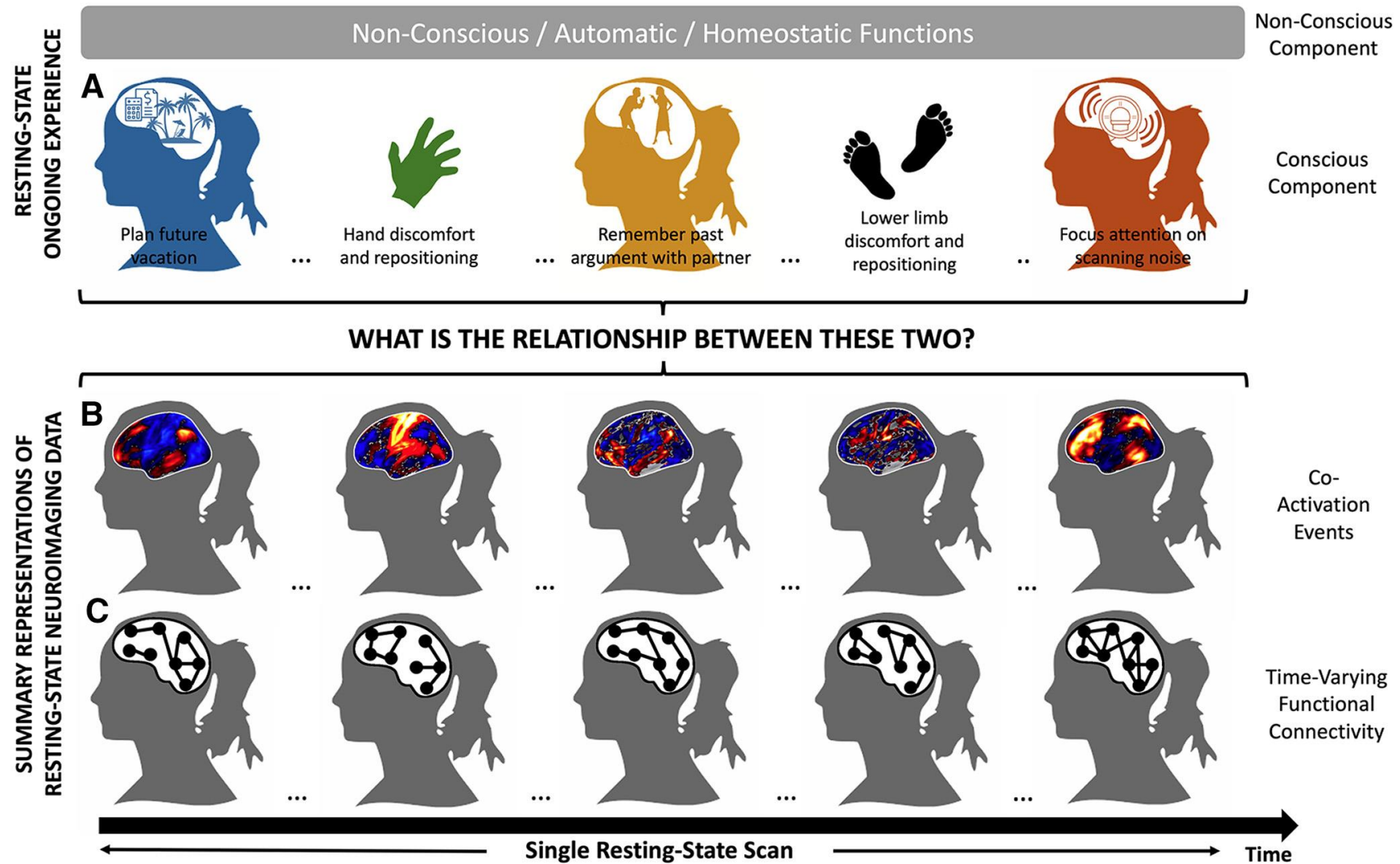


Default Mode Network – Brain Areas

The DMN preferentially activates when individuals focus on internal tasks such as **daydreaming**, **envisioning the future**, **retrieving memories**, and **gauging others' perspectives**.

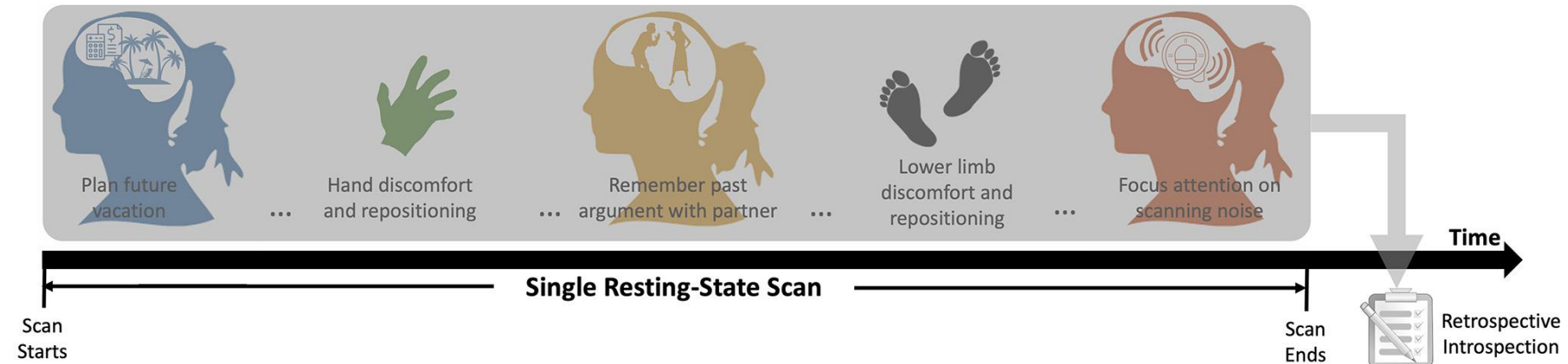


Processes associated with RS activity

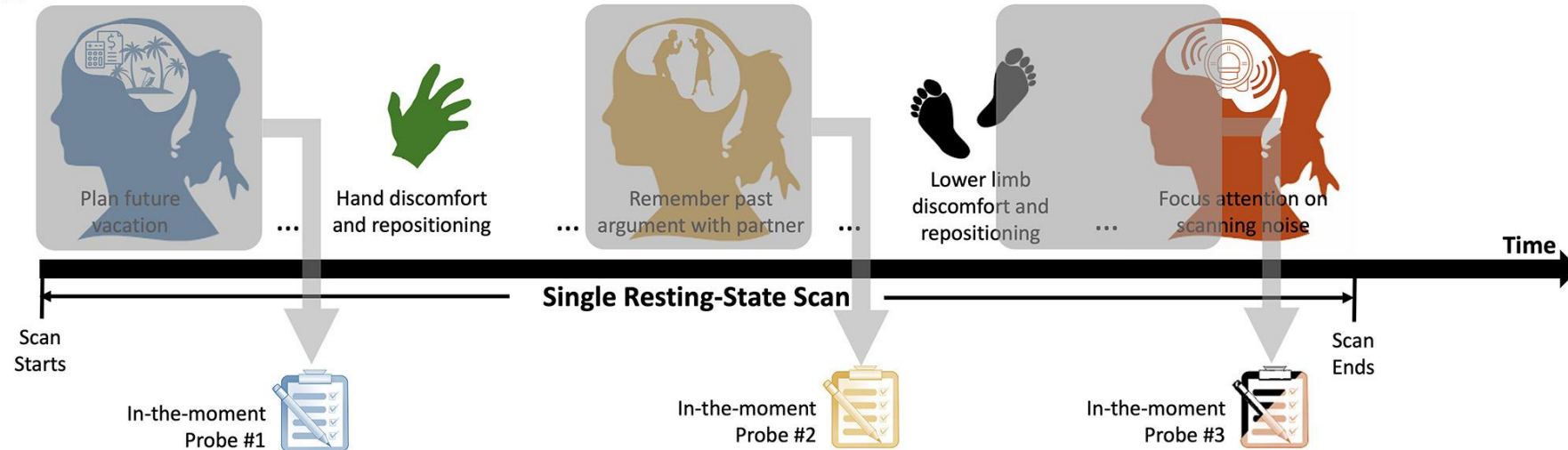


Processes associated with RS activity

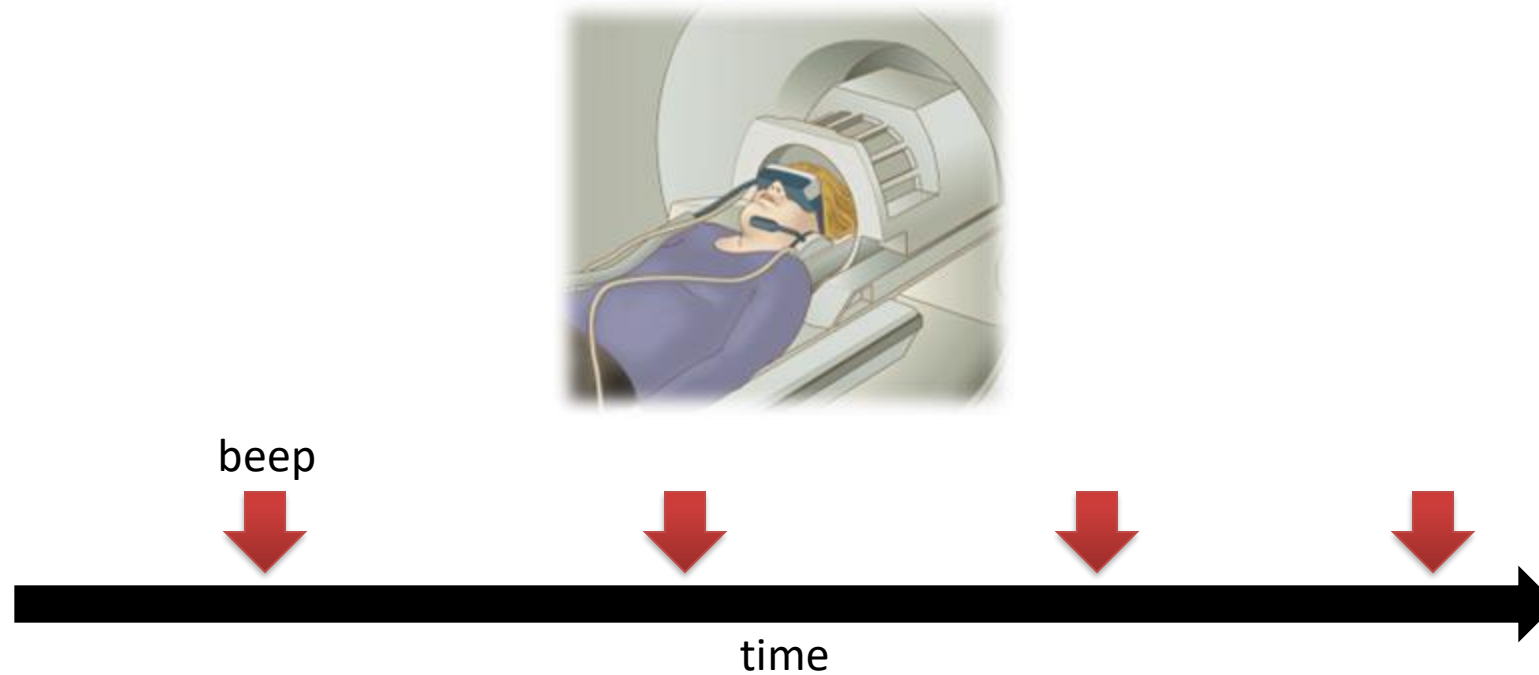
A Retrospective Introspection



B In-the-moment Introspection



Processes associated with DMN activity



The participant entered the scanner for a 25-min session with resting-state instructions “please relax, without falling asleep and do keep your eyes open.” At four random times, the participant received a beep through a headphone. Immediately after each beep, the participant jotted a few notes about the ongoing experience on a clipboard positioned on the lap.

Processes associated with DMN activity

Jack: Jack is rubbing with his thumb the fabric that holds the writing board, and he feels the snagging of the fabric on his thumb. This is more a sense in his thumb than of the fabric. He is involved with determining whether this is wool or synthetic, but that is a part of the sensation, not a cognitive act.

Sensory awareness

Processes associated with DMN activity

Otto: Otto is saying in inner speech “I just turned 30.” He doesn’t know why he is saying that or to whom, but he is clear that that is what he is experiencing and that he is emphasizing the word “turned.”

Inner speaking

Processes associated with DMN activity

Susan: Susan innerly sees the actress Sigourney Weaver in a cryogenic tank from the movie Alien. She sees Sigourney's face from above, below the glass window of the tank –the rest of Sigourney's body is vaguely or blurrily present. Mostly Susan is searching for the word used in the movie: cryogenic chamber, cryogenic tank, etc., waiting for the right word to appear. This is primarily a state of suspended animation, waiting for the word – she does not see or hear pieces of words, etc.

Inner seeing

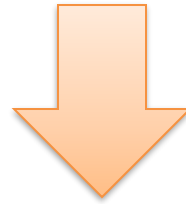
Processes associated with DMN activity

Jane: Jane is aware of the fidgetiness of her upper trunk, arms, and eyes; they want to move. She is also simultaneously looking at the gray plastic piece of the scanner, noticing especially the dark graynes sofit.

Sensory awareness

Processes associated with DMN activity

Neural activation patterns largely
consistent across time



Large individual differences in inner phenomena

The resting state itself may differ substantially
from one participant to the next

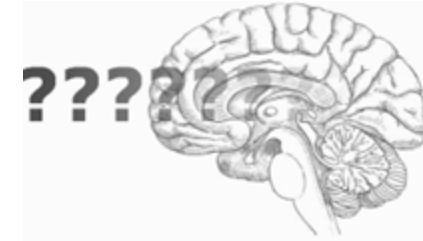
Processes associated with DMN activity

TABLE 1 | Characterizations of experiences when the default mode is active.

	Characterization	Source
1	"unconstrained verbally mediated thoughts"	Shulman et al., 1997, p. 648
2	"semantic knowledge retrieval, representation in awareness, and directed manipulation of represented knowledge for organization, problem-solving, and planning"	Binder et al., 1999, p. 80
3	"active retrieval of past experiences and planning of future experiences"	Andreasen et al., 1995, p. 1576
4	"retrieval and manipulation of past events, both personal and general, in an effort to solve problems and develop future plans"	Greicius et al., 2003, p. 257
5	"enhanced watchfulness toward the external environment (e.g., waiting for upcoming task-relevant stimuli or attending to scanner noise and incidental light)"	Gilbert et al., 2007, p. 43
6	"inner thought, self-reflective thinking in terms of planning for the future, or simulation of behavior. . . interrupted. . . into a . . . extrospective. . . state of mind. . . characterized. . . as increased attention and readiness,. . . sensorimotor planning for future routes of action in response to potential changes in the inner and outer environment"	Fransson, 2005, p. 26
7	"not focused on the external environment,. . . internally focused tasks including autobiographical memory retrieval, envisioning the future, and conceiving the perspectives of others"	Buckner et al., 2008, p. 1
8	"spontaneous, internally directed cognitive processes"	Andrews-Hanna et al., 2010, p. 322
9	"spontaneous mental contents which are unrelated to perception and coordinate[d]. . . so that they are maintained in the face of competing sensory information"	Smallwood et al., 2012, p. 67
10	"an ultimate state of inspection of the self"	Wicker et al., 2003, p. 229
11	"stable, unified perspective of the organism relative to its environment (a 'self ')"	Gusnard and Raichle, 2001, p. 692

Functional Significance of the Default Mode Network

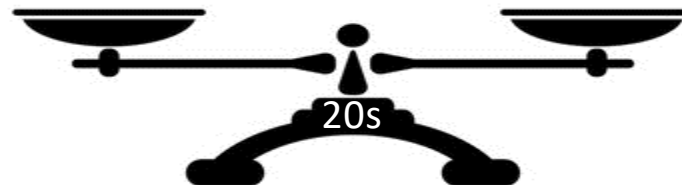
The function of the DMN is still unclear.



In humans, the DMN has been hypothesized to be involved in **introspection** or **self-referential thoughts**. In particular, it may be related to generation of *spontaneous thoughts*, *mind wandering* and *creativity*.

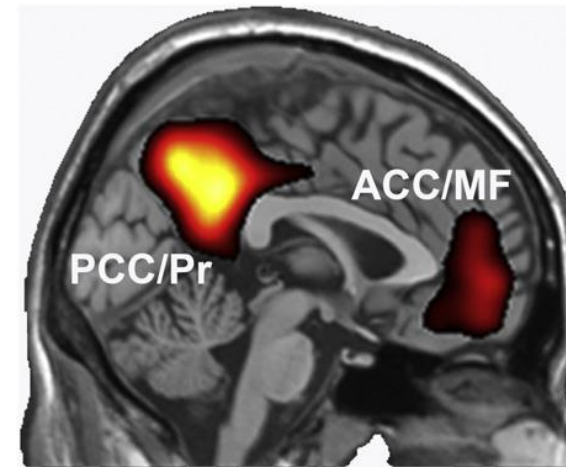
Alternatively, the DMN may represent **underlying physiological processes going on in the brain** that are **unrelated to any particular thought or thoughts**.

Internal Awareness VS External Awareness



Functional Significance of the Default Mode Network

In studies requiring participants to make judgments about their own feelings or about their own characters (no resting state) the engagement of the two medial core components of the DMN is observed (medial prefrontal and posterior cingulate cortex).



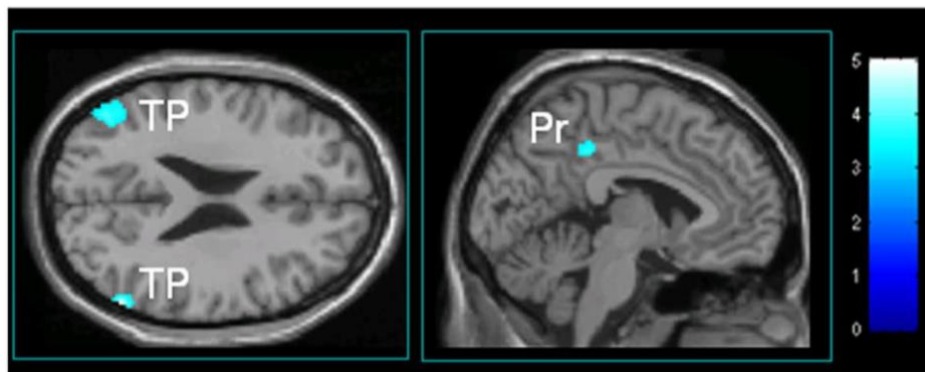
Components of the DMN have also been activated in social cognition—thinking about other people or what other people are thinking about (theory of mind – *internal simulation?*).

Functional Significance of the Default Mode Network

A positive relationship was observed between conscious perception of low-intensity somatosensory stimuli and immediately preceding levels of baseline activity in medial thalamus and the lateral frontoparietal network, respectively.



Increased baseline activity in the **medial thalamus (Th)**, **dorsolateral prefrontal cortex (DLPF)**, **intraparietal sulcus/posterior parietal cortex (IPS)**, and **aACC** 3 sec before stimulus presentation predicts perception of low-intensity sensory stimuli.



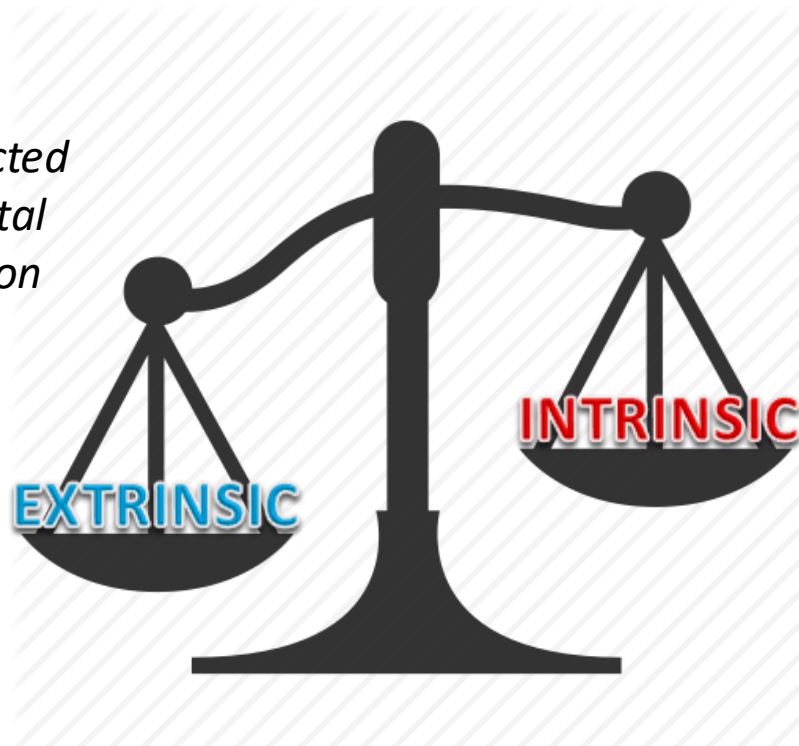
Decreased baseline activity in the **default brain network** encompassing **posterior cingulate/precuneus (Pr)** and **bilateral temporoparietal junctions (TP)** exerts a facilitatory effect upon perception of subsequent somatosensory stimuli.

Functional Significance of the Default Mode Network

Two complementary states of system imbalance are possible, where one system can be in a hyperfunctional state, while the other is hypoactive.

Extrinsic system

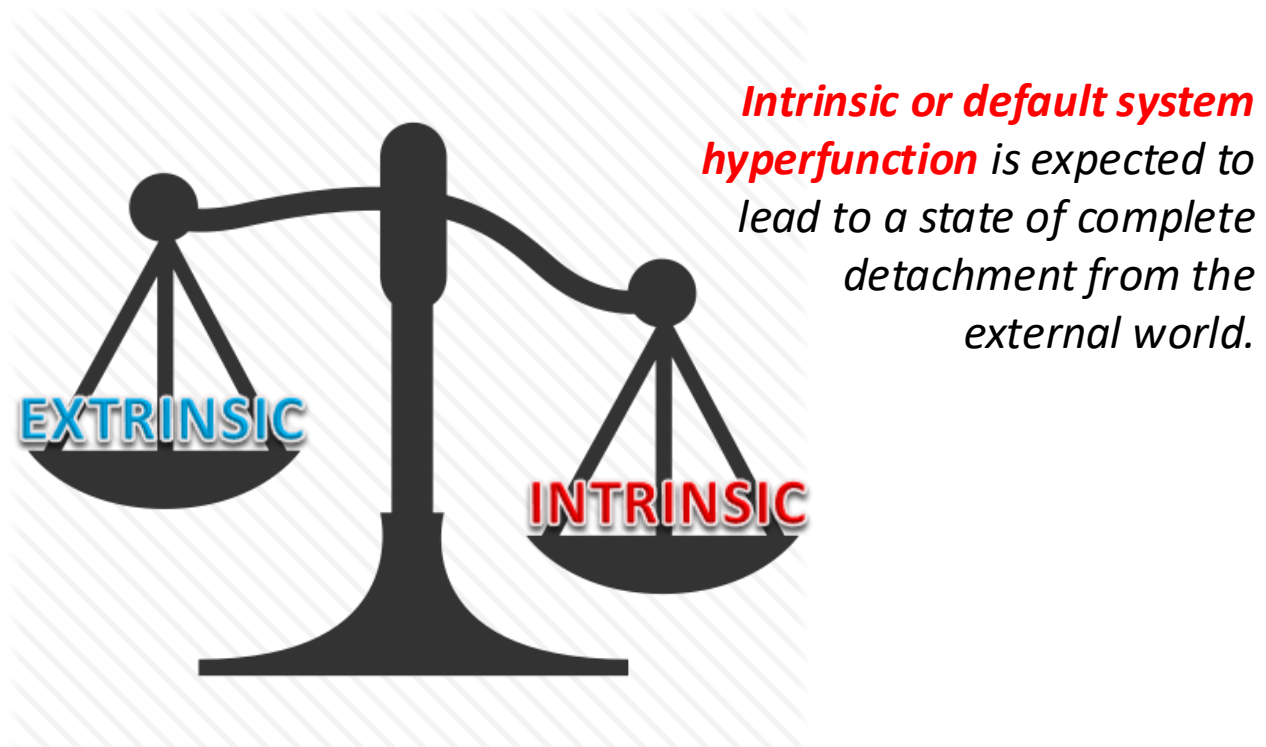
hyperfunction is expected to lead to a state of total sensorimotor absorption or “lost self.”



A state where both extrinsic and intrinsic systems are hypofunctional is predicted to lead to markedly impaired consciousness as seen in disorders of consciousness.

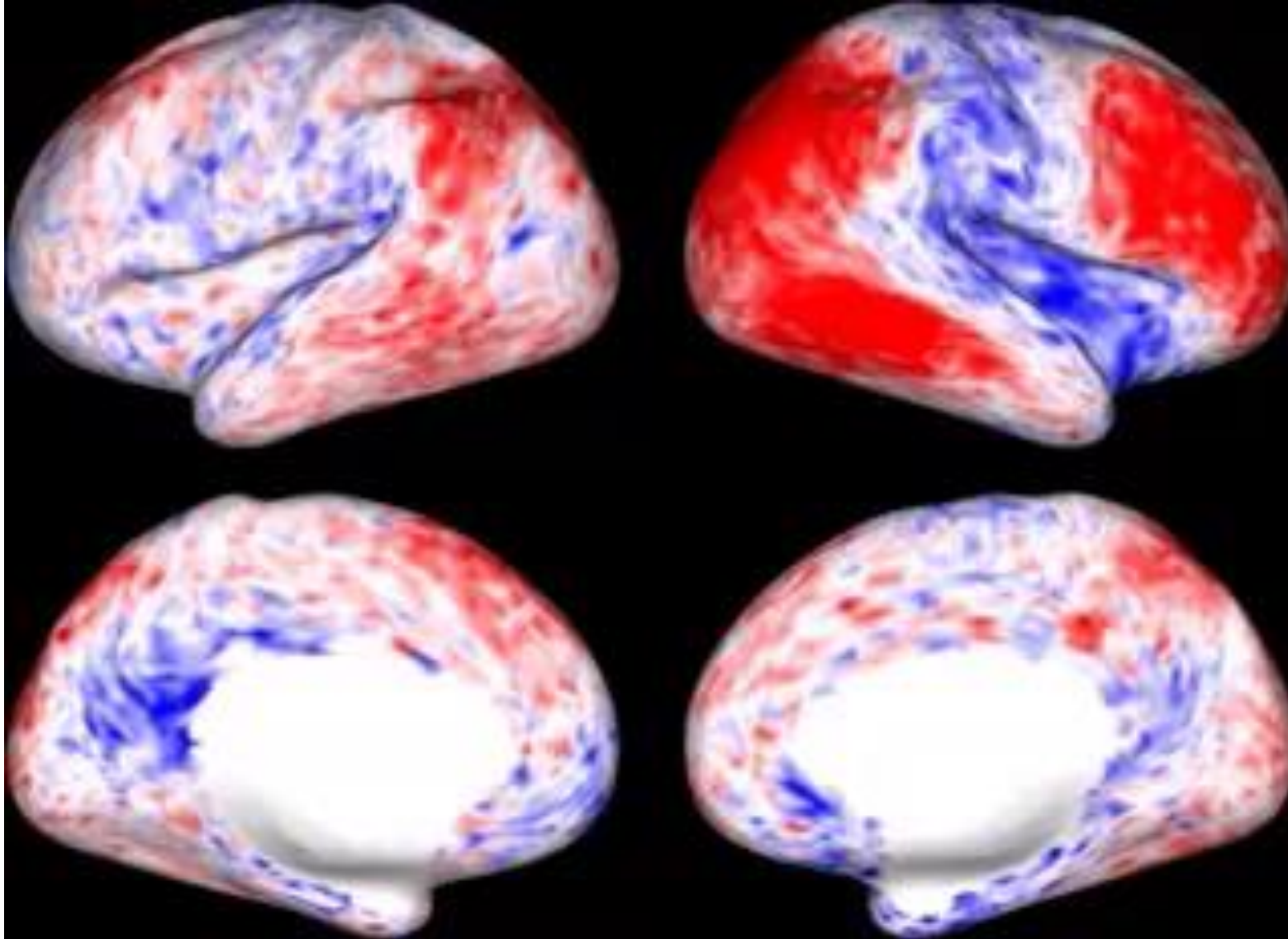
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Functional Significance of the Default Mode Network



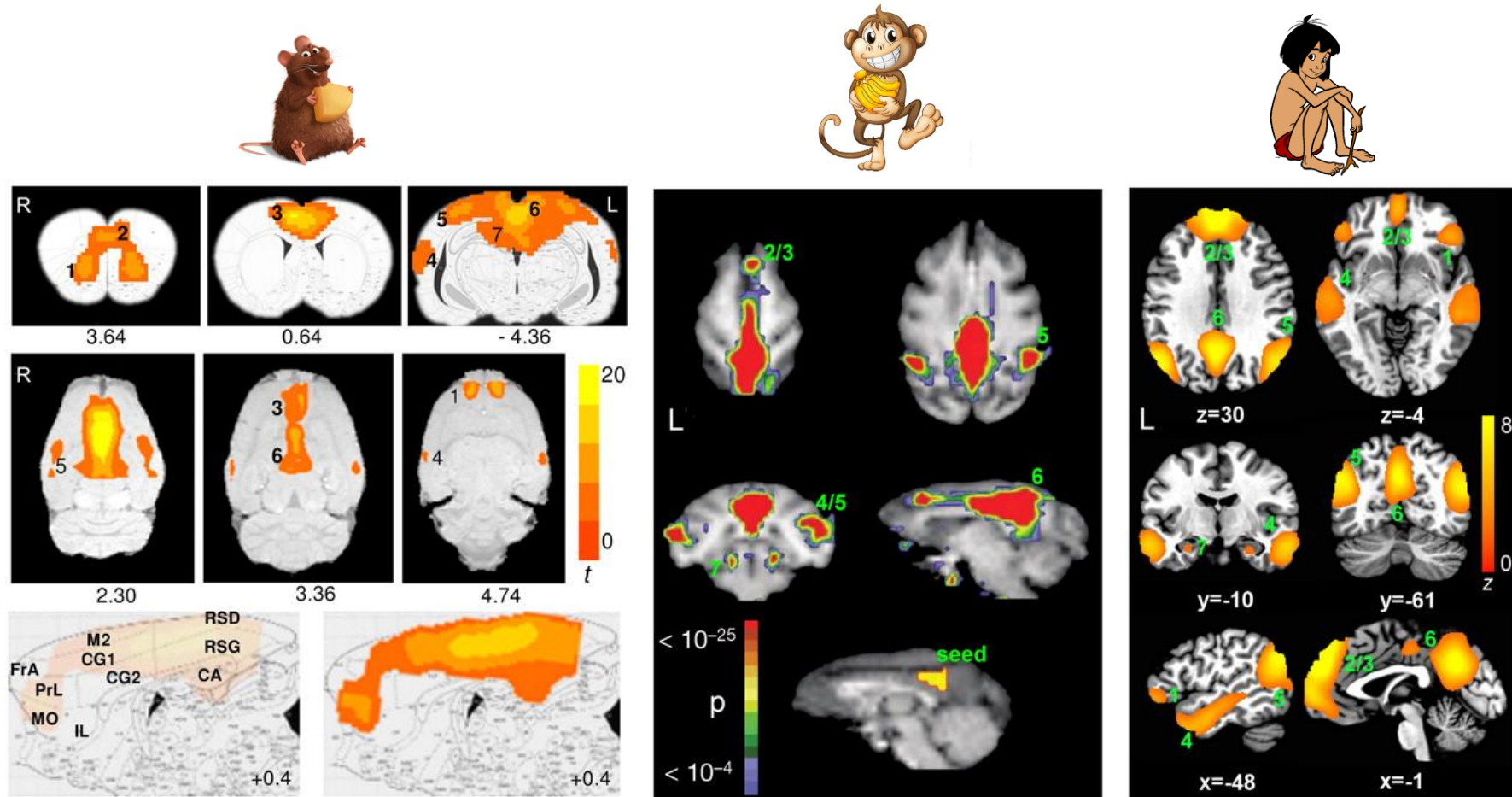
Functional Significance of the Default Mode Network

Switching between the “default” network and the “attention” network seem to occur spontaneously every 20 s on average.

Activation of the “**attention**” network has been correlated with subjects’ reports of being “**strongly externally aware**” (i.e., perception of environmental sensory stimuli).

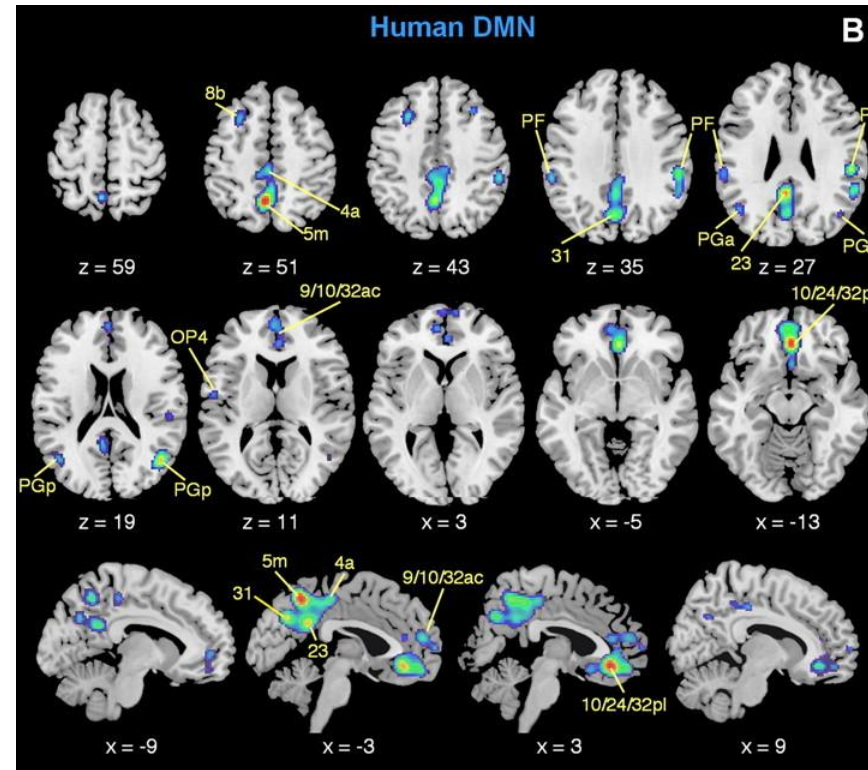
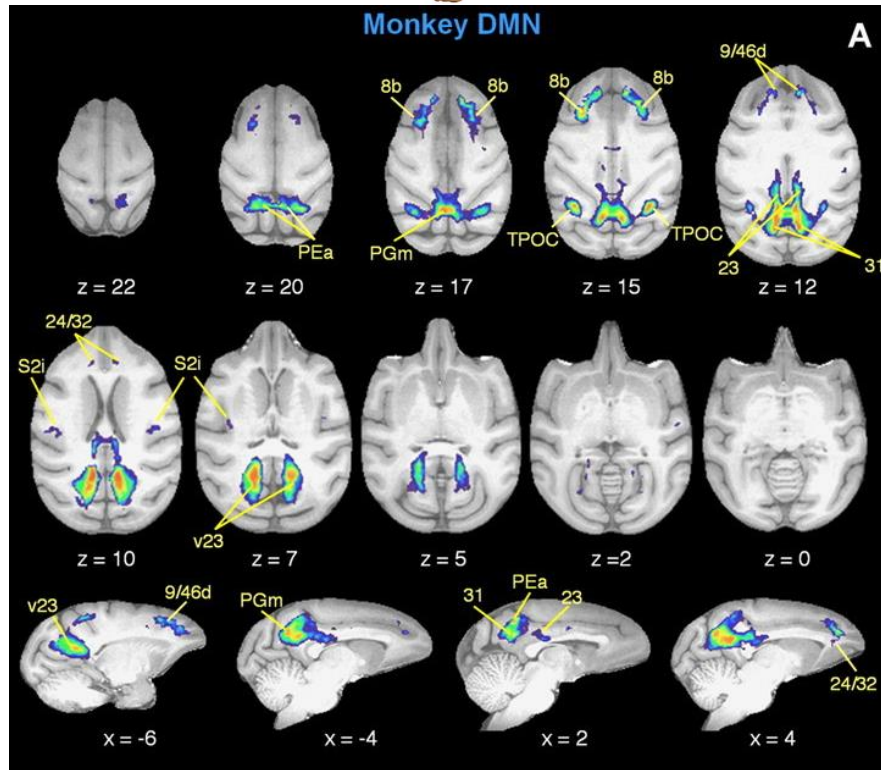
Activation of the “**default**” system has been correlated with reports of being “**strongly internally aware**” (i.e., stimuli independent thoughts such as inner speech or autobiographical memories).

Default Mode Network in Non-Human Mammals



Rats and Monkeys possess a DMN that is broadly similar to the DMNs of humans. A well-organized, intrinsically coherent DMN appears to be a fundamental feature in the mammalian brain.

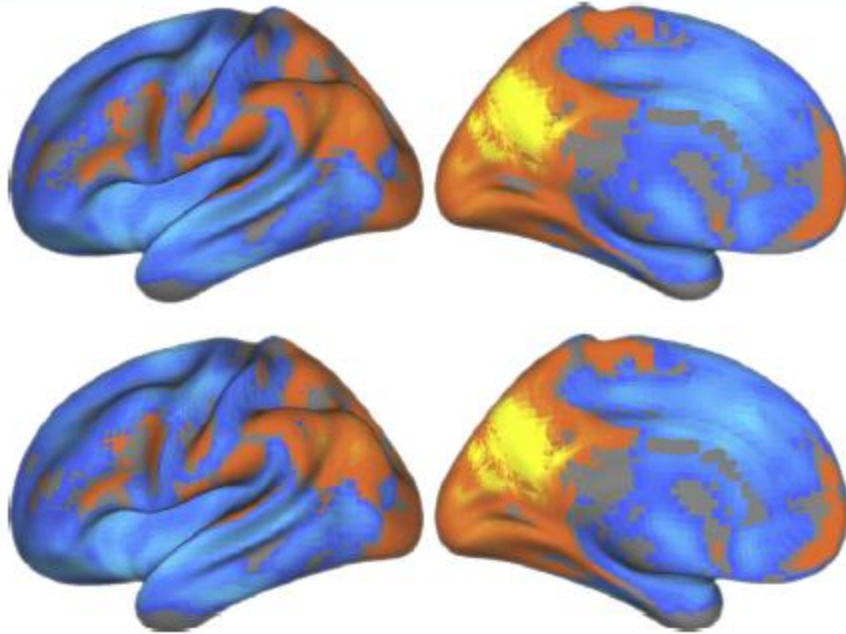
Default Mode Network in Non-Human Mammals



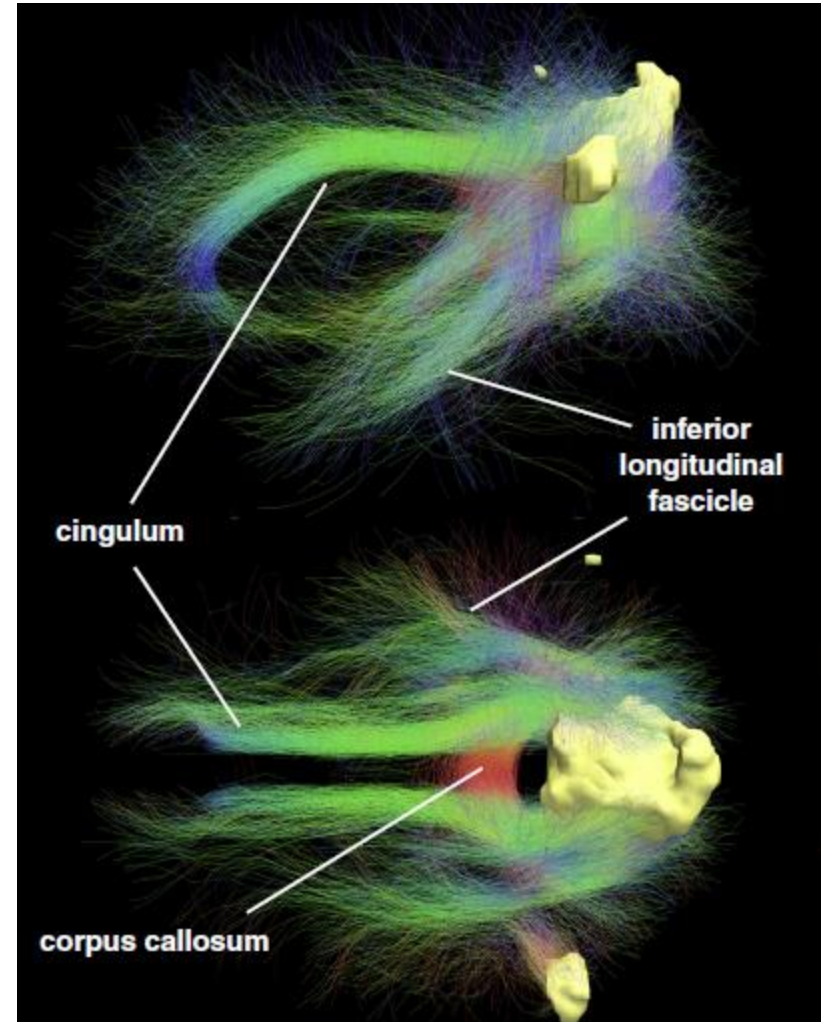
“Like humans, monkeys increase activity during passive epochs in heteromodal and limbic association regions, suggesting that they also default to internal modes of processing when not actively interacting with the environment.”

DMN and the Brain Structural Networks

probabilistic tracking

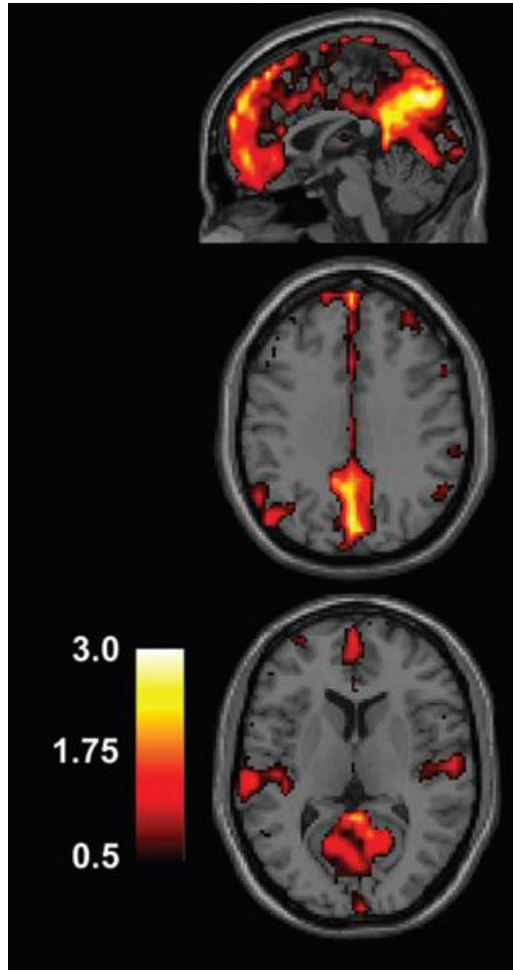


The similarity between functional and structural connectivity is highest in regions of the default mode network.



Cortical Hubs in the Human Brain

Adult Brain

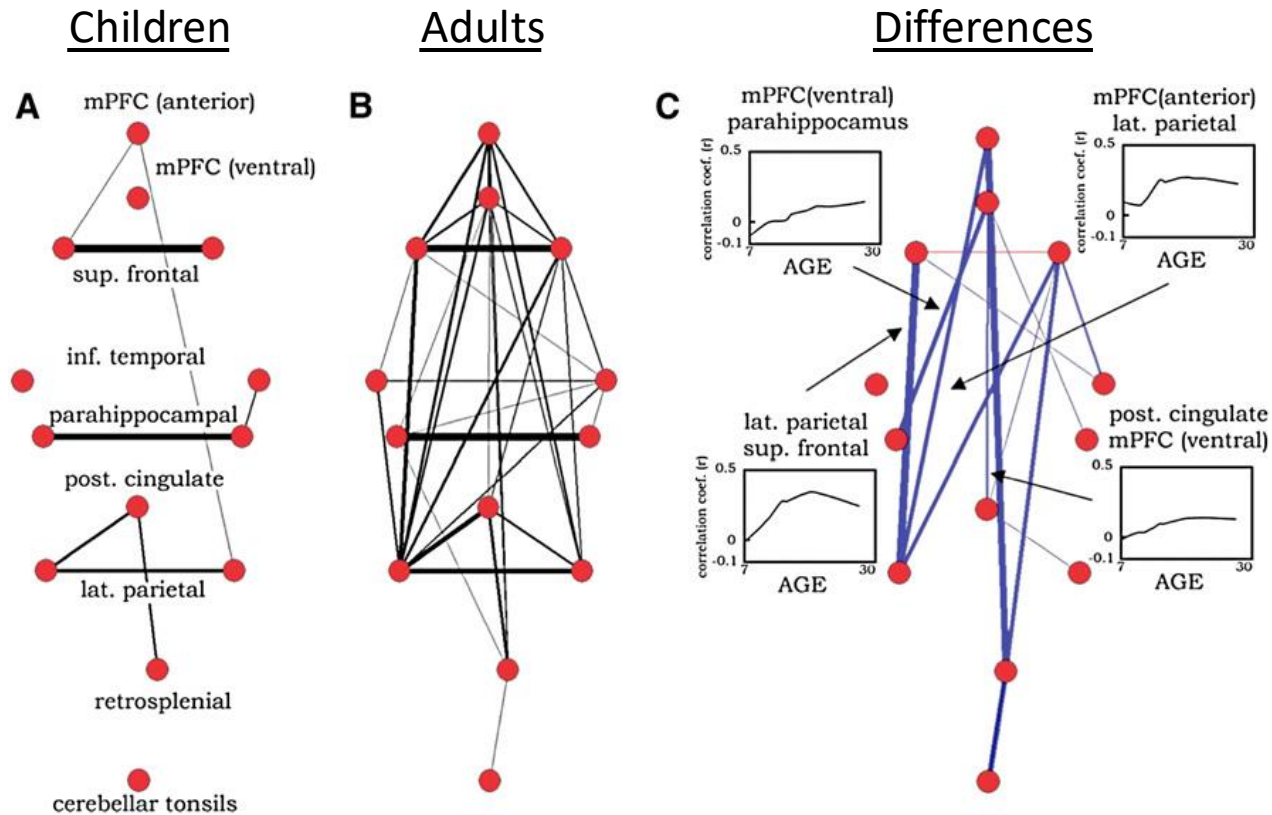


Spatial distribution of cortical hubs
based on the degree centrality measure

Prominent locations for network hubs in adults include precuneus/posterior cingulate cortex, medial PFC, anterior cingulate cortex, bilateral parietal lobule, and bilateral insula.

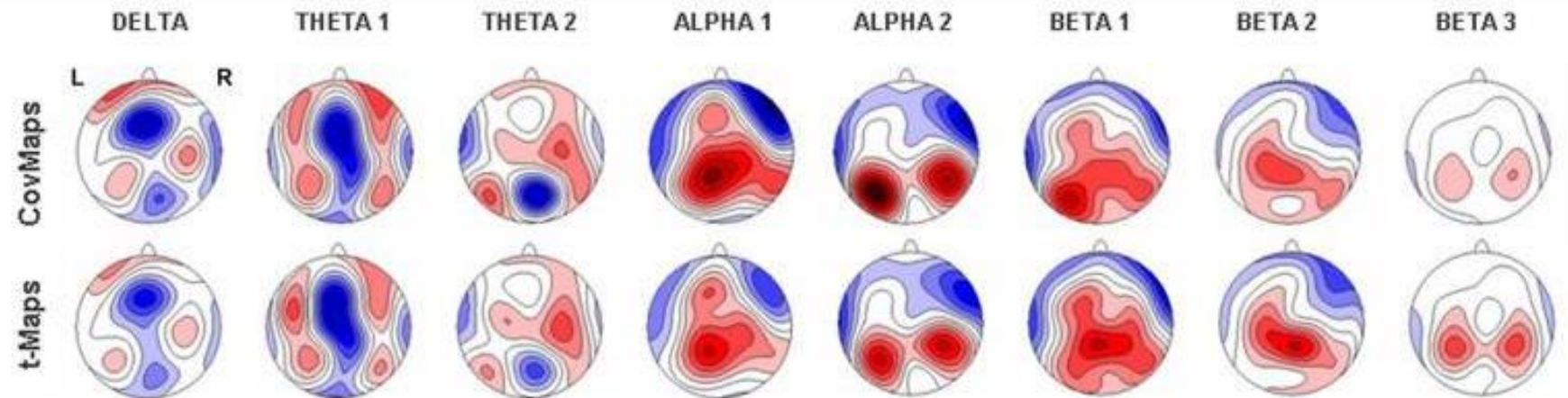
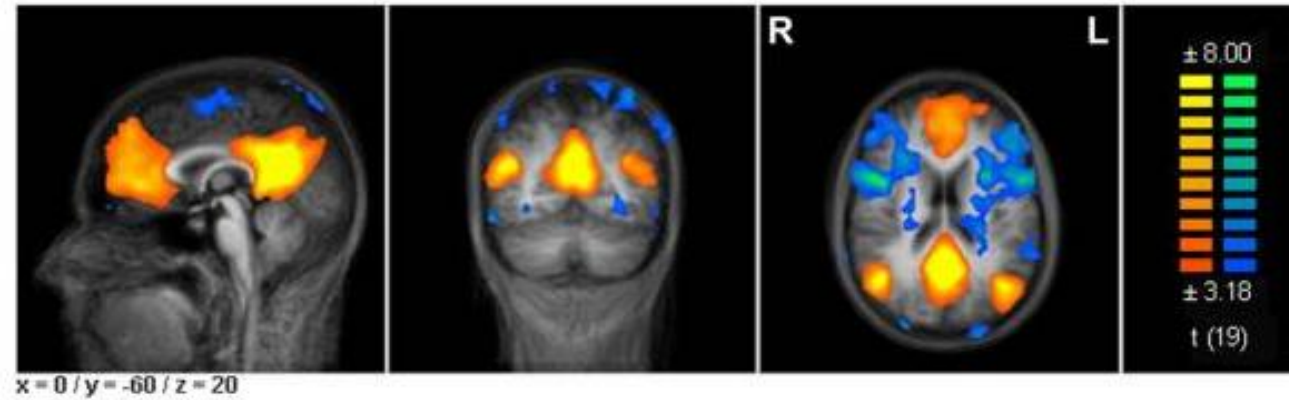
In infants, the majority of cortical hubs are located in the homomodal cortex, foremost in the auditory, visual and sensorimotor areas, and to a lesser extent in the PFC.

Development of the default mode network



The level of connectivity in the DMN in children is a subject of debate and intense investigation. The main finding so far is that **children** tend to show **diffuse correlations near seed locations** and **few long-distance correlations**. Long-distance correlations begin to emerge during **adolescence**.

EEG correlates of the Default Mode Network



Modulation of the Default Mode Network

Attention to external stimuli: A shift of attention toward specific stimuli or tasks is associated with a reduced connectivity within the DMN.

Meditation: Long-term meditation practitioners may show reduced activation and reduced functional connectivity in the DMN.

Psychedelic drugs: Psilocybin has been shown to be associated with a reduction of the blood flow to areas of the DMN (posterior cingulate cortex and medial prefrontal cortex).

Sleep: Sleep deepening is associated with a decrease in connectivity between the posterior cingulate cortex and medial prefrontal cortex.

Sleep deprivation: Sleep deprivation results in a decrease in connectivity within the DMN. A decreased anti-correlation between the DMN and the TPN can be also observed.

Pathophysiology of the Default Mode Network

The DMN overlaps areas involved in major brain disorders. In fact, alterations of the DMN have been observed in several pathological conditions. Discerning precisely which aspects of the network are affected by different disorders may provide new diagnostic tools.

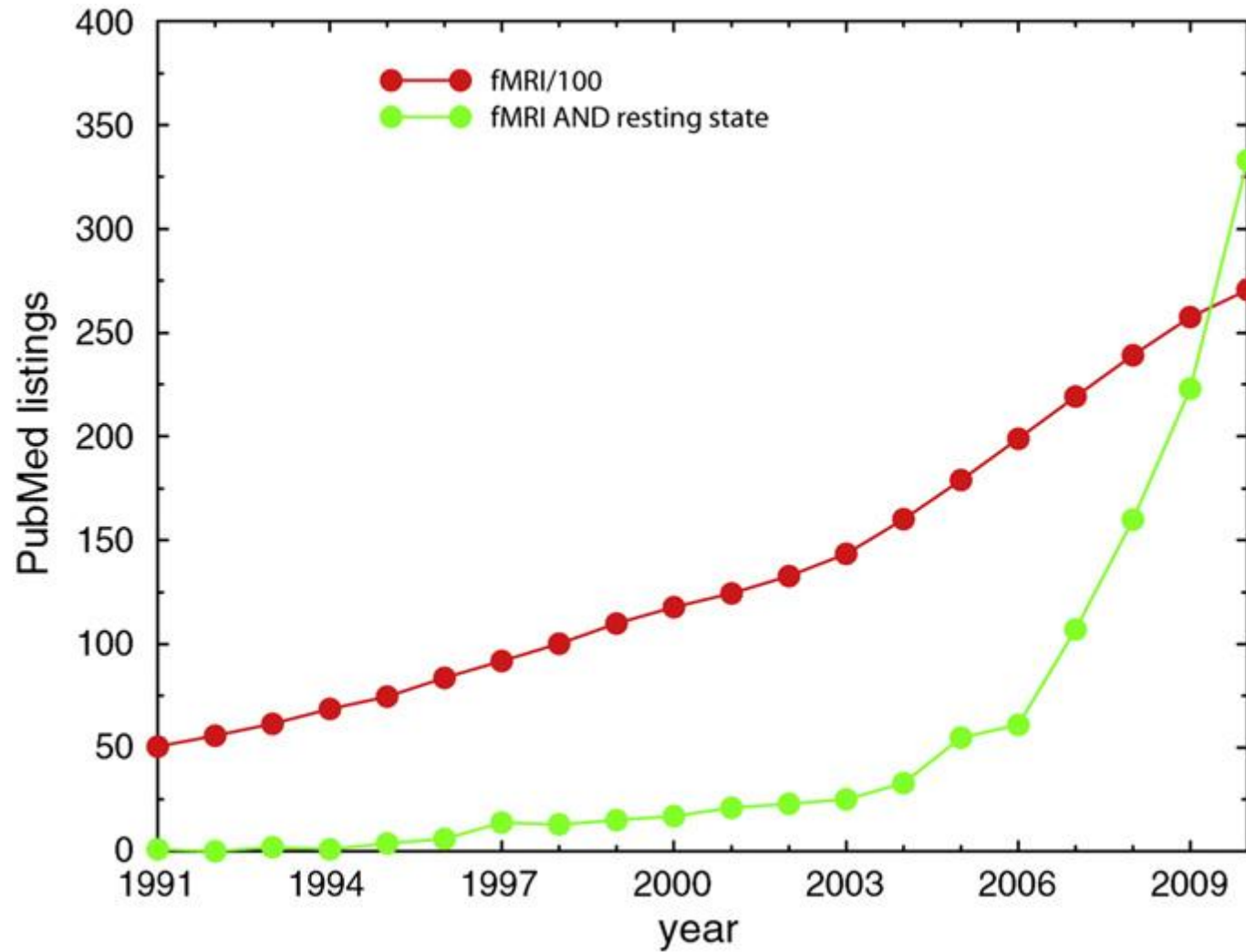
Some examples...

Alzheimer's Disease: Brain areas that show marked atrophy in Alzheimer's overlap very closely with major centers of the DMN.

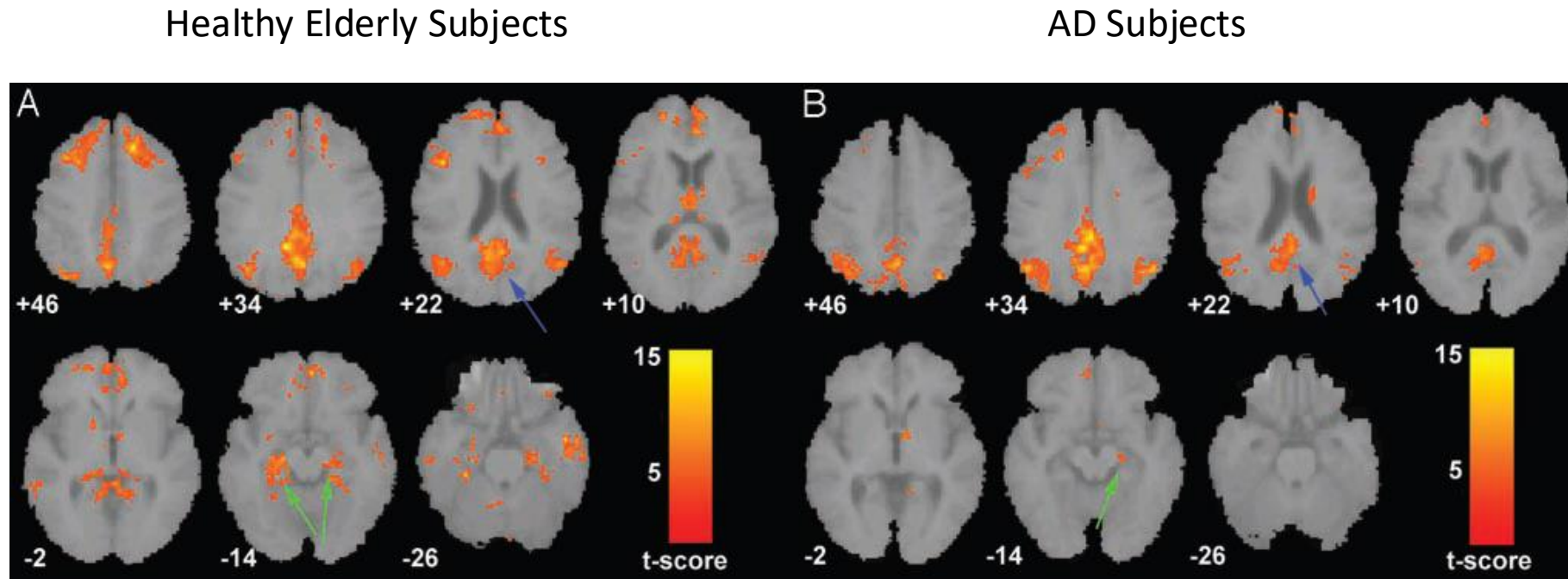
Depression: Hyperconnectivity in the DMN seems associated with rumination in Depression.

Autism: Patients may fail to reduce DMN activity during cognitively demanding tasks.

Works about «Resting State»

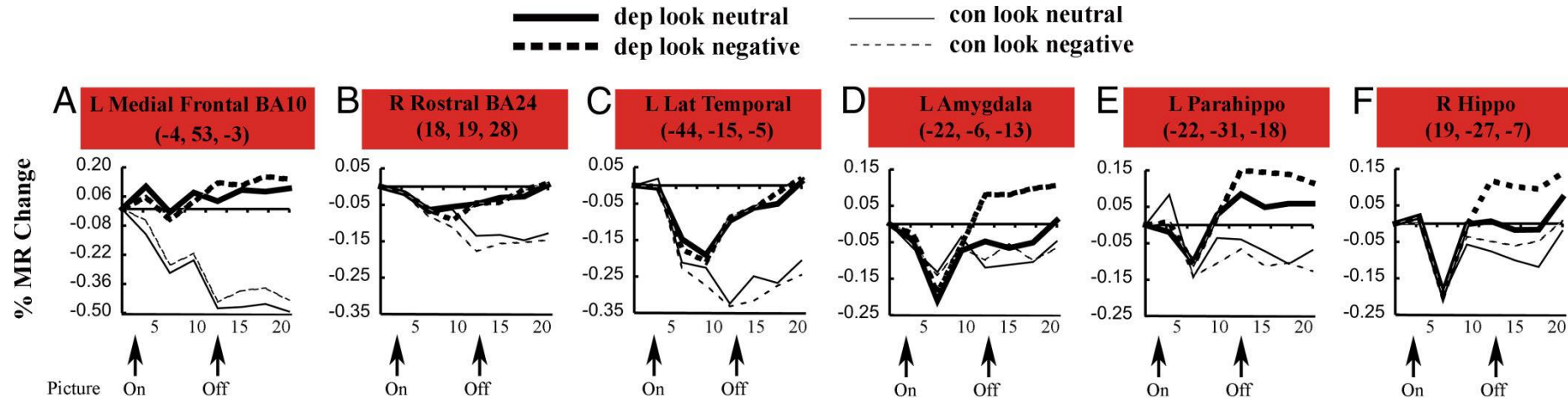


The DMN and Self-Referential Processes in AD



The AD group showed decreased resting-state activity in the posterior cingulate and hippocampus. Of note a posterior cingulate hypometabolism is commonly detected in positron emission tomography (PET) studies of early AD.

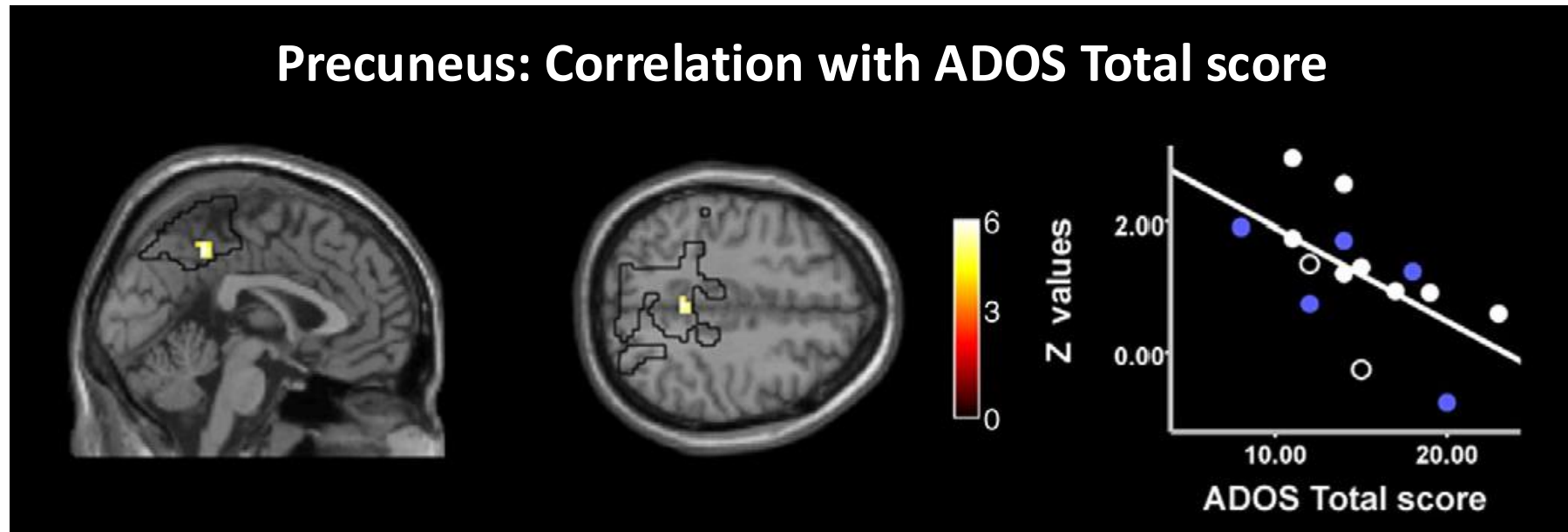
The DMN and Self-Referential Processes in Depression



Patients and controls were asked to examine negative pictures passively and also to reappraise them actively. In widely distributed elements of the DMN, depressed, but not control subjects, exhibited a failure to reduce activity while both looking at negative pictures and reappraising them.

Moreover, looking at negative pictures elicited a significantly greater increase in activity in other DMN regions (amygdala, parahippocampus, and hippocampus) in depressed than in control subjects.

The DMN and Self-Referential Processes in Autism



Compared to controls, ASD patients showed decreased FC between areas of the DMN.

FC in these regions was inversely correlated with the severity of patients' social and communication deficits (Autism Diagnostic Observational Schedule).

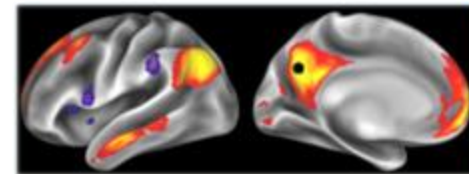
Association between ASD deficits and DMN under-connectivity?

*Methods for Exploration
of the Resting State Brain Activity*

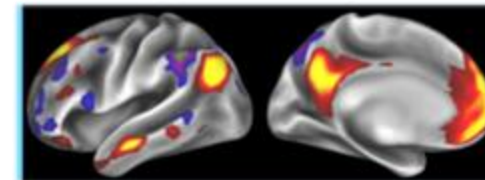
Common Approaches for the Study of Resting-State Data

Three main approaches are commonly used to study resting state functional MRI data (rs-fMRI) :

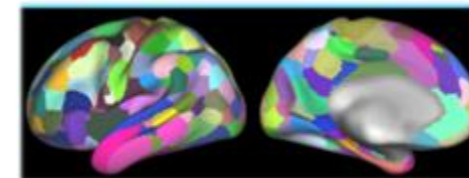
- ✓ Map-Based Methods
 - Seed-based Functional Connectivity (FC) Analysis



- Independent Component Analysis (ICA)



- ✓ Graph-Based Methods
 - Connectivity Matrices and Graph Analysis



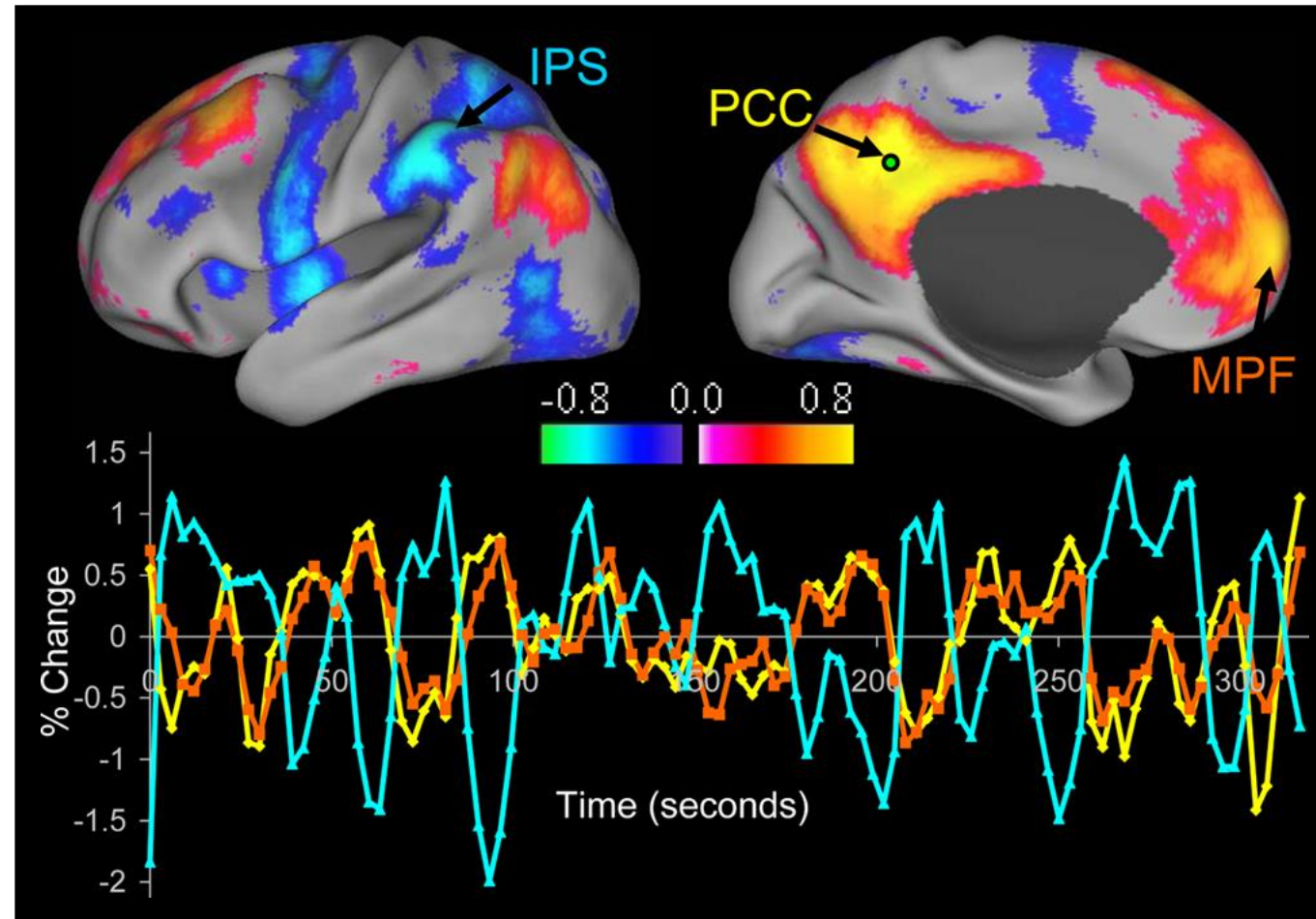
Differences between ROI-based and ICA-based methods

Seed-based correlation mapping requires the a priori identification of a target region of interest and requires extensive preprocessing to minimize the influence of non-neuronal sources of variance.

In contrast, ***ICA*** provides a direct means of separating artifact from neural signals but is less suited to investigating targeted regions of interest.

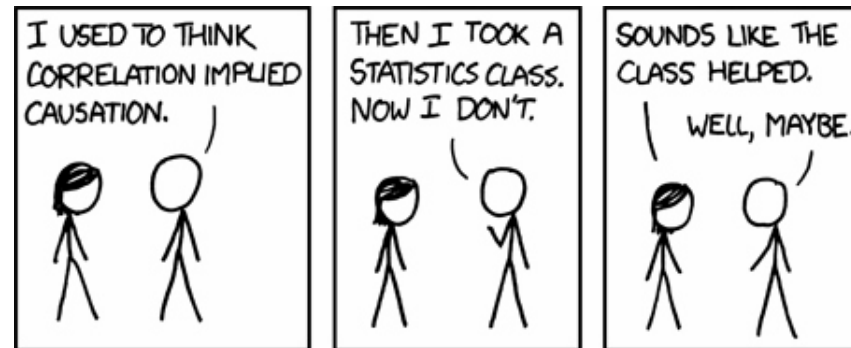
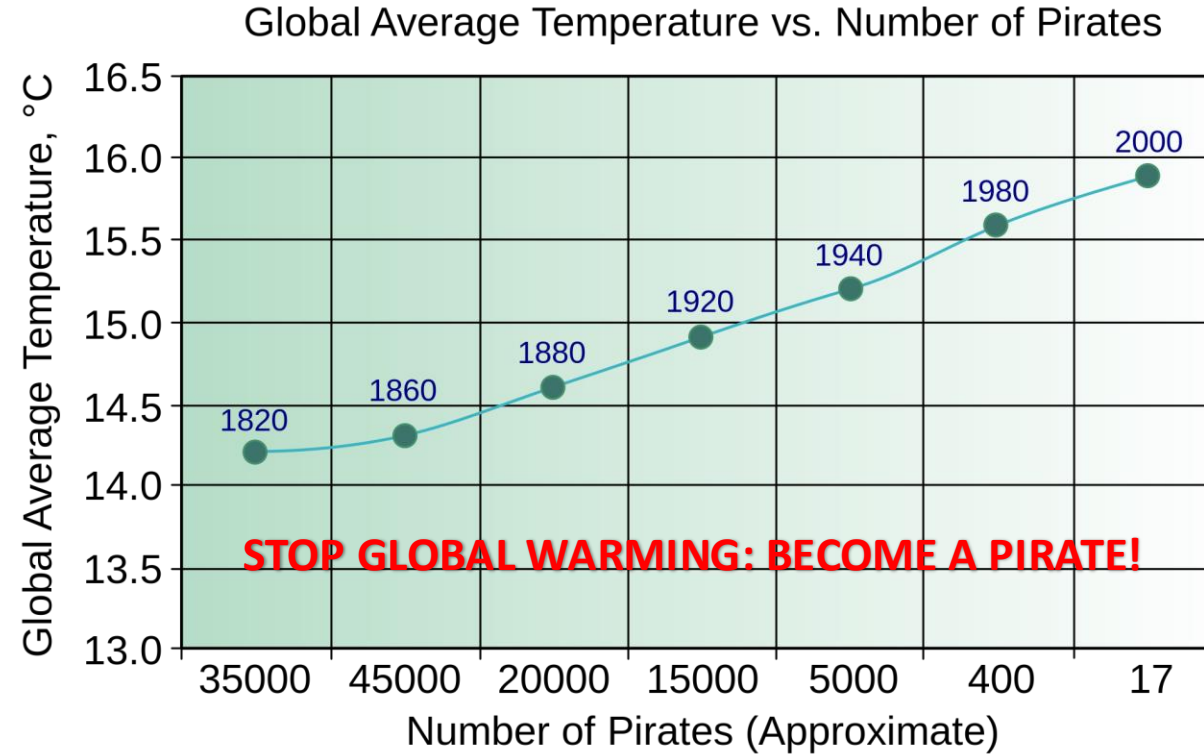
The topography of the *DMN* appears more or less the same regardless of which method is used.

Seed-Based Functional Connectivity Analysis



Signal from a certain voxel or cluster of voxels (**seed region of interest**, seed-ROI) is used to calculate the temporal correlation (usually *Pearson's correlation coefficient*) with other voxels of the brain. Usually the seed-ROI is placed on precuneus / posterior cingulate cortex (PCC) or on anterior cingulate / medial prefrontal cortex for the study of DMN.

Correlation is NOT Causation



Functional Connectivity

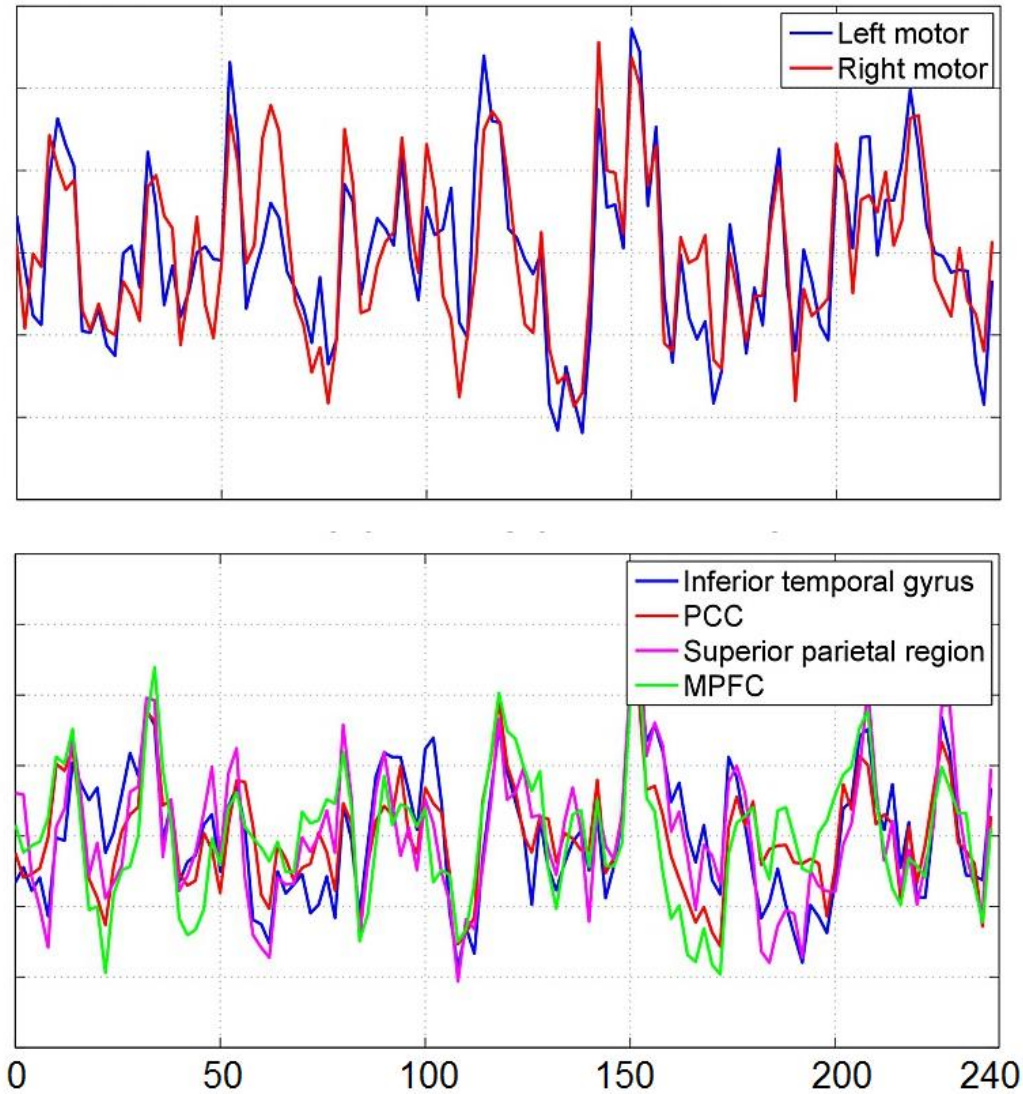
The term “**Functional Connectivity**” (FC) indicates analyses of temporal covariance. Differently, the term “**Effective Connectivity**” (EC) commonly indicates analyses of causal relationships between different brain areas, and typically implies a temporal influence of an area over another region.

The most common approach for FC analyses is based on the calculation of the Pearson correlation coefficient. The Pearson coefficient is a measure of the linear correlation between two variables X and Y, giving a value between +1 and -1 inclusive, where 1 is total positive correlation, 0 is no correlation, and -1 is total negative correlation.

$$\rho(A, B) = \frac{\text{cov}(A, B)}{\sigma_A \sigma_B} \quad \Rightarrow \quad \rho(A, B) = \frac{1}{N-1} \sum_{i=1}^N \left(\frac{\overline{A_i} - \mu_A}{\sigma_A} \right) \left(\frac{B_i - \mu_B}{\sigma_B} \right)$$

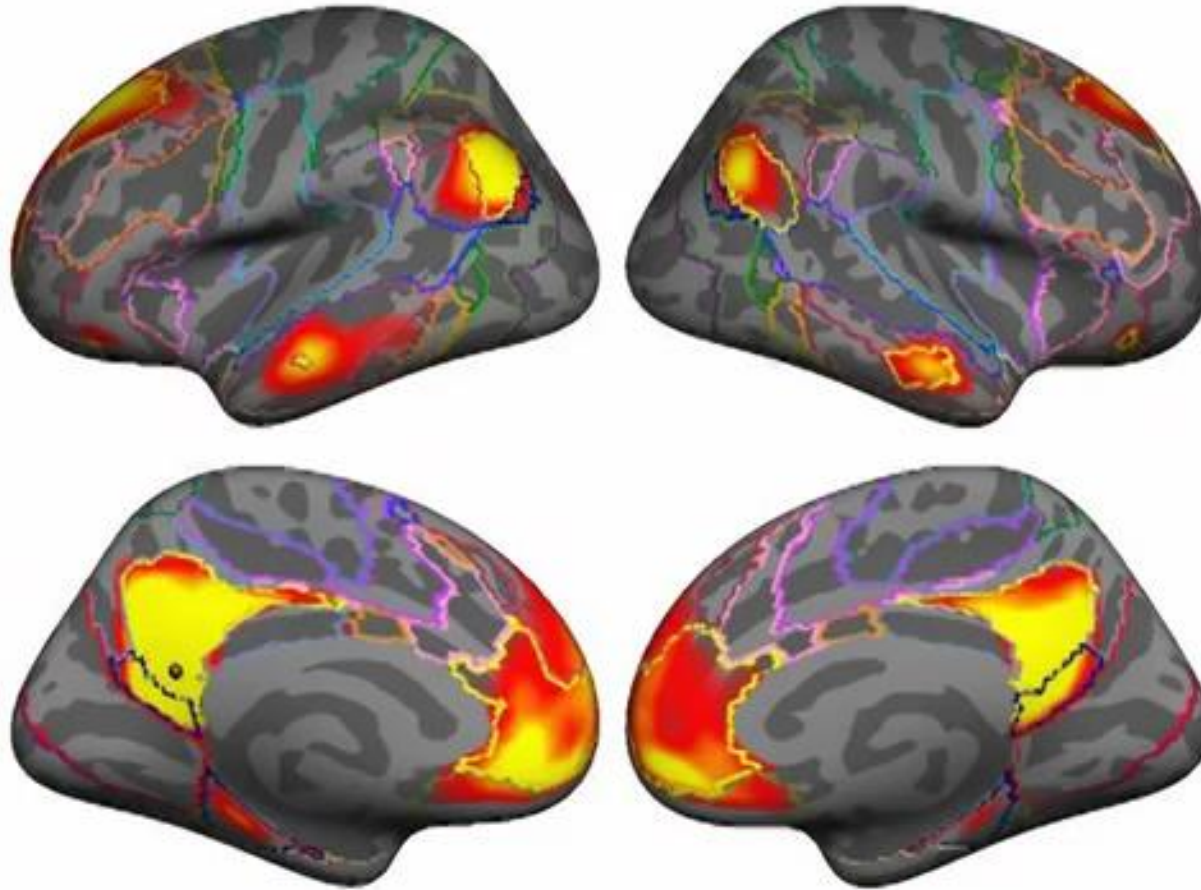
cov = covariance; σ_A = standard deviation of A; μ_A = mean of A

Regional Synchronization

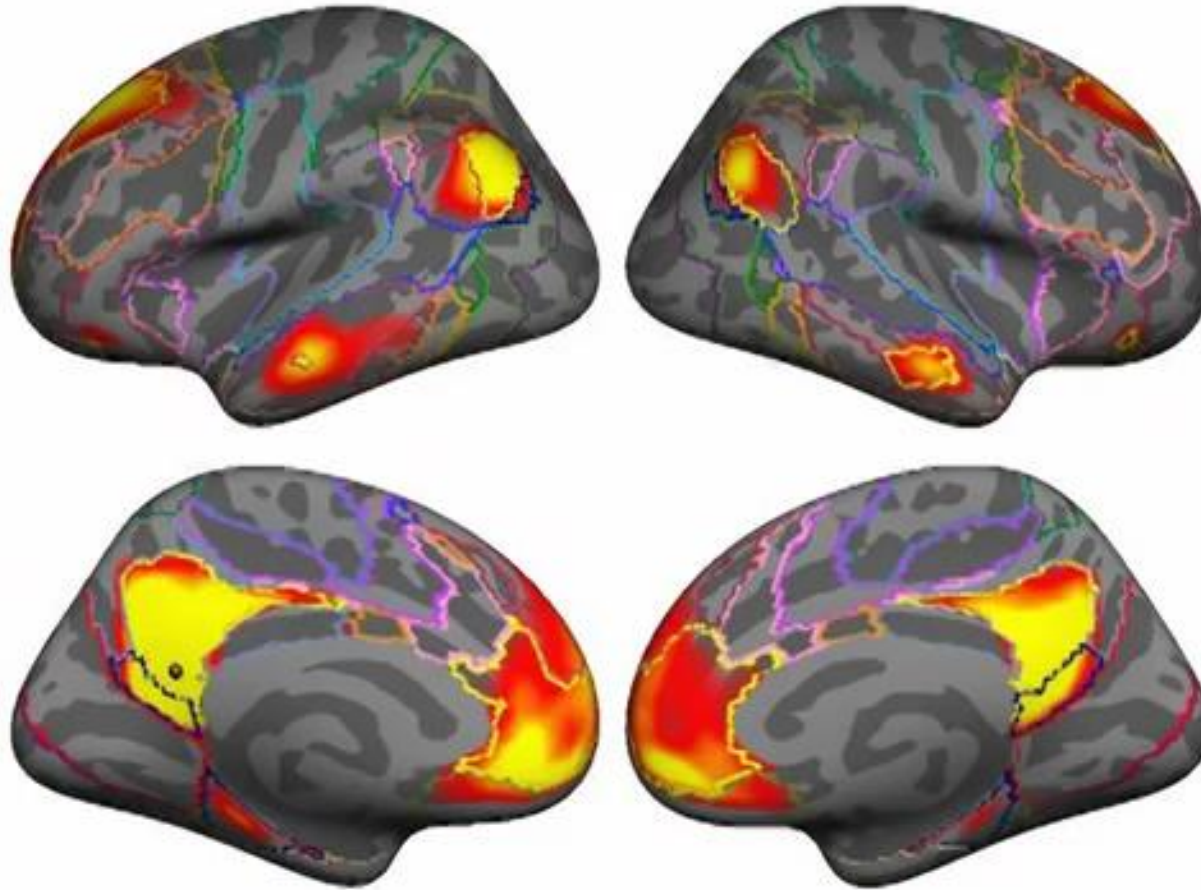


PCC = posterior cingulate cortex; MPFC = medial prefrontal cortex

Functional Connectivity: PCC

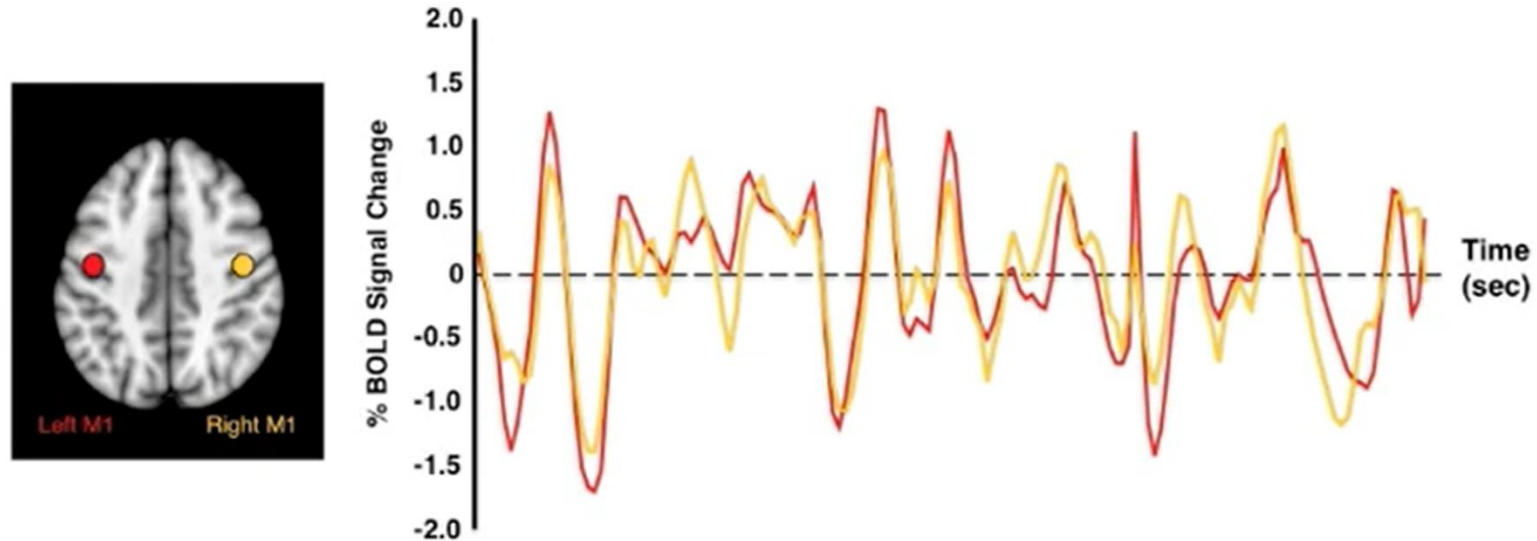


Functional Connectivity: ACC



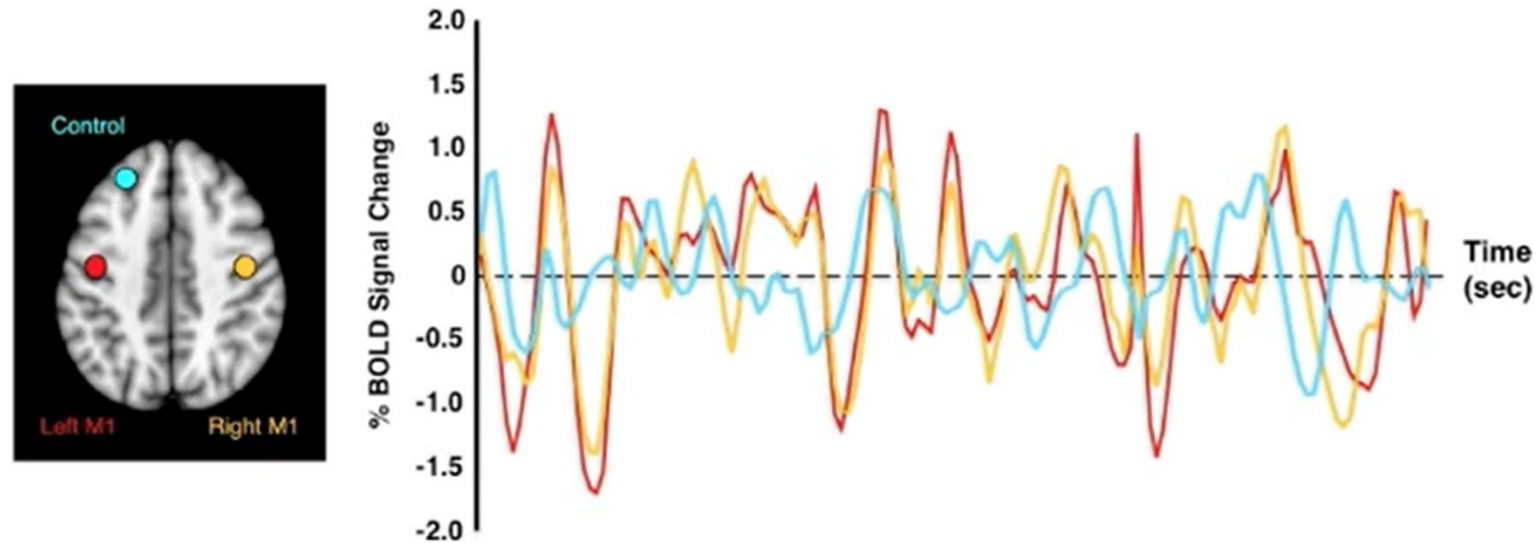
Seed-Based Functional Connectivity Analysis

The brain is functionally organized at rest

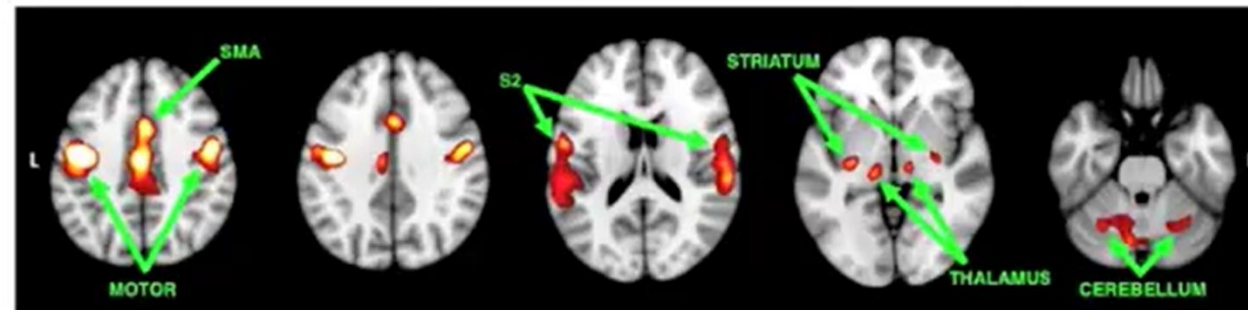


Seed-Based Functional Connectivity Analysis

The brain is functionally organized at rest



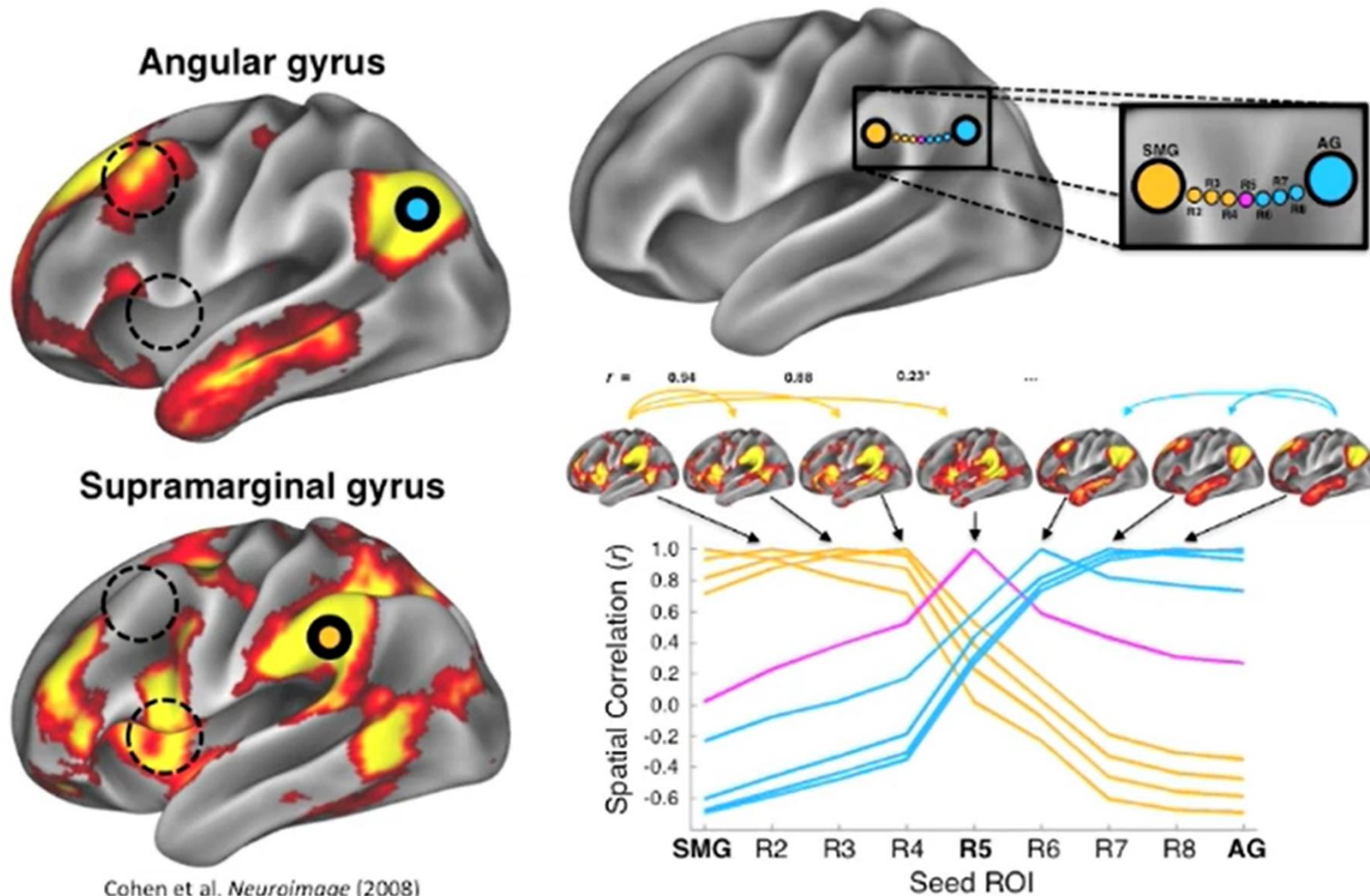
Left M1 Seed RSFC Map



Initial observations by Biswal et al. *MRM* (1995)

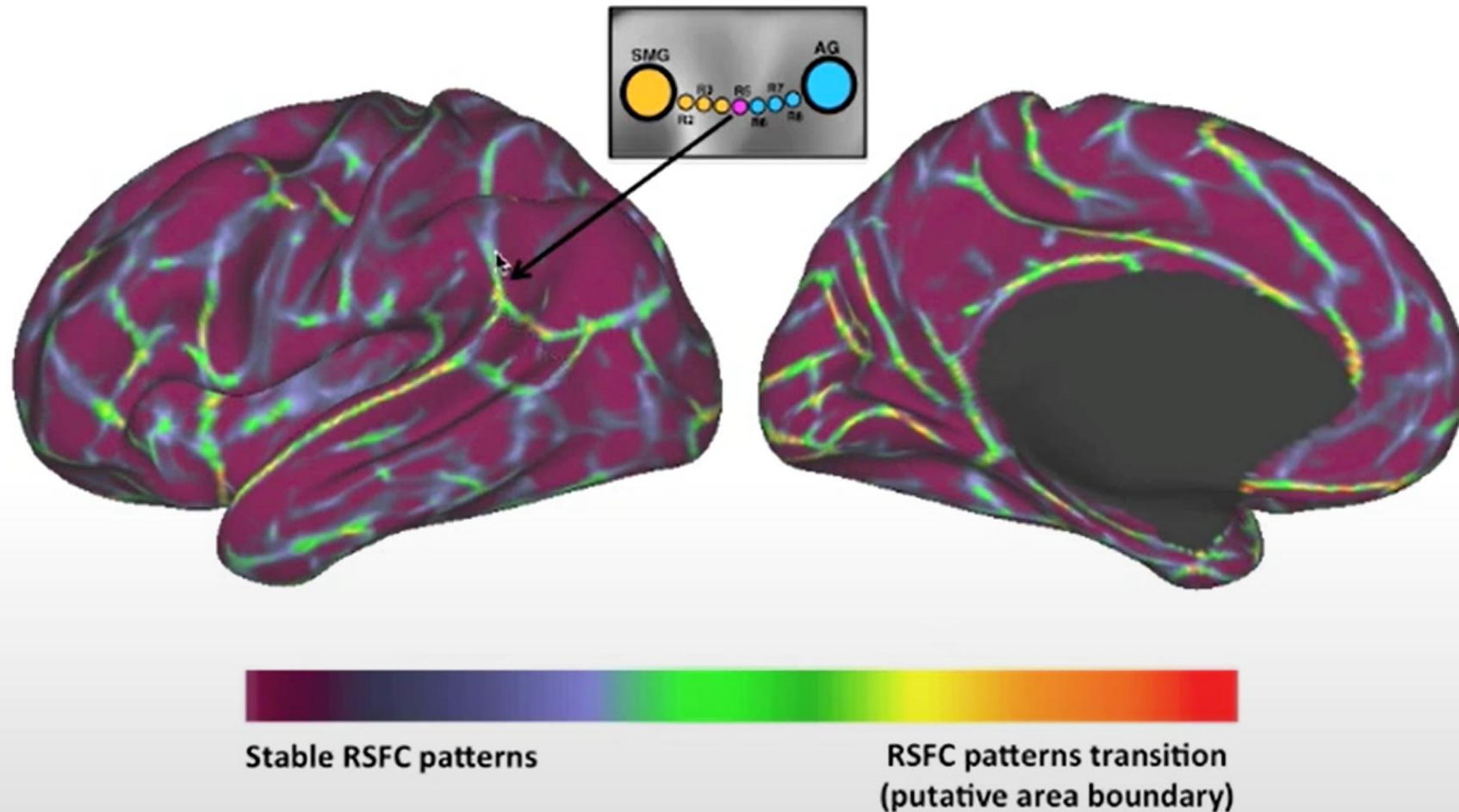
Seed-Based Functional Connectivity Analysis

Adjacent brain areas can exhibit distinct patterns of functional correlations

Cohen et al. *Neuroimage* (2008)

Seed-Based Functional Connectivity Analysis

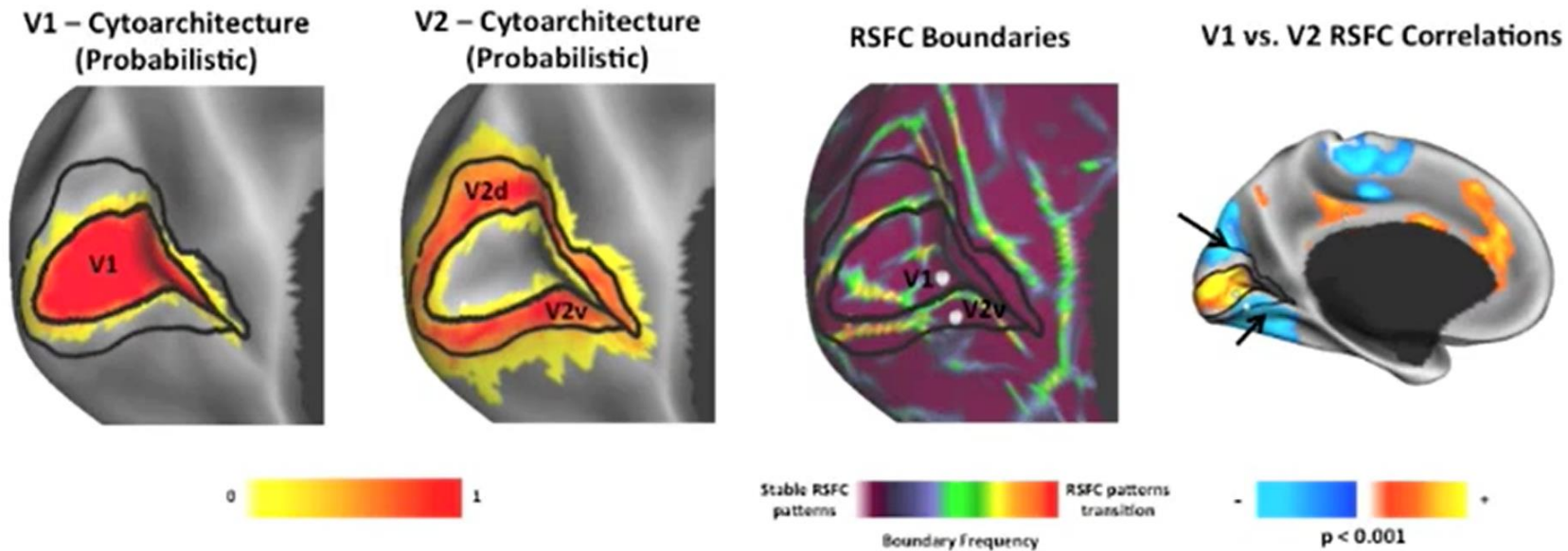
This approach can be applied to locations across the brain to identify putative areal boundaries



Seed-Based Functional Connectivity Analysis

Maps of probabilistic cytoarchitecture can be used to define the V1/V2 border in humans

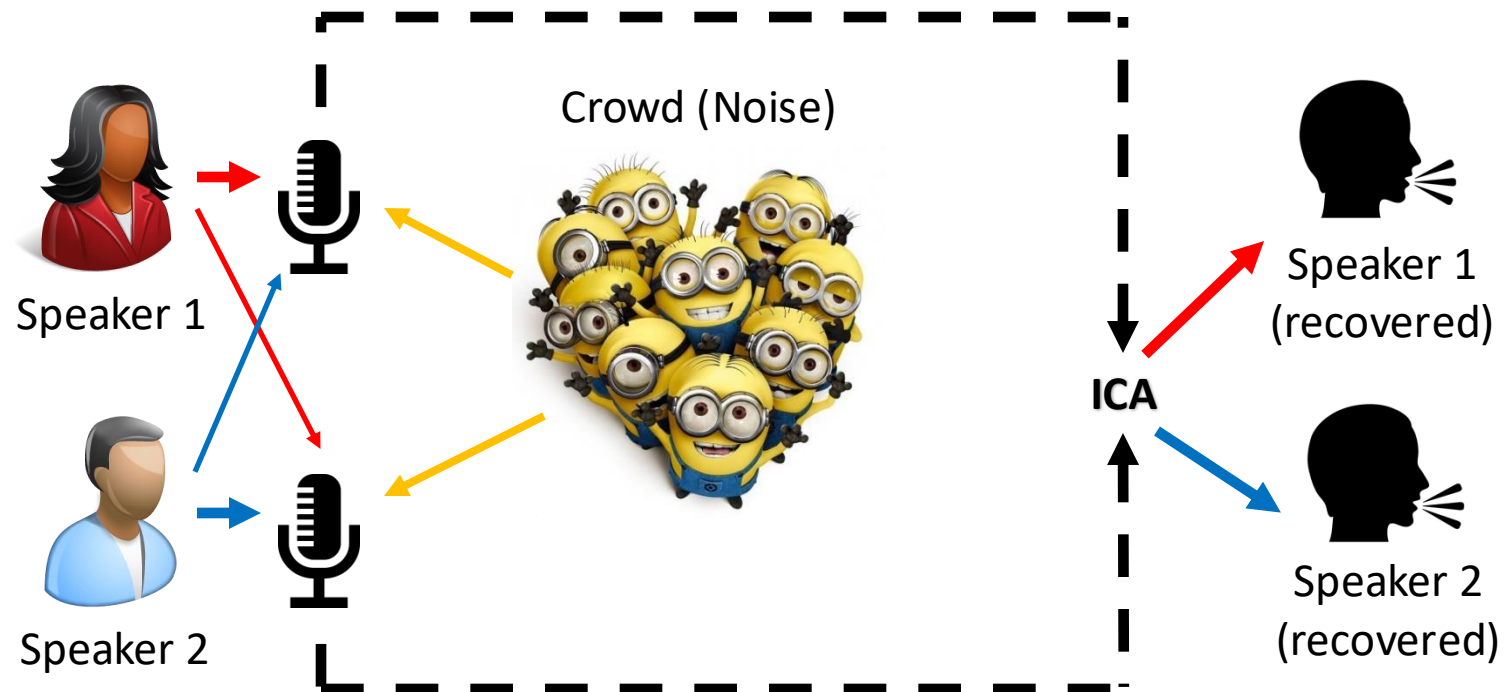
The location of the RSFC-defined area boundary is consistent with the V1/V2 area division defined by cytoarchitecture



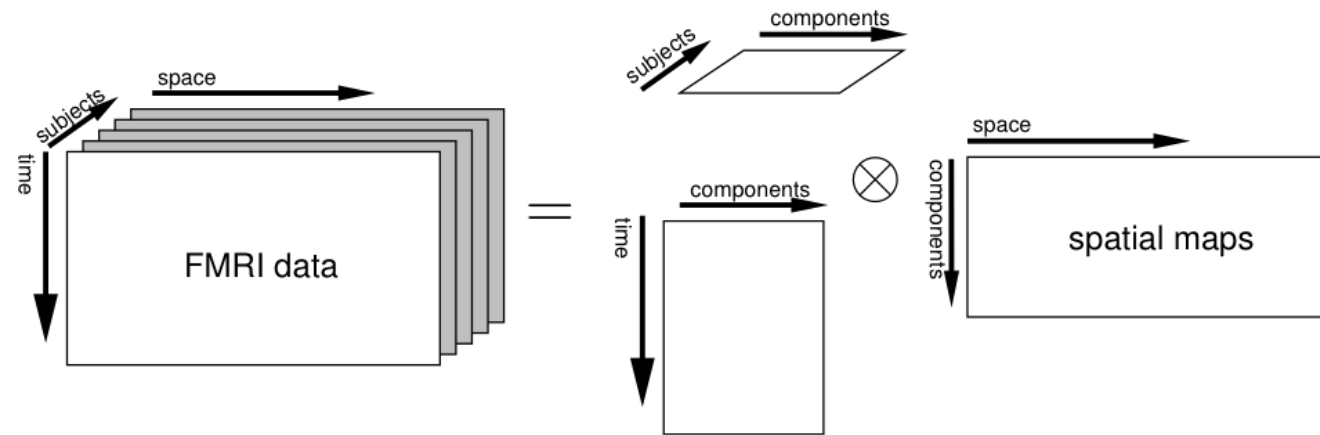
Fischl, Amunts, Zilles, others (2008)

Independent Component Analysis

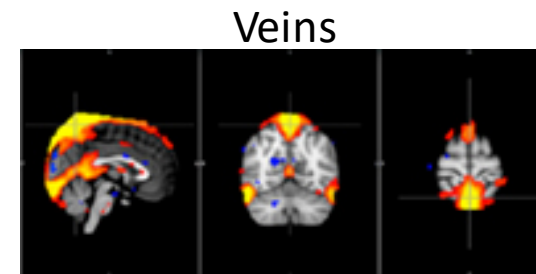
Independent component analysis (ICA) is a computational method for separating a multivariate signal into additive subcomponents. The ICA separation is based on two assumptions: (1) The source signals are independent of each other; (2) The values in each source signal have non-gaussian distributions. ICA is a special case of **blind source separation** (BSS).



Independent Component Analysis

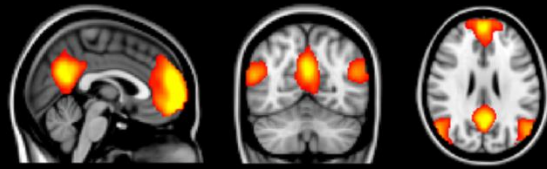


ICA separates a signal into non-overlapping spatial and time components and for this reason can be used in the detection of resting state networks. It is highly data-driven and allows for better removal of noisy components of the signal (motion, scanner drift, etc). It also has been shown to reliably extract default mode network as well as many other networks with very high consistency.

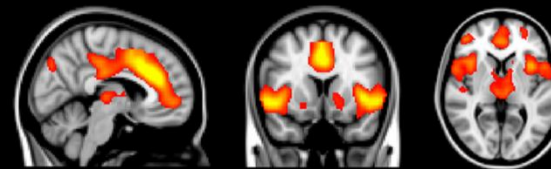


Resting State Networks

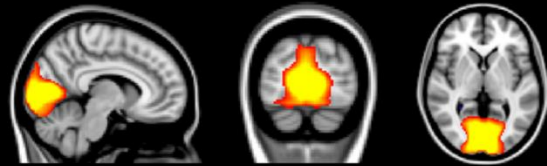
Default Mode



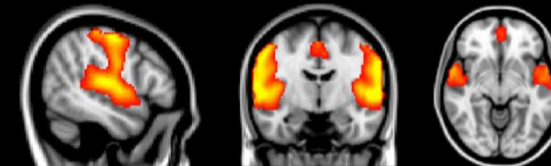
Salience



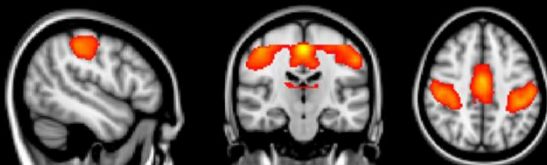
Primary Visual



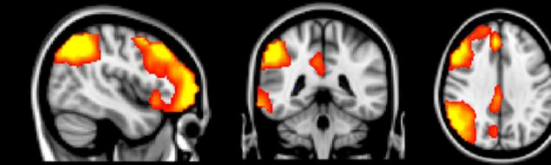
Auditory



Sensory-Motor

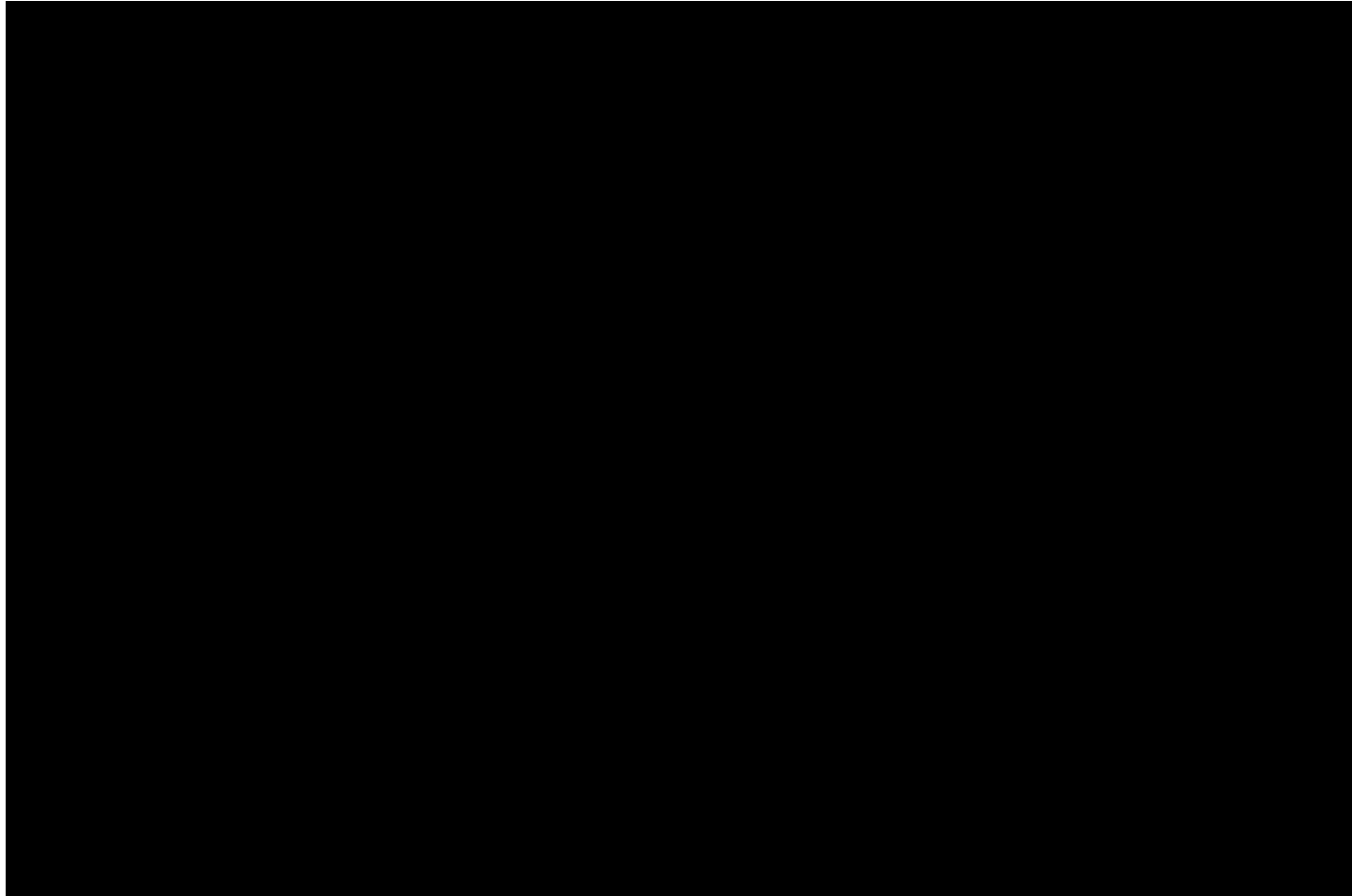


Right Fronto-Parietal

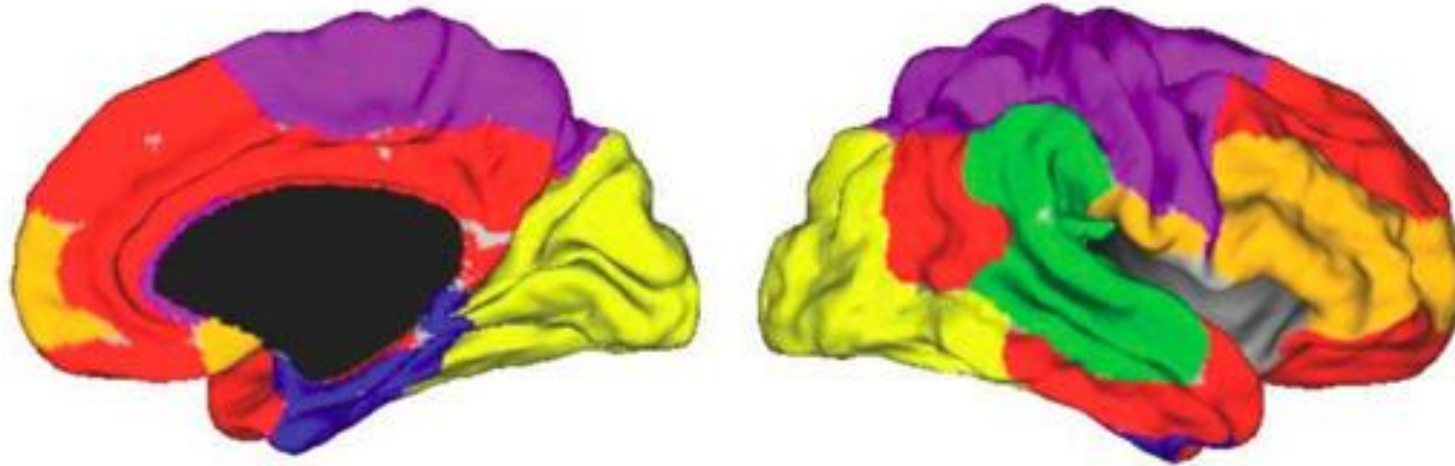


Etc...

Resting State Networks



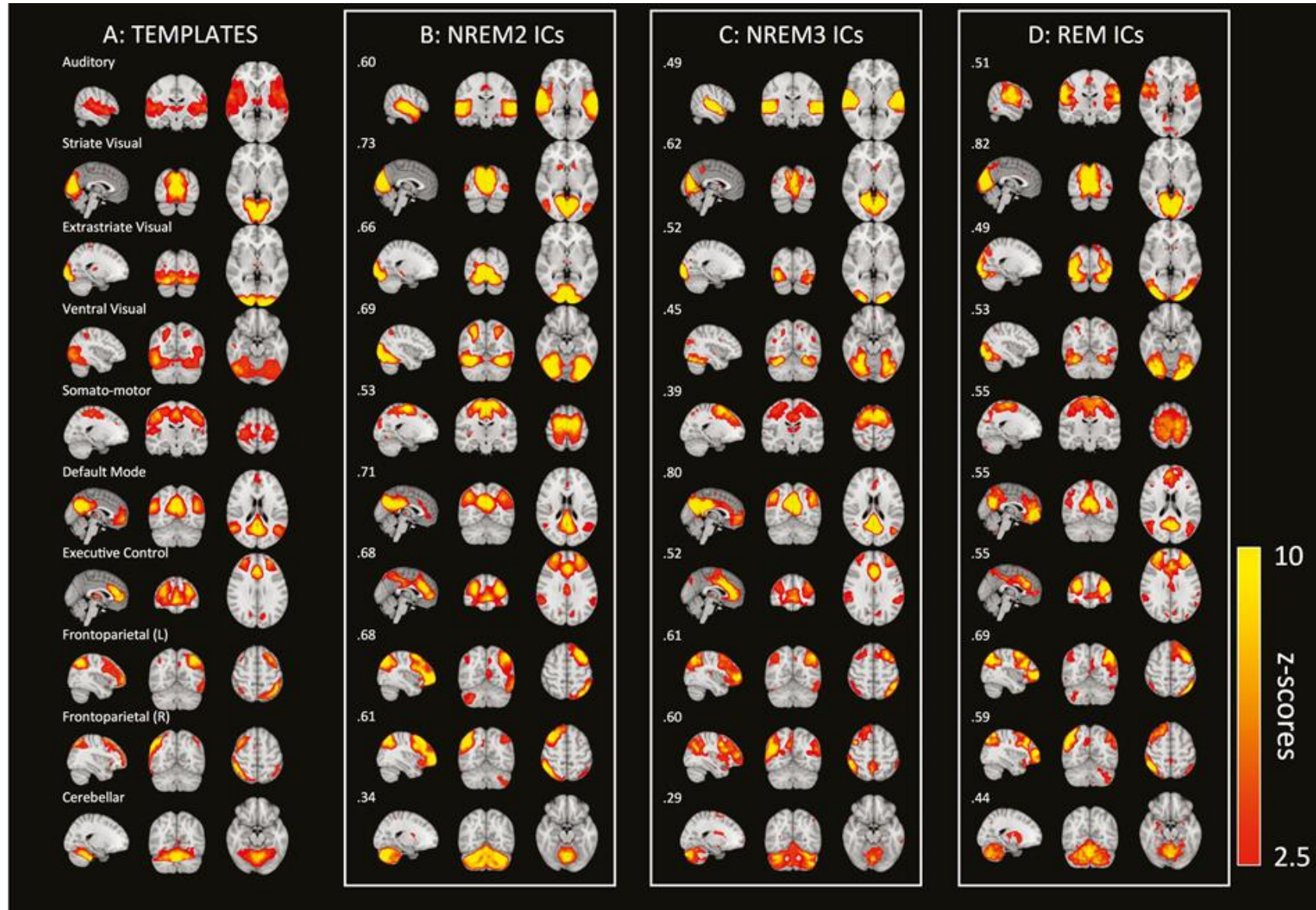
Resting State Networks



Resting-state networks consist of **anatomically separated**, but **functionally connected** regions displaying a high level of correlated BOLD signal activity. These networks are found to be quite consistent across studies, despite differences in the data acquisition and analysis techniques. Most of these resting-state components represent functional networks of regions that are known to **share and support specific cognitive functions**

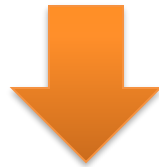
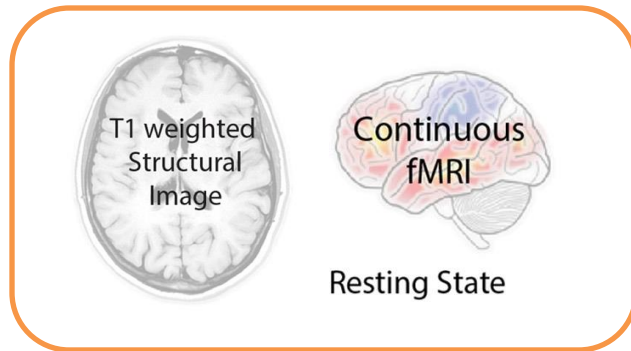
	RSN 1	default mode network
	RSN 2	attention network
	RSN 3	visual recognition
	RSN 4	auditory network
	RSN 5	sensory-motor
	RSN 6	subcortical network

Resting State Networks



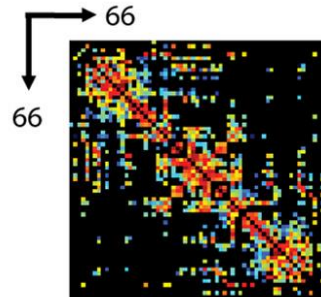
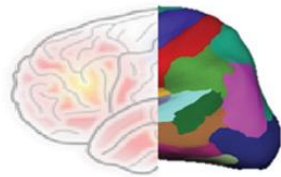
Graph-Analysis – Typical Data Processing

Data Acquisition



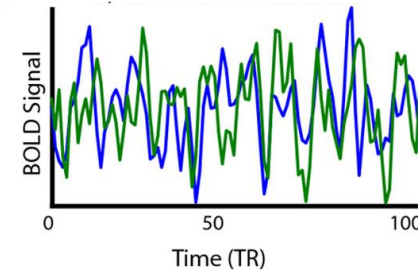
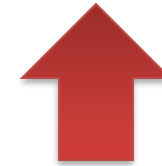
Brain Parcellation

Can be based on individual anatomical or functional data, group-level data or standard atlases



Correlation Matrix

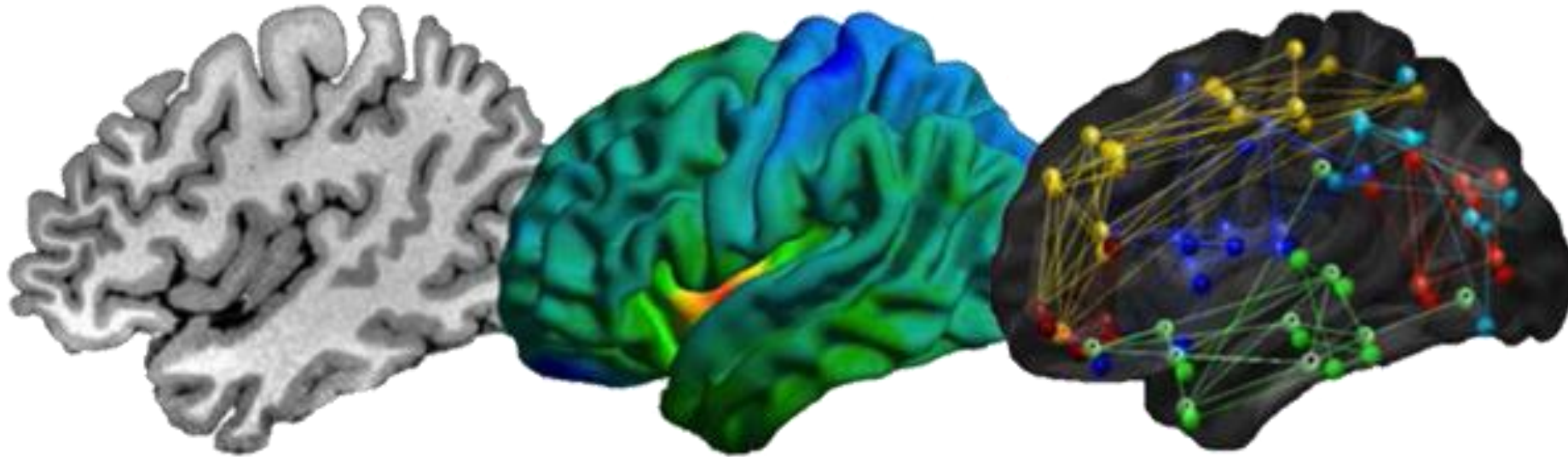
For most analyses a threshold is applied (e.g., $r^2 > 0.3$) to obtain a binary matrix (connection present/absent)



Signal Extraction and Connectivity Computation

Simplest Case: Pearson's correlation coefficient between each pair of ROIs

Graph-Analysis – Typical Data Processing



Graph-Analysis – Examples of Relevant Parameters

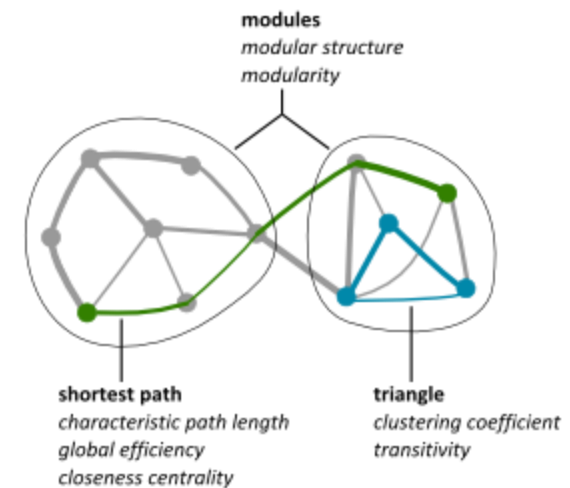
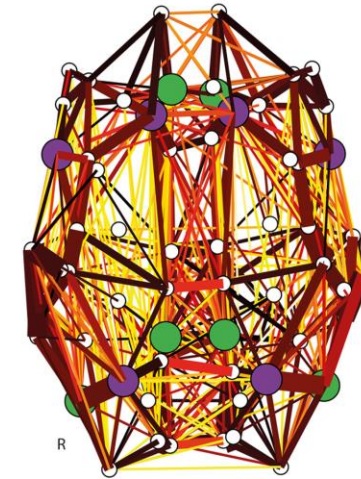
Graphs are mathematical structures used to model pairwise relations between objects. A graph is made up of vertices or nodes and edges or lines that connect them.

Global efficiency: *Is a measure of how efficiently a network exchanges information. It is the average inverse shortest path length in the network, and is inversely related to the characteristic path length.*

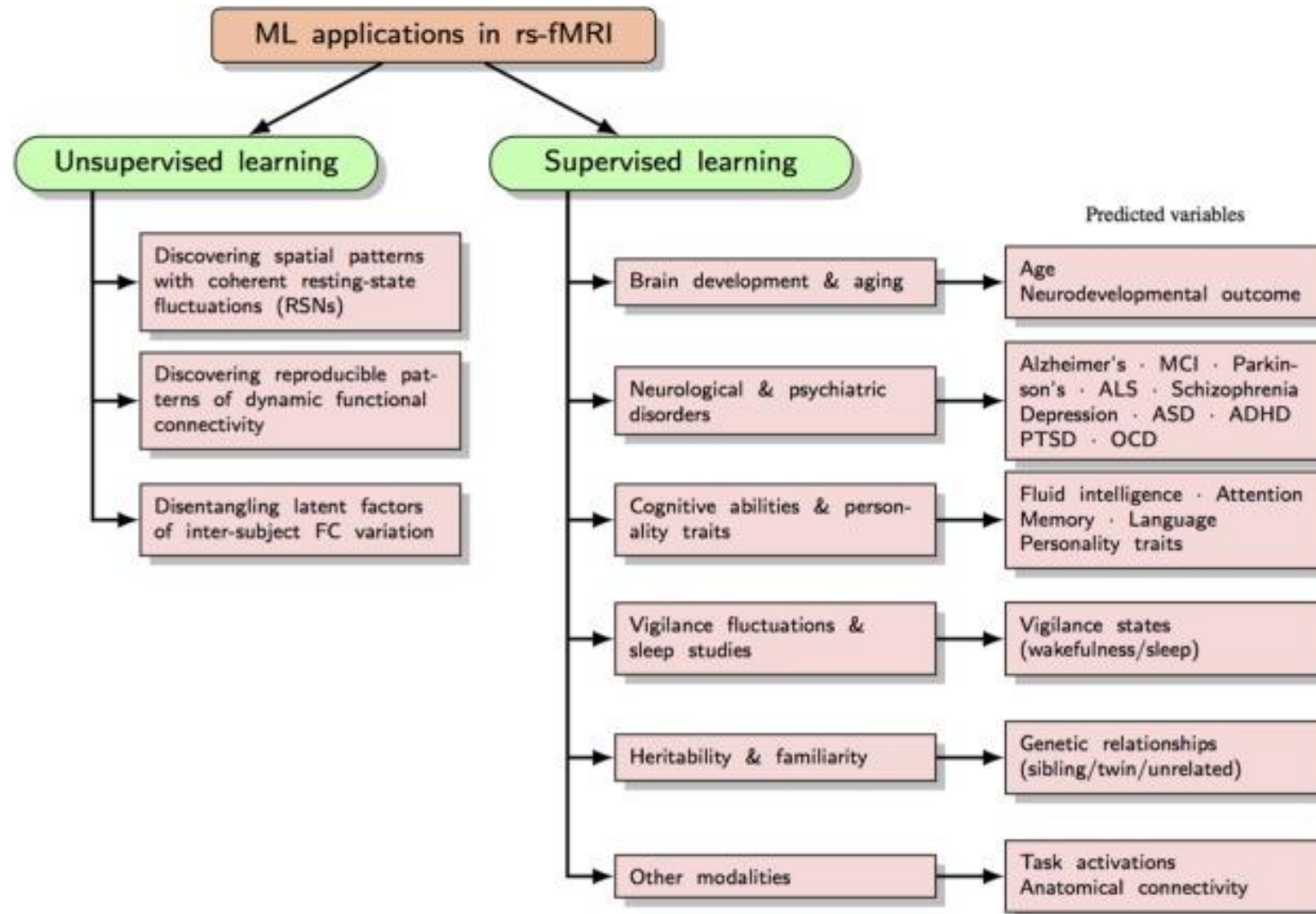
Clustering coefficient: *Is a measure of the degree to which nodes in a graph tend to cluster together. It is the fraction of triangles around a node and is equivalent to the fraction of node's neighbors that are neighbors of each other.*

Characteristic path length: *is the average shortest path length in the network (lengths of shortest paths between all pairs of nodes).*

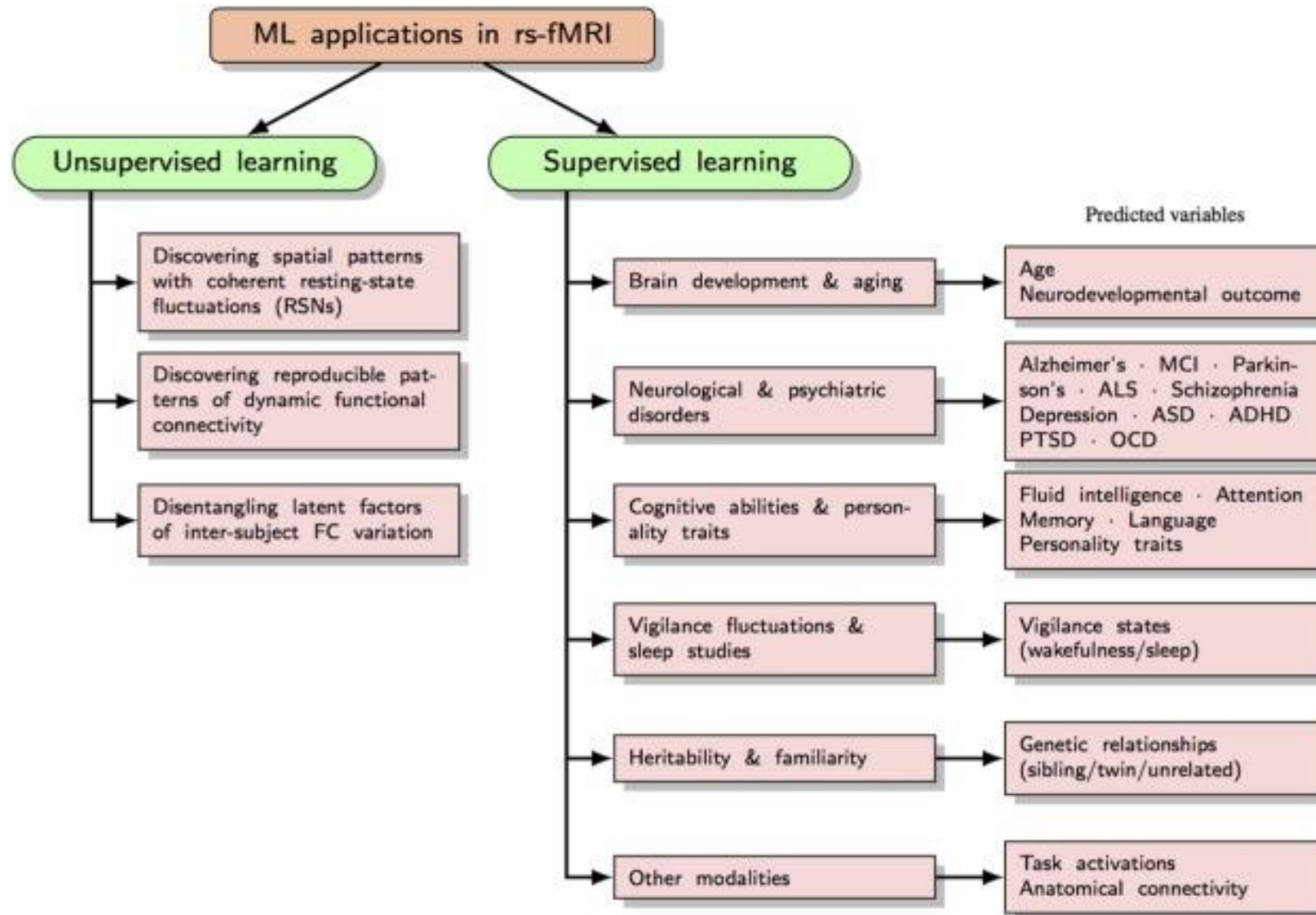
Etc...



Machine Learning and Resting State fMRI



Machine Learning and Resting State fMRI



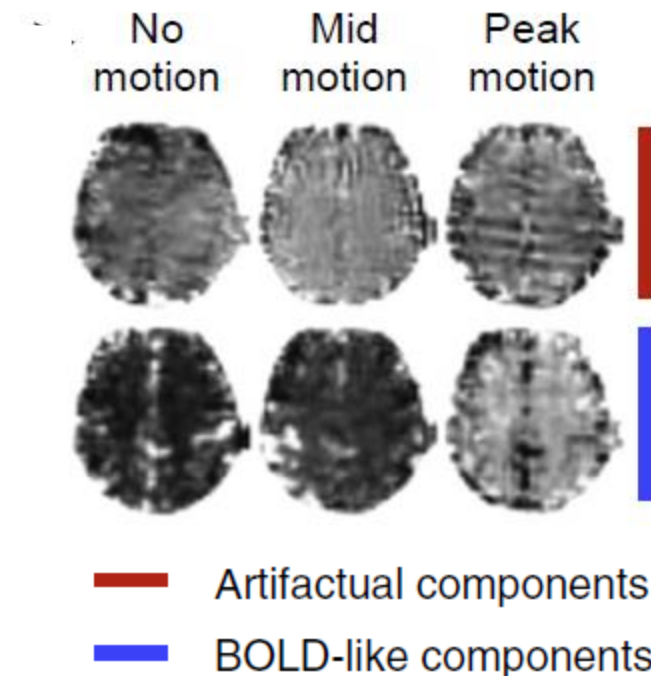
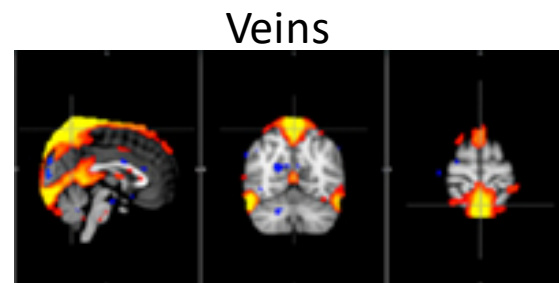
Methodological Limitations: RS fMRI confounds

Potential confounds arises from:

- Head/Body movements
- Cardiac and respiratory cycles,
- Variations in arterial CO₂ concentration
- Blood pressure/cerebral autoregulation
- Vasomotion

Resting-state fMRI confounds: Motion

Movement of the head causes the content of each voxel to change, and this will determine a change in the BOLD signal! This is particularly problematic at tissue interfaces and around large vessels.



Resting-state fMRI confounds: Motion

Functional connectivity conclusions may be erroneous when motion artifacts have a differential effect on resting BOLD signals for between group comparisons!

For Instance...



Children and patient populations are usually more uncomfortable in the MR environment than young healthy controls, resulting in a greater amount of head motion!

Resting-state fMRI confounds: Cardiac and Respiratory

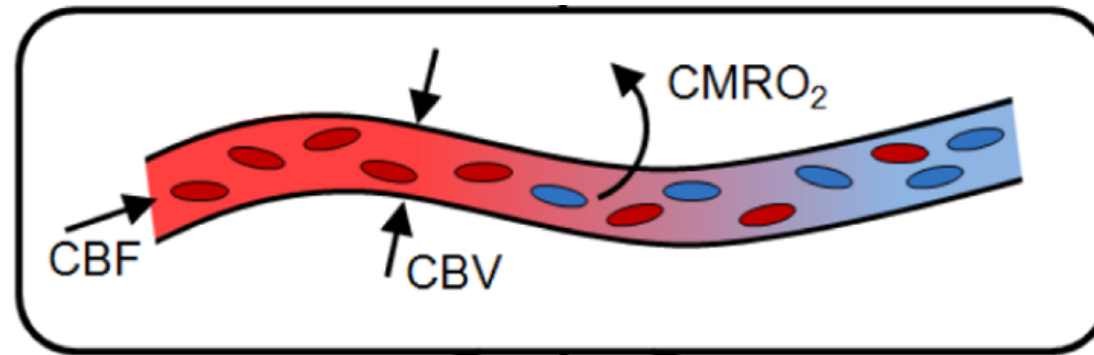
Cardiac and respiratory cycles are relatively high frequency (~ 1 Hz and ~ 0.3 Hz, respectively) compared to the low-frequency (< 0.1 Hz) BOLD fluctuations.

Bulk motion related to the cardiac and respiration cycles will lead to similar confounds as motion of the head itself.

Pulsations of the vessels caused by cardiac-related pressure changes will generate small movements in and around large blood vessels.

Resting-state fMRI confounds: Cardiac and Respiratory

Changes in **arterial CO_2 concentration** and **blood pressure** (both under sympathetic and parasympathetic nervous system control) can determine low-frequency BOLD fluctuations.



Regional low frequency oscillations may artificially inflate functional connectivity measures or cause spurious connectivity patterns where BOLD connectivity is not present or low.

Resting-state fMRI confounds: CO₂ Concentration

Hypercapnia arising from breath-holds increases BOLD signal whilst hypocapnia arising from hyperventilation reduces it.

An **increase in CO₂ concentration** determines an increase in CBF as well as in the BOLD signal: the **increased CBF reduces deoxyhemoglobin concentration!**

The response of the vasculature to CO₂ is very sensitive, thus any changes in respiration rate and resulting CO₂ variations will have consequences for resting BOLD signal fluctuations.

Resting-state fMRI confounds: Blood Pressure

Cerebral autoregulation is the intrinsic dynamic ability of cerebral vessels to maintain steady-state CBF despite fluctuations in arterial blood pressure.

The arterial and arteriolar systems allow to maintain stable CBF by **changing vascular tone** through vasoconstriction and vasodilation processes.

Blood pressure variations are a key **source of CBF fluctuations** due to the delay in dynamic autoregulation.

Fluctuations in arterial blood pressure are likely to have a great influence on resting-state BOLD signal variations.

Resting-state fMRI confounds: Changes in Cognitive State

Implicit to the use of connectivity estimates over extended resting-state sessions is the assumption that the brain state of subjects will be homogeneous.

Brain states associated with considerable neurophysiological changes should be monitored and avoided.

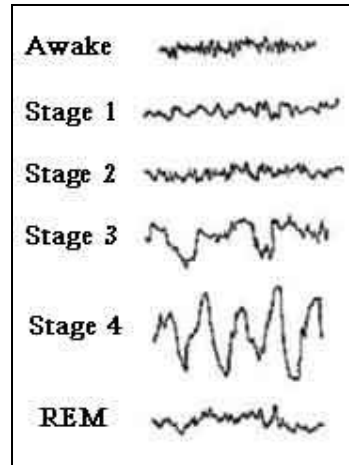
Vigilance shifts indeed occur during rest, therefore it is in the interest of researchers to maintain a steady brain state during fMRI acquisition (e.g., wakefulness).

Fluid Boundaries Between Wake and Sleep: Implications

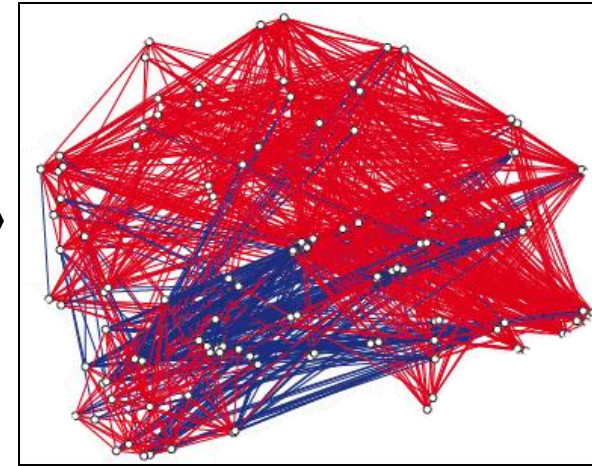
Simultaneous EEG-fMRI recordings



Sleep Scoring



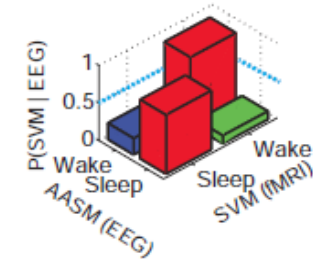
SVM Classifier



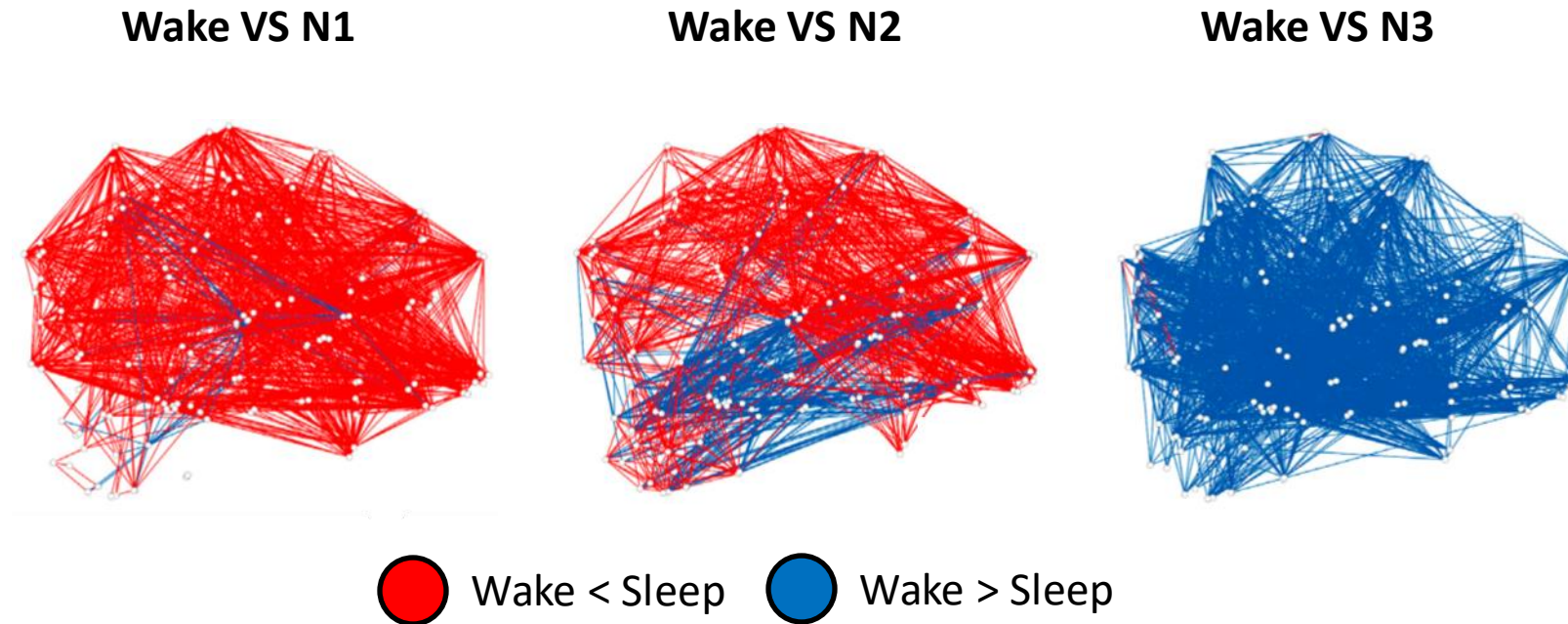
Based on fMRI connectivity matrices

Accuracy corresponding to 75%

Sleep classification was conservative



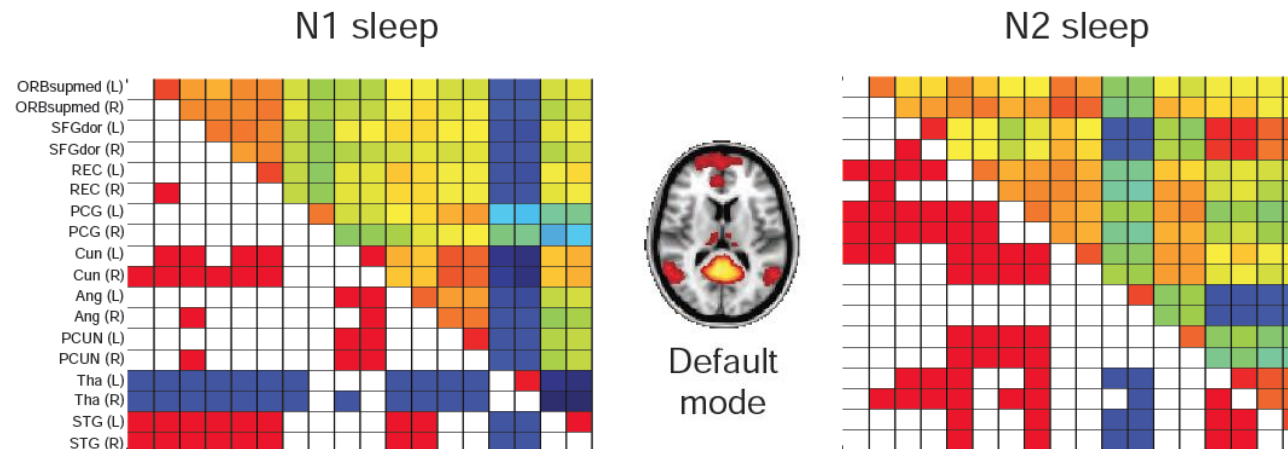
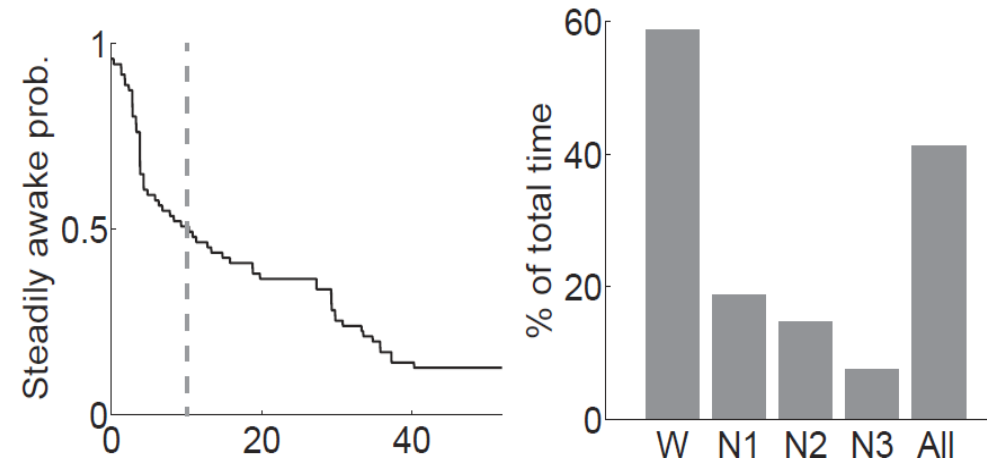
Fluid Boundaries Between Wake and Sleep: Implications



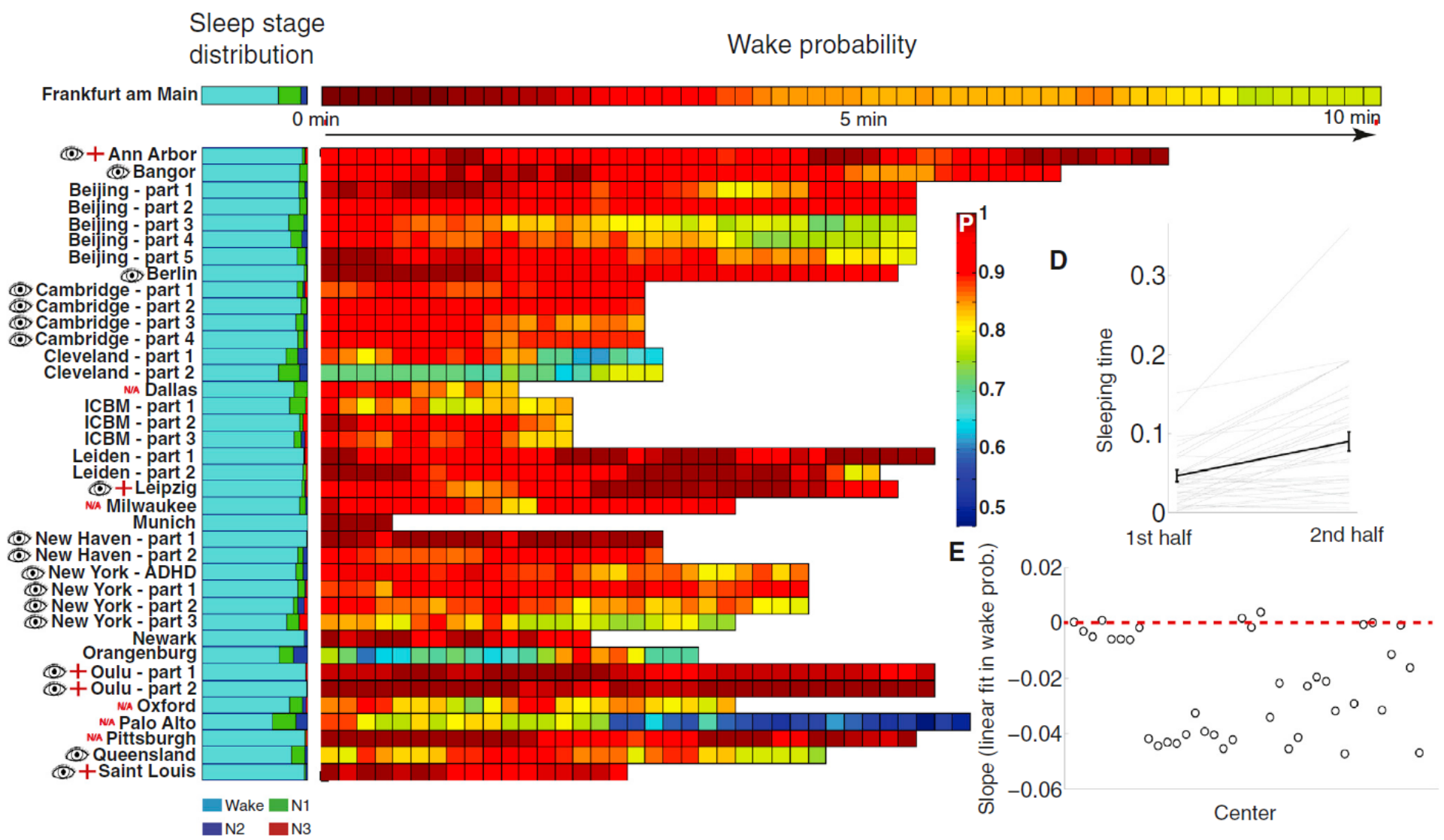
The net reduction in connectivity observed during deep sleep reflect a reduction in brain interregional cross talk, i.e., a loss of global integration well in line with a phenotypically concomitant reduction in consciousness. Modularity, a spatial measure of functional segregation, generally increased across the brain with deepening sleep.

Fluid Boundaries Between Wake and Sleep: Implications

- ✓ EEG-fMRI data collection
- ✓ Classifier trained to detect sleep
- ✓ 1147 fMRI datasets Analyzed
- ✓ Transition to sleep in $\frac{1}{3}$ within 3'

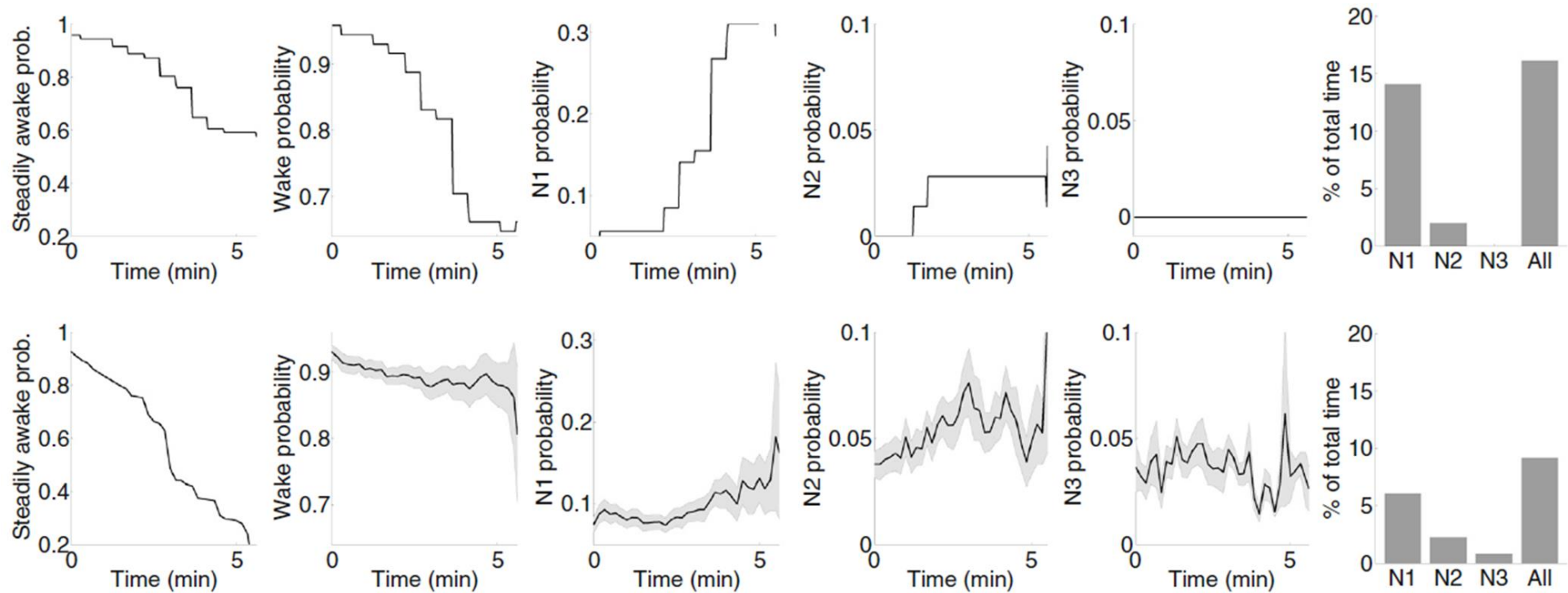


Fluid Boundaries Between Wake and Sleep: Implications



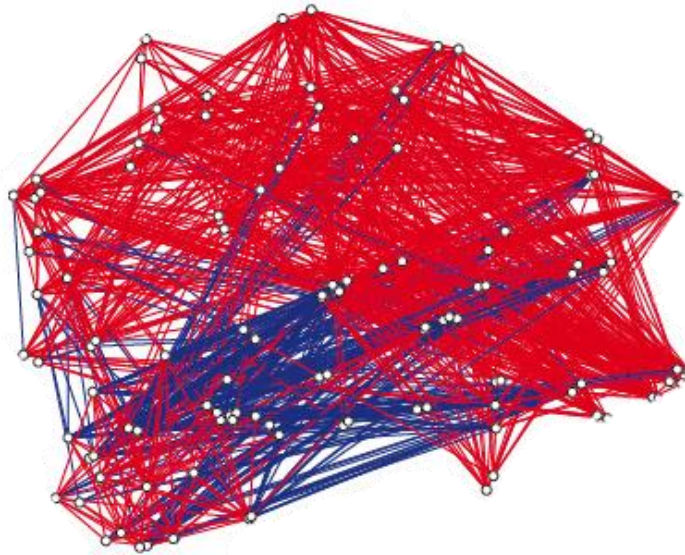
Fluid Boundaries Between Wake and Sleep: Implications

Frankfurt Dataset



Connectome Dataset

Fluid Boundaries Between Wake and Sleep: Implications

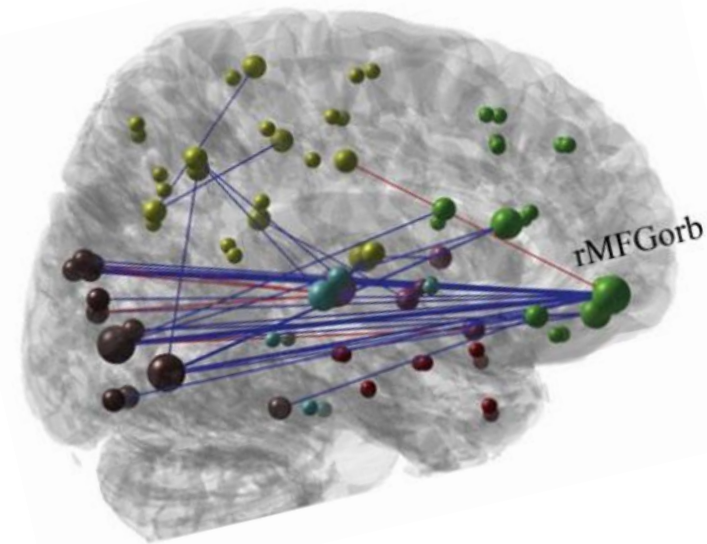


Wake < N2 sleep

Wake > N2 sleep

Tagliazucchi and Laufs, Neuron, 2014

?



social anxiety disorder
VS
healthy controls

*Pipeline for the Analysis
Of Resting-State fMRI Data*

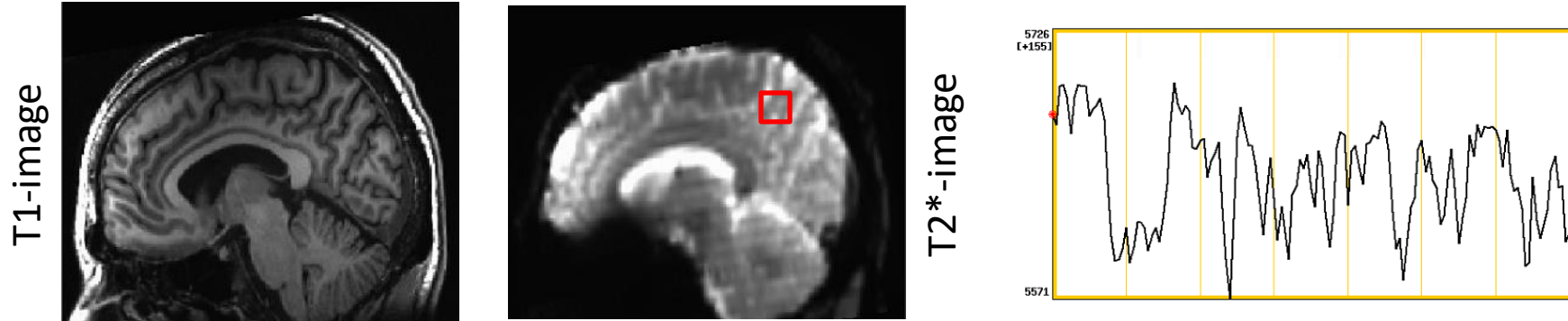
Example: Study Paradigm

3T scanner, Discovery MR750
GE Medical Systems



- 5 min eyes-open EPI resting-state scan: repetition time: 2000 ms, echo time: 25 ms, flip angle: 60, voxel size: 1.8 x 1.8 x 3.5 mm, plane: 128 x 128, number of slices: 40, total number of volumes: 150.
- high-resolution FSPGR T1 anatomical scan: repetition time: 8.2 ms, echo time: 3.2 ms, inversion time: 450 ms, flip angle: 12, voxel size: 1 x 1 x 1 mm, plane: 256 x 256, number of slices: 156.

Example: Data Preparation



Elimination of Initial Dummy Volumes (fMRI signal stabilization)

```
3dTcat -prefix GB_resting_cut GB_resting+orig'[5..149]'
```

Slice temporal alignment

```
3dTshift -prefix GB_resting_tshft GB_resting_cut+orig
```

Spatial alignment (Head Motion Correction)

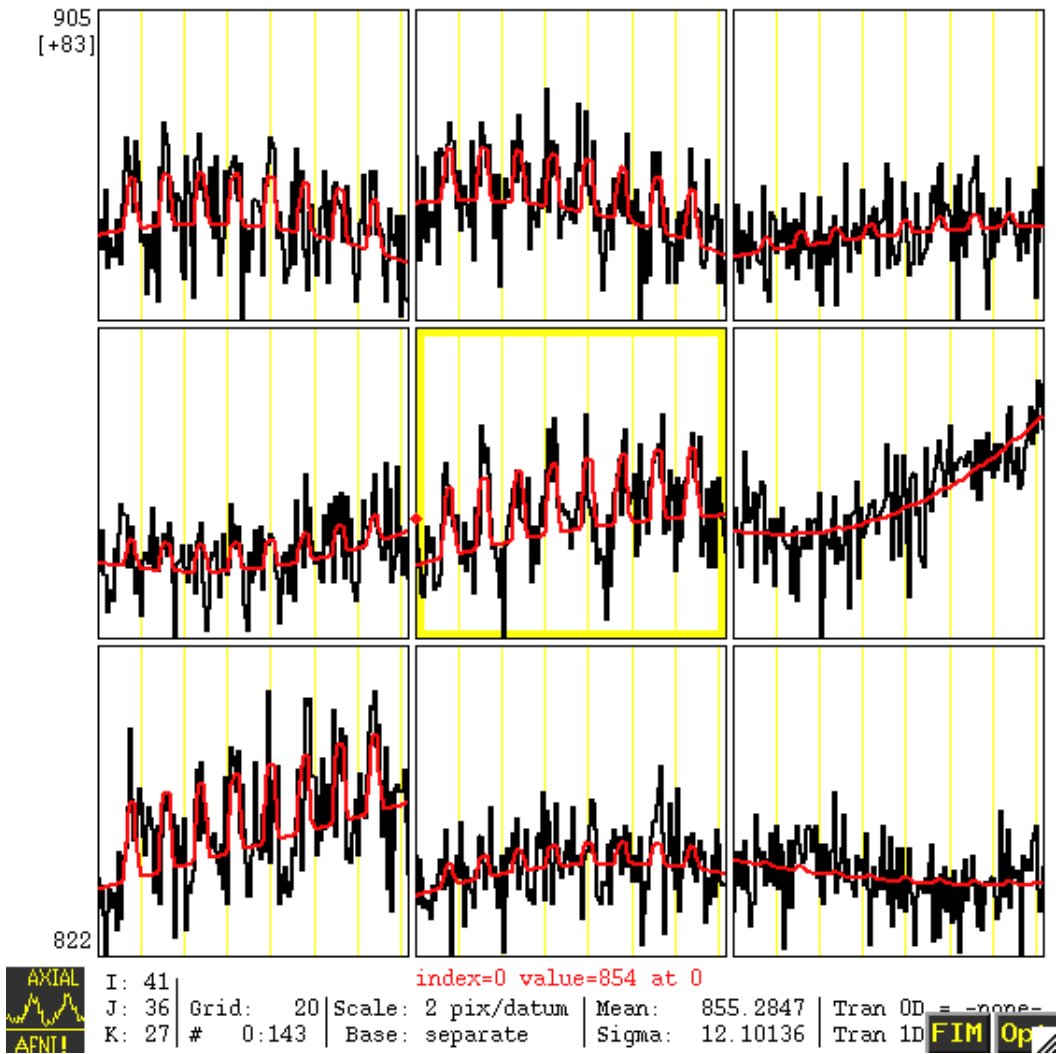
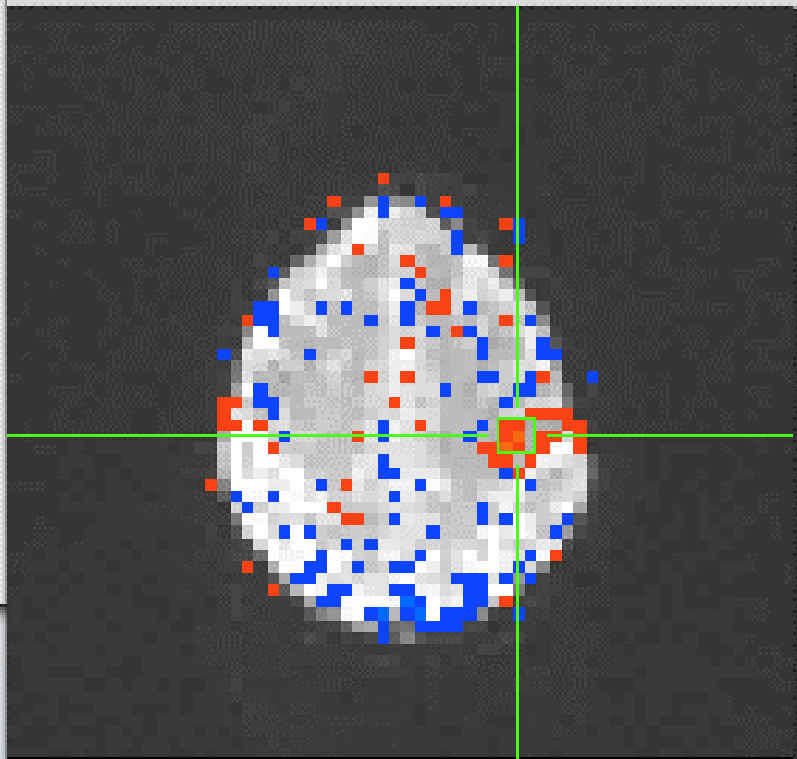
```
3dvolreg -verbose -maxit 50 -dfile GB_resting.out -base  
GB_resting_cut+orig'[0]' -prefix GB_resting_vr GB_resting_tshft+orig
```

Elimination of Artifactual Spikes

```
3dDespike -prefix GB_resting_despike GB_resting_vr+orig
```

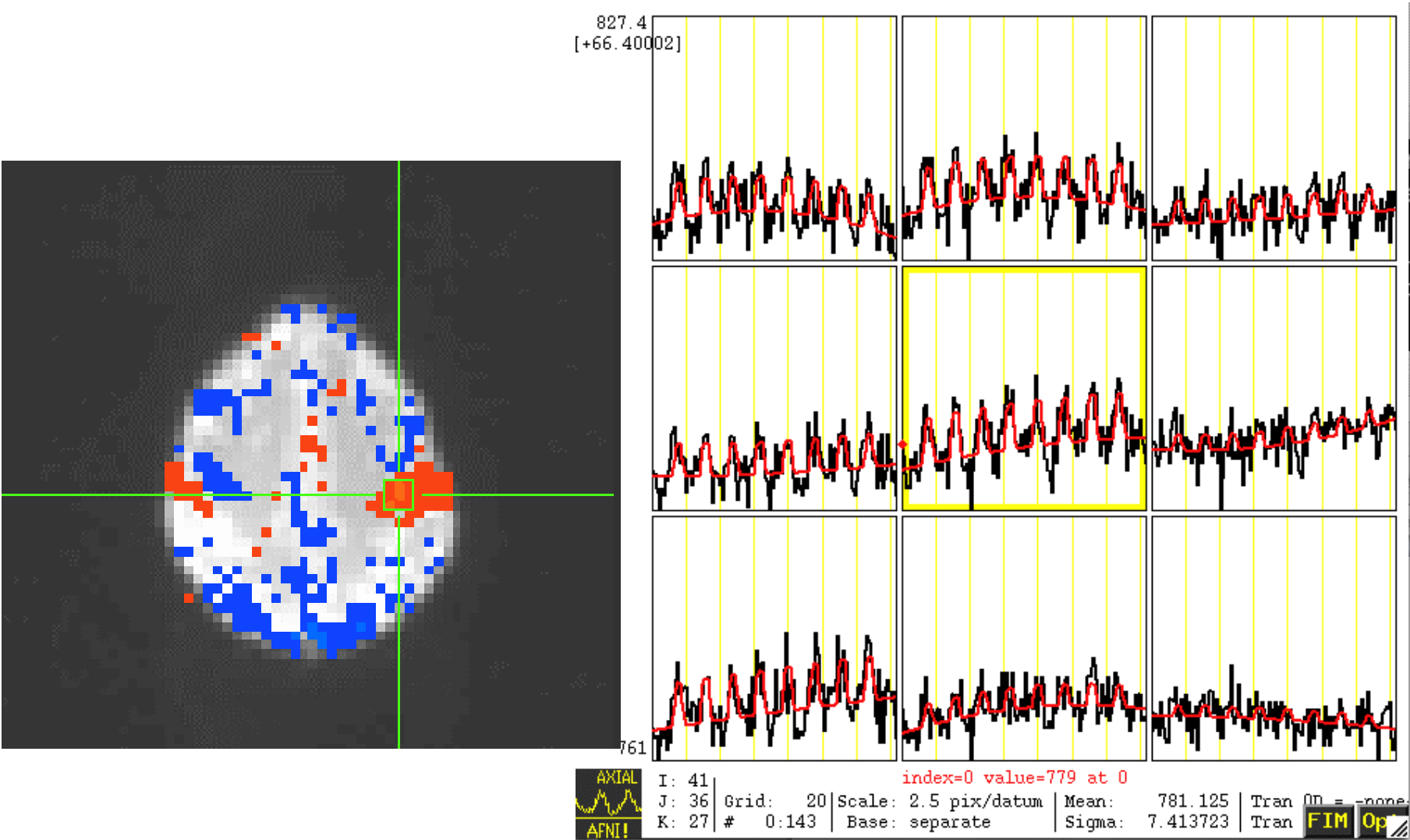

Example: Spatial Smoothing

Model Fit: No smoothing



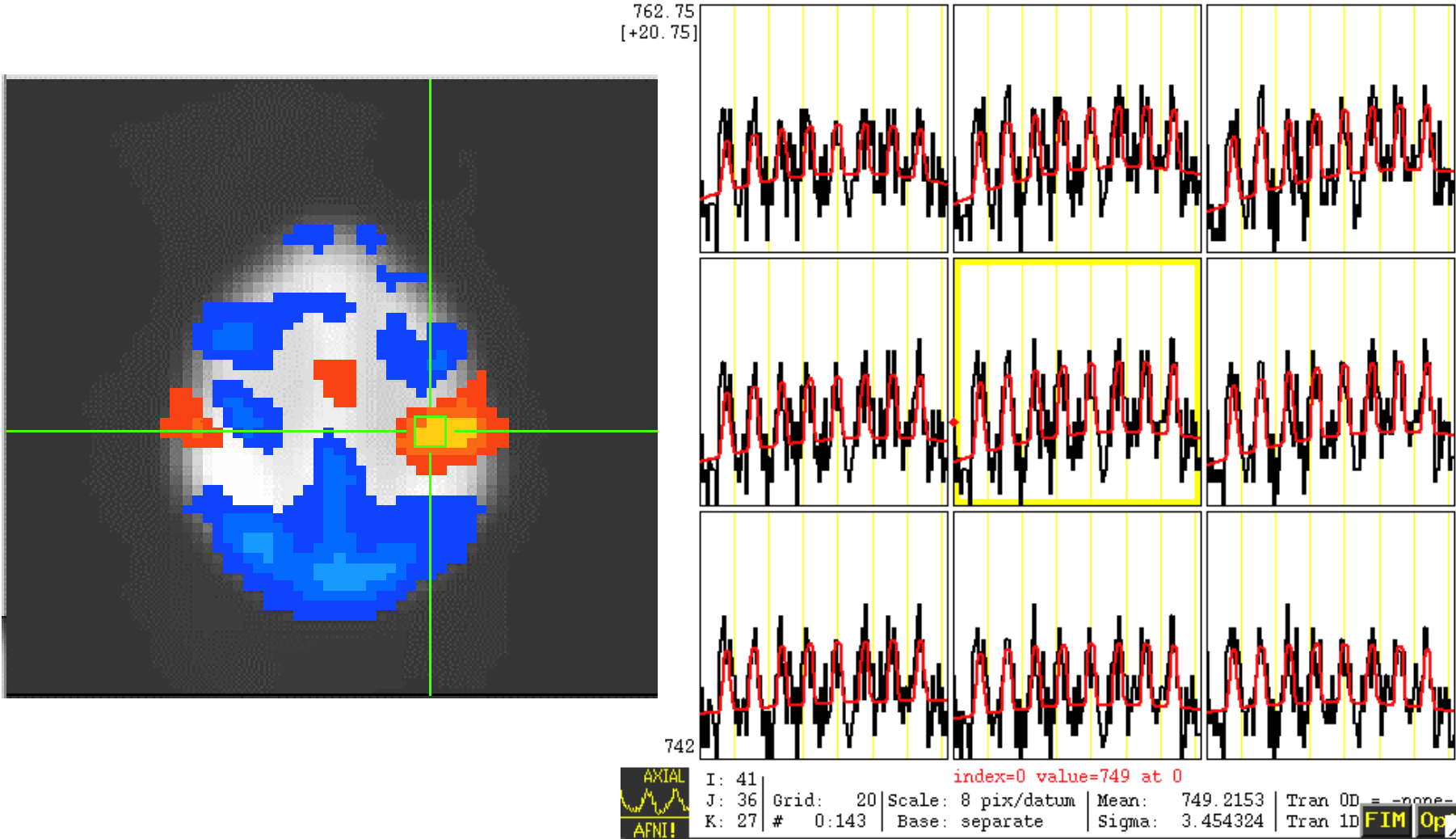
Example: Spatial Smoothing

Model Fit: 4mm smooth



Example: Spatial Smoothing

Model Fit: 12 mm smooth



Example: Basic Preprocessing

Spatial Smoothing (6mm FWHM)

```
3dmerge -doall -1blur_fwhm 6 -prefix GB_resting_fwhm6  
GB_resting_despike+orig
```

Spatial Transformation in Common Space (Talairach Space)

```
adwarp -apar GB_anat+tlrc -dpar GB_resting_fwhm6+orig -dxyz 2 -prefix  
GB_resting_fwhm6
```

Generation of a Brain Mask (Exclusion of voxels outside brain parenchima)

```
3dAutomask -SI 130 -dilate 4 -prefix GB_resting_mask GB_anat+tlrc
```

Calculation of Signal Average for each voxel (baseline)

```
3dTstat -prefix GB_resting_mean GB_resting_fwhm6+tlrc
```

Signal normalization using the mean (baseline correction)

```
3dcalc -fscale -a GB_resting_fwhm6+tlrc -b GB_resting_mean+tlrc -c  
GB_resting_mask_res+tlrc -prefix GB_resting_norm -expr '(a/b*100)*c'
```

Example: Preparation for Data Cleaning

Extraction of Signal from a White Matter ROI

```
3dmaskave -quiet -mrange 1 1 -mask ROI_CLEAN_fr+tlrc  
GB_resting_norm+tlrc > ROI_WM.1D
```

Extraction of Signal from a Ventricular ROI

```
3dmaskave -quiet -mrange 2 2 -mask ROI_CLEAN_fr+tlrc  
GB_resting_norm+tlrc > ROI_LV.1D
```

Deconvolution including stimuli of no interest

```
3dDeconvolve -jobs 4 -input GB_resting_norm+tlrc \  
-polort 2 \  
-num_stimts 8 \  
-stim_file 1 'GB_resting.out[1]' -stim_file 2 'GB_resting.out[2]' -stim_file 3  
'GB_resting.out[3]' \  
-stim_file 4 'GB_resting.out[4]' -stim_file 5 'GB_resting.out[5]' -stim_file 6  
'GB_resting.out[6]' \  
-stim_file 7 'ROI_WM.1D' -stim_file 8 'ROI_LV.1D' \  
-cbucket out_cbucket -x1D out_x1D
```

6 rigid body head motion
parameters (derived by
retrospective motion
correction)

} Timeseries extracted from one WM
region and one CSF region (vs Global
Signal Regression used in the past)

Example: Data Cleaning

Extraction of waveforms representing signal of no-interest

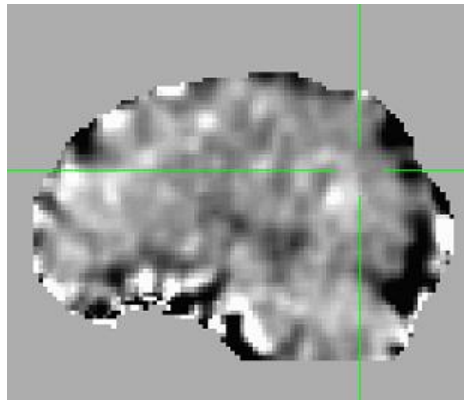
```
3dSynthesize -cbucket out_cbucket+tlrc -matrix out_x1D -select all -prefix  
out_effectsofnointerest
```

Removal of signal of no-interest from data

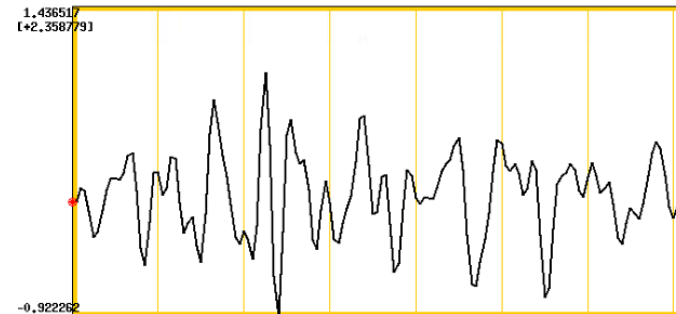
```
3dcalc -datum float -a GB_resting_norm+tlrc -b out_effectsofnointerest+tlrc -  
expr 'a-b' -prefix GB_resting_norm_clean
```

Low-pass filter at 0.1 Hz

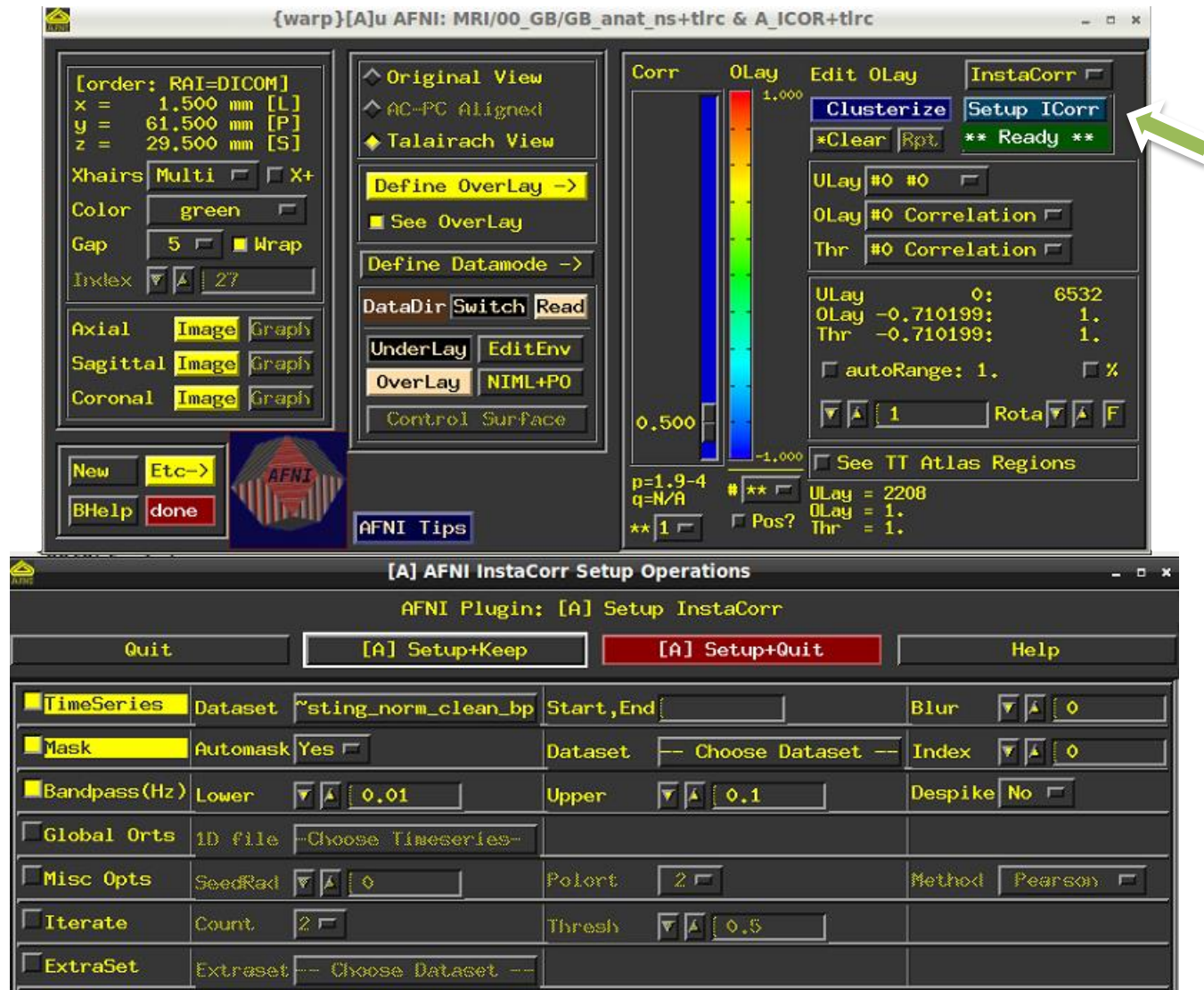
```
3dFourier -prefix GB_resting_norm_clean_bp -lowpass 0.1  
GB_resting_norm_clean+tlrc
```



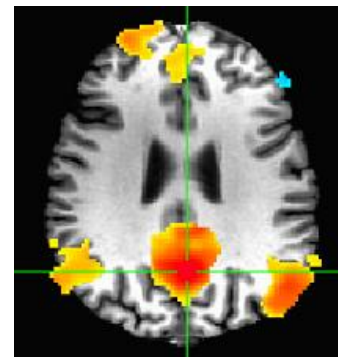
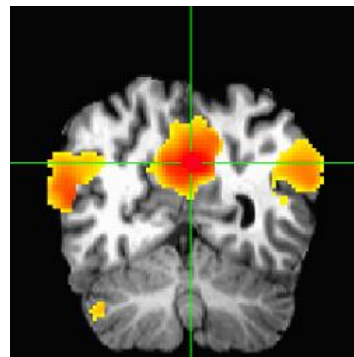
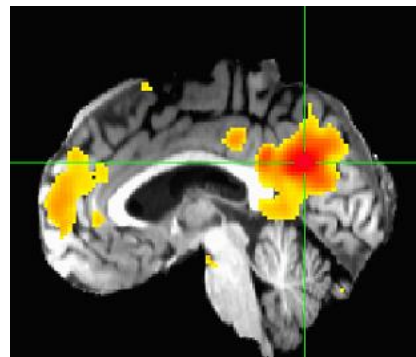
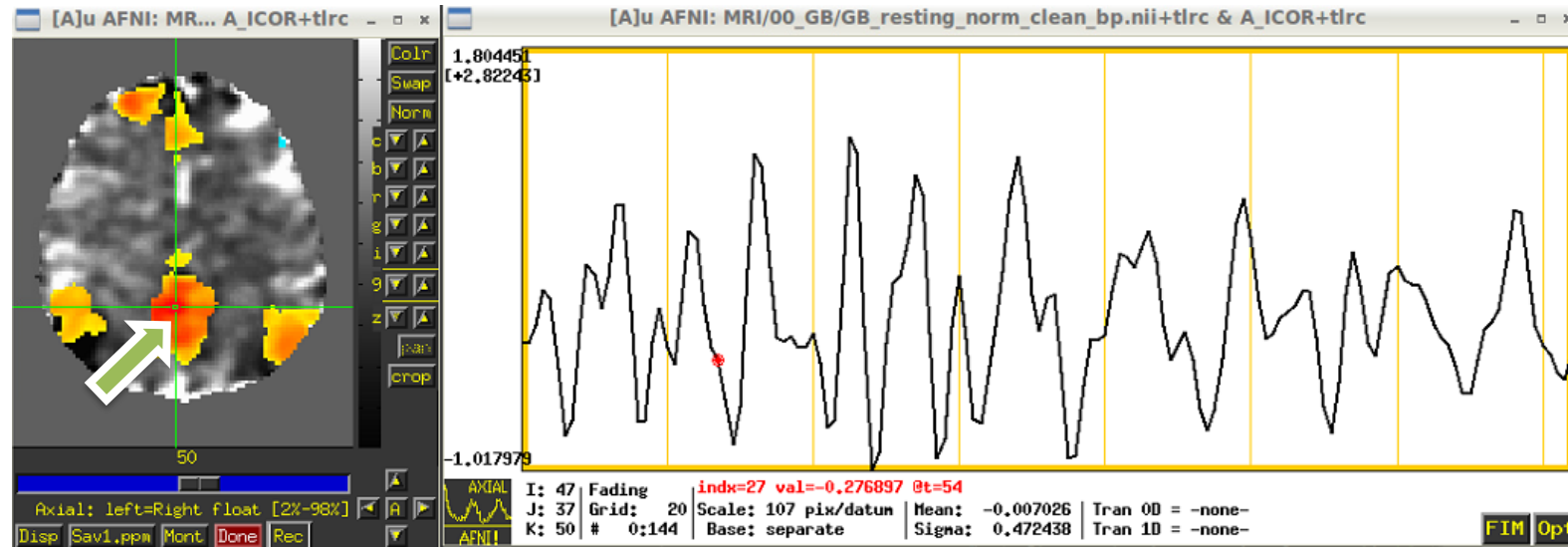
BOLD modulations of neural origin are absent above this frequency, whereas artifacts of various origin, including cardio-pulmonary pulsations, are not



Seed-Based FC Analysis: FC Calculation



Seed-Based FC Analysis: Results



Conclusions

Key-Points

- Resting-State paradigms can be used to study the brain functional organization: several networks can be identified!
- The Default Mode Network is the main functional network at rest, and its activity is reduced during active tasks
- Changes in the DMN and in other resting-state networks seem to be associated with specific clinical conditions.
- Resting-State networks are investigated using seed-based functional connectivity or independent component analysis
- Several factors can modulate Resting-State connectivity: physiological noise may lead to spurious correlations!

Suggested Reading

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